

IR-11 Solids

CONTENTS

- IR-11.1 Introduction
 - IR-11.1.1 General
 - IR-11.1.2 Stoichiometric and non-stoichiometric phases
- IR-11.2 Names of solid phases
 - IR-11.2.1 General
 - IR-11.2.2 Mineral names
- IR-11.3 Chemical composition
 - IR-11.3.1 Approximate formulae
 - IR-11.3.2 Phases with variable composition
- IR-11.4 Point defect (Kröger–Vink) notation
 - IR-11.4.1 General
 - IR-11.4.2 Indication of site occupancy
 - IR-11.4.3 Indication of crystallographic sites
 - IR-11.4.4 Indication of charges
 - IR-11.4.5 Defect clusters and use of quasi-chemical equations
- IR-11.5 Phase nomenclature
 - IR-11.5.1 Introduction
 - IR-11.5.2 Recommended notation
- IR-11.6 Non-stoichiometric phases
 - IR-11.6.1 Introduction
 - IR-11.6.2 Modulated structures
 - IR-11.6.3 Crystallographic shear structures
 - IR-11.6.4 Unit cell twinning or chemical twinning
 - IR-11.6.5 Infinitely adaptive structures
 - IR-11.6.6 Intercalation compounds
- IR-11.7 Polymorphism
 - IR-11.7.1 Introduction
 - IR-11.7.2 Use of crystal systems
- IR-11.8 Final remarks
- IR-11.9 References

IR-11.1 INTRODUCTION

IR-11.1.1 **General**

This chapter deals with some aspects of terminology, nomenclature and notation for solids. However, in cases where detailed structural information is to be conveyed, fully systematic names can be difficult to construct. An attempt to deal with this problem is described in Ref. 1.

IR-11.1.2 **Stoichiometric and non-stoichiometric phases**

In binary and multi-component systems, intermediate crystalline phases (stable or metastable) may occur. Thermodynamically, the composition of any such phase is variable. In some cases, such as sodium chloride, the possible variation in composition is very small. Such phases are called stoichiometric. However, in other phases appreciable variations in composition can occur, for example in wustite (nominally FeO).

These are called non-stoichiometric phases. In general, it is possible to define an ideal composition relative to which the variations occur. This composition, called the stoichiometric composition, is usually that in which the ratio of the numbers of different atoms corresponds to the ratio of the numbers of normal crystallographic sites of different types in the ideal (ordered) crystal.

This concept can be used even when the stoichiometric composition is not included in the homogeneity range of the phase. The term 'non-stoichiometric' does not mean phases with complex formulae, but those with *variable composition*, for which the term *solid mixture* is an alternative. Formerly, the term *solid solution* was used, but this term is recommended to apply only in the following sense.²⁻⁴ *Mixture* is used to describe a solid phase containing more than one substance, when all substances are treated in the same way. *Solution* is used to describe a liquid or solid phase containing more than one substance when, for convenience, one (or more) of the substances, called the *solvent*, is treated differently from the other substances, called *solutes*. For non-stoichiometric phases, each atom or group of atoms that contributes to the non-stoichiometry is treated equally, so the term *mixture* is appropriate.

IR-11.2 NAMES OF SOLID PHASES

IR-11.2.1 **General**

Names for stoichiometric phases, such as NaCl, are derived simply as in Chapter IR-5, whereas formulae are derived as presented in Chapter IR-4. Although NaCl in the solid state consists of an infinite network of units, $(\text{NaCl})_\infty$, the compound is named sodium chloride and represented symbolically as NaCl.

However, for non-stoichiometric phases and solid solutions, formulae are preferable to names, since strictly systematic names tend to be inconveniently cumbersome. They should be used only when unavoidable (*e.g.* for indexing) and should be constructed in the following style.

Examples:

1. iron(II) sulfide (iron deficient)
2. molybdenum dicarbide (carbon excess)

IR-11.2.2 Mineral names

Mineral names should be used only to designate actual minerals and not to define chemical composition. Thus, the name calcite refers to a particular mineral (contrasted with other minerals of similar composition) and is not a term for the chemical compound the composition of which is properly expressed by the name calcium carbonate.

A mineral name may, however, be used to indicate the structure type. Where possible, a name that refers to a general group should replace a more specific name. For example, large numbers of minerals have been named that are all spinels, but which have widely differing atomic constituents. In this case, the generic name '*spinel* type' should be used rather than the more specific names chromite, magnetite, *etc.* The generic mineral name, printed in italics, should be accompanied by a representative chemical formula. This is particularly important for *zeolite* types.⁵

Examples:

1. FeCr_2O_4 (*spinel* type)
2. BaTiO_3 (*perovskite* type)

IR-11.3 CHEMICAL COMPOSITION**IR-11.3.1 Approximate formulae**

The formula used in any given case depends upon how much information is to be conveyed. A general notation, which can be used even when the mechanism of the variation in composition is unknown, is to put the sign $\sim$ (read as *circa*, or 'approximately') before the formula.

Examples:

1. $\sim\text{FeS}$
2. $\sim\text{CuZn}$

If it is desirable to give more information, one of the notations described below may be used.

IR-11.3.2 Phases with variable composition

For a phase where the variation in composition is caused solely or partially by isovalent substitution, the symbols of the atoms or groups that replace each other may be separated by a comma and placed together between parentheses. If possible, the formula is written so that the limits of the homogeneity range, when one or the other of the two atoms or groups is lacking, are represented.

Examples:

1. (Cu,Ni) denotes the complete range of compositions from pure Cu to pure Ni.
2. K(Br,Cl) comprises the range from pure KBr to pure KCl.

Phases for which substitution also results in vacant positions are denoted in the same way.

Examples:

3. $(\text{Li}_2, \text{Mg})\text{Cl}_2$ denotes a solid mixture intermediate in composition between LiCl and MgCl_2 .
4. $(\text{Al}_2, \text{Mg}_3)\text{Al}_6\text{O}_{12}$ represents a solid mixture intermediate in composition between MgAl_2O_4 ($= \text{Mg}_3\text{Al}_6\text{O}_{12}$) and Al_2O_3 (*spinel* type) ($= \text{Al}_2\text{Al}_6\text{O}_{12}$).

In general, however, a notation in which there are variables which define composition should be used. The ranges of the variables can also be indicated. Thus, a phase involving substitution of atom A for B is written $A_{m+x}B_{n-x}C_p$ ($0 \leq x \leq n$). The commas and parentheses called for above are not then required.

Examples:

5. $\text{Cu}_x\text{Ni}_{1-x}$ ($0 \leq x \leq 1$) is equivalent to (Cu, Ni) .
6. $\text{KBr}_x\text{Cl}_{1-x}$ ($0 \leq x \leq 1$) is equivalent to $\text{K}(\text{Br}, \text{Cl})$.
7. $\text{Li}_{2-2x}\text{Mg}_x\text{Cl}_2$ ($0 \leq x \leq 1$) is equivalent to $(\text{Li}_2, \text{Mg})\text{Cl}_2$ but shows explicitly that one vacant cation position appears for every 2 Li^+ replaced by Mg^{2+} .
8. Co_{1-x}O indicates that there are vacant cation sites; for $x=0$ the formula corresponds to the stoichiometric composition CoO .
9. $\text{Ca}_x\text{Zr}_{1-x}\text{O}_{2-x}$ indicates that Zr has been partly replaced by Ca, resulting in vacant anion sites; for $x=0$ the formula corresponds to the stoichiometric composition ZrO_2 .

If the variable x is limited this may be shown by using δ or ε instead of x . A specific composition or composition range can be indicated by stating the actual value of the variable x (or δ , or ε). This value can be written in parentheses after the general formula. However, the value of the variable may also be introduced in the formula itself. This notation can be used both for substitutional and for interstitial solid solutions.⁶

Examples:

10. $\text{Fe}_{3x}\text{Li}_{4-x}\text{Ti}_{2(1-x)}\text{O}_6$ ($x = 0.35$), or $\text{Fe}_{1.05}\text{Li}_{3.65}\text{Ti}_{1.30}\text{O}_6$
11. LaNi_5H_x ($0 < x < 6.7$)
12. $\text{Al}_4\text{Th}_8\text{H}_{15.4}$
13. $\text{Ni}_{1-\delta}\text{O}$

IR-11.4 POINT DEFECT (KRÖGER–VINK) NOTATION

IR-11.4.1 General

As well as the chemical composition, information about point defects, site symmetry, and site occupancy can be given by using additional symbols. These symbols may also be used to write quasi-chemical equilibria between point defects.⁶

IR-11.4.2 Indication of site occupancy

In a formula, the main symbols indicate the species present at a certain site, defined with respect to empty space. This will generally be the symbol of an element. If a site is vacant this is denoted by the italicized symbol *V*. (In certain contexts other symbols, such as a square box, $\square$, are used for vacancies, but the use of italicized *V* is preferred, the element vanadium being written with the upright letter *V*).

The site and its occupancy in a structure of ideal composition are represented by right lower indexes. The first index indicates the type of site, and the second index (if used), separated from the first by a comma, indicates the number of atoms on this site. Thus, an atom A on a site normally occupied by A in the ideal structure is expressed by A_A ; an atom A on a site normally occupied by B is expressed A_B ; and $M_{M,1-x}N_{M,x}M_{N,x}N_{N,1-x}$ stands for a disordered alloy, where the ideal composition is M_MN_N with all M atoms on one type of crystallographic site and all N atoms on a second type of crystallographic site. An alternative description is $(M_{1-x}N_x)_M(M_xN_{1-x})_N$. A species occupying an interstitial site (*i.e.* a site which is unoccupied in the ideal structure) is indicated by the subscript 'i'.

Examples:

1. $Mg_{Mg,2-x}Sn_{Mg,x}Mg_{Sn,x}Sn_{Sn,1-x}$ shows that in Mg_2Sn some of the Mg atoms are present on Sn sites and *vice versa*.
2. $(Bi_{2-x}Te_x)_{Bi}(Bi_xTe_{3-x})_{Te}$ shows that in Bi_2Te_3 some of the Bi atoms are present on Te sites and *vice versa*.
3. $Na_{Na,1-x}V_{Na,x}Cl_{Cl,1-x}V_{Cl,x}$ shows that x Na and x Cl sites in NaCl are vacant, giving Schottky defects.
4. $Ca_{Ca,1}F_{F,2-x}V_{F,x}F_{i,x}$ shows that in CaF_2 , x F sites are vacant, while x F ions are situated on interstitial sites, creating Frenkel defects.
5. $(Ca_{0.15}Zr_{0.85})_{Zr}(O_{1.85}V_{0.15})_O$, or $Ca_{Zr,0.15}Zr_{Zr,0.85}O_{O,1.85}V_{O,0.15}$, shows that in CaO-stabilized ZrO_2 , 0.85 of the Zr sites are occupied by Zr, 0.15 of the Zr sites are occupied by Ca, and that, of the two oxygen sites, 1.85 sites are occupied by oxygen ions, leaving 0.15 sites vacant.
6. $V_{V,1}C_{C,0.8}V_{C,0.2}$ shows that 0.2 C-sites are vacant in vanadium carbide, VC.

The defect symbols can be used in writing quasi-chemical reactions.

Examples:

7. $Na_{Na} \rightarrow V_{Na} + Na(g)$ indicates the evaporation of a Na atom, leaving behind a sodium vacancy in the lattice.
8. $0.5Cl_2(g) + V_{Cl} \rightarrow Cl_{Cl}$ indicates the incorporation of a chlorine atom, from a dichlorine molecule, on a vacant chlorine site in the lattice.

IR-11.4.3 Indication of crystallographic sites

Crystallographic sites can be distinguished by subscripts, *e.g.* tet, oct and dod, denoting tetrahedrally, octahedrally and dodecahedrally coordinated sites, respectively. The use of

subscripts such as a, b, ... , which are not self-explanatory, is not approved. In some cases, such as oxides and sulfides, the number of subscripts can be reduced by defining specific symbols to indicate site symmetries, *e.g.* () for tetrahedral sites, [] for octahedral sites, { } for dodecahedral sites. To avoid confusion, such enclosing marks should be restricted to cases where they are not being used to express multiplication. The meaning of the symbols should be clearly stated in the text.

Examples:

1. $\text{Mg}_{\text{tet}}\text{Al}_{\text{oct},2}\text{O}_4$ or $(\text{Mg})[\text{Al}_2]\text{O}_4$ denotes a normal spinel.
2. $\text{Fe}_{\text{tet}}\text{Fe}_{\text{oct}}\text{Ni}_{\text{oct}}\text{O}_4$ or $(\text{Fe})[\text{FeNi}]\text{O}_4$ denotes NiFe_2O_4 (*inverse spinel type*).

IR-11.4.4 Indication of charges

Charges are indicated by a right upper index. When formal charges are given, the usual convention holds: one unit of positive charge is indicated by a superscript +, n units of positive charge by a superscript $n+$, one unit of negative charge by a superscript $-$, n units of negative charge by a superscript $n-$. Thus A^{n+} denotes n units of formal positive charge on an atom of symbol A. In defect chemistry, charges are defined preferably with respect to the ideal unperturbed crystal. In this case, they are called *effective charges*. One unit of positive effective charge is shown by a superscript dot, $\bullet$, (not to be confused with the radical dot described in Section IR-4.6.2) and one unit of negative effective charge by a prime, $'$; n units of effective charge are indicated by superscript $n\bullet$ or n' . The use of double dots $\bullet\bullet$ or double primes $''$ in the case of two effective charges is also allowed. Thus $\text{A}^{2\bullet}$ and $\text{A}^{\bullet\bullet}$ indicate that an atom of symbol A has two units of effective positive charge. Sites that have no effective charge relative to the unperturbed lattice may be indicated explicitly by a superscript cross, *i.e.* $^{\times}$.

Examples:

1. $\text{Li}_{\text{Li},1-2x}\text{Mg}_{\text{Li},x}^{\bullet}\text{V}_{\text{Li},x}'\text{Cl}_{\text{Cl}}$ and $\text{Li}_{\text{Li},1-2x}^{\times}\text{Mg}_{\text{Li},x}^{\bullet}\text{V}_{\text{Li},x}'\text{Cl}_{\text{Cl}}^{\times}$ are equivalent expressions for a substitutional solid solution of MgCl_2 in LiCl .
2. $\text{Y}_{\text{Y},1-2x}\text{Zr}_{\text{Y},2x}^{\bullet}\text{O}_{\text{i},x}''\text{O}_3$ and $\text{Y}_{\text{Y},1-2x}^{\times}\text{Zr}_{\text{Y},2x}^{\bullet}\text{O}_{\text{i},x}''\text{O}_3^{\times}$ are equivalent expressions for an interstitial solid solution of ZrO_2 in Y_2O_3 .
3. $\text{Ag}_{\text{Ag},1-x}\text{V}_{\text{Ag},x}'\text{Ag}_{\text{i},x}^{\bullet}\text{Cl}_{\text{Cl}}$ indicates that a fraction x of the Ag^+ ions is removed from the Ag sites to interstitial sites, leaving the silver site vacant.

Formal charges may be preferred in cases where the unperturbed crystal contains an element in more than one oxidation state.

Examples:

4. $\text{La}_{\text{La},1-3x}^{2+}\text{La}_{\text{La},2+2x}^{3+}\text{V}_{\text{La},x}(\text{S}^{2-})_4$ ($0 < x < 1/3$)
5. $\text{Cu}_{\text{Cu},2-x}^{+}\text{Fe}_{\text{Cu},x}^{3+}\text{Tl}_{\text{Ti}}^{+}\text{Se}_{\text{Se},1+2x}^{2-}\text{Se}_{\text{Se},1-2x}^{-}$ ($0 < x < 1/2$) shows that Fe^{3+} partly replaces Cu^{+} in $\text{Cu}^{+}_2\text{Tl}^{+}\text{Se}^{2-}_2\text{Se}^{-}$.

Free electrons are denoted by e' , free holes by $h^{\bullet}$. As crystals are macroscopically neutral bodies, the sums of the formal charges and of the effective charges must be zero.

Key aspects of the Kröger–Vink point defect notation are summarized in Table IR-11.1.

Table IR-11.1 Examples^a of defect notation in $M^{2+}(X^-)_2$ containing a foreign ion Q

interstitial M^{2+} ion	$M_i^\bullet$	M atom vacancy	V_M^x
interstitial X^- ion	X_i'	X atom vacancy	V_X^x
M^{2+} ion vacancy	V_M''	normal M^{2+} ion	M_M^x
X^- ion vacancy	$V_X^\bullet$	normal X^- ion	X_X^x
interstitial M atom	M_i^x	Q^{3+} ion at M^{2+} site	$Q_M^\bullet$
interstitial X atom	X_i^x	Q^{2+} ion at M^{2+} site	Q_M^x
interstitial M^+ ion	$M_i^\bullet$	Q^+ ion at M^{2+} site	Q_M'
M^+ ion vacancy	V_M'	free electron	e'
		free hole	$h^\bullet$

^a Consider an ionic compound $M^{2+}(X^-)_2$. The formal charge on M is 2+, the formal charge on X is 1-. If an atom X is removed, one negative unit of charge remains on the vacant X site. The vacancy is neutral with respect to the ideal MX_2 lattice and is therefore indicated by V_X or V_X^x . If the electron is also removed from this site, the resultant vacancy is effectively positive, *i.e.* $V_X^\bullet$. Similarly, removal of an M atom leaves V_M , removal of an M^+ ion leaves V_M' , removal of an M^{2+} ion leaves V_M'' . If an impurity with a formal charge of three positive units Q^{3+} is substituted on the M^{2+} site, its effective charge is one positive unit. Therefore it is indicated by $Q_M^\bullet$.

IR-11.4.5 Defect clusters and use of quasi-chemical equations

Pairs or more complicated clusters of defects can be present in a solid. Such a defect cluster is indicated between parentheses. The effective charge of the cluster is indicated as an upper right index.

Examples:

1. $(Ca_K^\bullet V_K')^x$ denotes a neutral defect pair in a solid solution, for example of $CaCl_2$ in KCl.
2. $(V_{Pb}'' V_{Cl}^\bullet)'$ or $(V_{Pb} V_{Cl})'$ indicates a charged vacancy pair in $PbCl_2$.

Quasi-chemical reactions may be written for the formation of such defect clusters.

Examples:

3. $Cr_{Mg}^\bullet + V_{Mg}'' \rightarrow (Cr_{Mg} V_{Mg})'$ describes the association reaction of a Cr^{3+} impurity in MgO with magnesium vacancies.
4. $2Cr_{Mg}^\bullet + V_{Mg}'' \rightarrow (Cr_{Mg} V_{Mg} Cr_{Mg})^x$ gives another possible association reaction in the system of Example 3.
5. $Gd_{Ca}^\bullet + F_i' \rightarrow (Gd_{Ca} F_i)^x$ describes the formation of a dipole between a Gd^{3+} impurity and a fluorine interstitial in CaF_2 .

IR-11.5 PHASE NOMENCLATURE

IR-11.5.1 Introduction

The use of the Pearson notation⁷ (see also Section IR-3.4.4) is recommended for the designation of the structures of metals and solid solutions in binary and more complex

systems. The use of Greek letters, which do not convey the necessary information, and of the *Strukturbericht* designations, which are not self-explanatory, is not acceptable.

IR-11.5.2 **Recommended notation**

The Pearson symbol consists of three parts: first, a lower-case italic letter (*a, m, o, t, h, c*) designating the crystal system; second, an italic capital letter (*P, S, F, I, R*) designating the lattice setting and, finally, a number designating the number of atoms or ions in the conventional unit cell. Table IR-3.1 summarizes the system.

Examples:

1. Cu, symbol (*cF4*), indicates copper of cubic symmetry, with face-centred lattice, containing 4 atoms per unit cell.
2. NaCl, symbol (*cF8*), indicates a cubic face-centred lattice with 8 ions per unit cell.
3. CuS(*hP12*), indicates a hexagonal primitive lattice with 12 ions per unit cell.

If required, the Pearson symbol can be followed by the space group and a prototype formula.

Example:

4. $\text{CaMg}_{0.5}\text{Ag}_{1.5}(\text{hP12}, P6_3/mmc)$ (MgZn_2 type).

IR-11.6 NON-STOICHIOMETRIC PHASES

IR-11.6.1 **Introduction**

Several special problems of nomenclature for non-stoichiometric phases have arisen with the improvements in the precision with which their structures can be determined. Thus, there are references to homologous series, non-commensurate and semi-commensurate structures, Vernier structures, crystallographic shear phases, Wadsley defects, chemical twinned phases, infinitely adaptive phases and modulated structures. Many of the phases that fall into these classes have no observable composition ranges although they have complex structures and formulae; an example is $\text{Mo}_{17}\text{O}_{47}$. These phases, despite their complex formulae, are essentially stoichiometric and possession of a complex formula must not be taken as an indication of a non-stoichiometric compound (*cf.* Section IR-11.1.2).

IR-11.6.2 **Modulated structures**

Modulated structures possess two or more periodicities in the same direction of space. If the ratio of these periodicities is a rational number, the structures are called *commensurate*;

if the ratio is irrational, the structures are called *non-commensurate* or *incommensurate*. Commensurately modulated structures exist in many stoichiometric and non-stoichiometric compounds; they may be regarded as superstructures and be described by the usual rules. Non-commensurately modulated structures occur in several stoichiometric compounds (and some elements), usually in a limited temperature range, *e.g.* U, SiO₂, TaS₂, NbSe₃, NaNO₂, Na₂CO₃ and Rb₂ZnBr₄.

Many modulated structures can be regarded as being composed of two or more substructures. The substructure with the shortest periodicity often represents a simple *basic structure*, while the other periodicities cause modulations of the basic structure. The basic structure often remains unchanged within a certain composition range, while the other substructures take up the change in stoichiometry. If this change takes place continuously, a non-stoichiometric phase with a non-commensurate structure results. If the change occurs discontinuously, a series of (essentially stoichiometric) *homologous compounds* with commensurate structures (superstructures of the basic structure) may result or, in the intermediate case, a series of compounds with *semi-commensurate* or *Vernier* structures.

Examples:

1. Mn_nSi_{2n-m}

The structure is of the *TiSi₂* type which has two atom substructures, the Mn array being identical to that of the Ti array in TiSi₂ and the Si₂ array being identical to that of the Si₂ array in TiSi₂. Removal of Si leads to a composition Mn_nSi_{2n-m} in which the Mn array is completely unchanged. The Si atoms are arranged in rows and, as the Si content falls, the Si atoms in the rows spread out. In this case there will be a Vernier relationship between the Si atom rows and the static Mn positions which will change as the composition varies, giving rise to non-commensurate structures.

2. YF_{2+x}O

The structure is of the *fluorite* type with extra sheets of atoms inserted into the parent YX₂ structure. When these are ordered, a homologous series of phases results. When they are disordered, there is a non-commensurate, non-stoichiometric phase, while partial ordering will give a Vernier or semi-commensurate effect. Other layer structures can be treated in the same way.

Misfit structures consist of two or more different, often mutually non-commensurate, units which are held together by electrostatic or other forces; no basic structure can be defined. The composition of compounds with misfit structures is determined by the ratio of the periodicities of their structural units and by electroneutrality.

Examples:

3. Sr_{1-p}Cr₂S_{4-p} with $p = 0.29$, where chains of compositions Sr₃CrS₃ and Sr_{3-x}S lie in tunnels of a framework of composition Cr₂₁S₃₆; the three units are mutually non-commensurate.

4. LaCrS₃, which is built from non-commensurate sheets of (LaS)⁺ and (CrS₂)⁻.

IR-11.6.3 Crystallographic shear structures

Crystallographic shear planes (CS planes) are planar faults in a crystal that separate two parts of the crystal which are displaced with respect to each other. The vector describing the displacement is called the crystallographic shear vector (CS vector). Each CS plane causes the composition of the crystal to change by a small increment because the sequence of crystal planes that produces the crystal matrix is changed at the CS plane. (From this it follows that the CS vector must be at an angle to the CS plane. If it were parallel to the plane, the succession of crystal planes would not be altered and no composition change would result. A planar boundary where the displacement vector is parallel to the plane is more properly called an *antiphase boundary*.)

Because each CS plane changes the composition of the crystal slightly, the overall composition of a crystal containing a population of CS planes will depend upon the number of CS planes present and their orientation. If the CS planes are disordered, the crystals will be non-stoichiometric, the stoichiometric variation being due to the CS plane 'defect'. If the CS planes are ordered into a parallel array, a stoichiometric phase with a complex formula results. In this case, a change in the separation of the CS planes in the ordered array will produce a new phase with a new composition. The series of phases produced by changes in the spacing between CS planes forms an *homologous* series. The general formula of a particular series will depend upon the type of CS plane in the array and the separation between the CS planes. A change in the CS plane may change the formula of the homologous series.

Examples:

1. $\text{Ti}_n\text{O}_{2n-1}$

The parent structure is TiO_2 (*rutile* type). The CS planes are the (121) planes. Ordered arrays of CS planes can exist, producing an homologous series of oxides with formulae Ti_4O_7 , Ti_5O_9 , Ti_6O_{11} , Ti_7O_{13} , Ti_8O_{15} and Ti_9O_{17} . The series formula is $\text{Ti}_n\text{O}_{2n-1}$, with n between 4 and 9.

2. $(\text{Mo,W})_n\text{O}_{3n-1}$

The parent structure is WO_3 . The CS planes are the (102) planes. Ordered arrays of CS planes can form, producing oxides with formulae Mo_8O_{23} , Mo_9O_{26} , $(\text{Mo,W})_{10}\text{O}_{29}$, $(\text{Mo,W})_{11}\text{O}_{32}$, $(\text{Mo,W})_{12}\text{O}_{35}$, $(\text{Mo,W})_{13}\text{O}_{38}$ and $(\text{Mo,W})_{14}\text{O}_{41}$. The series formula is $(\text{Mo,W})_n\text{O}_{3n-1}$, with n between 8 and 14.

3. $\text{W}_n\text{O}_{3n-2}$

The parent structure is WO_3 . The CS planes are the (103) planes. Ordered arrays of CS planes can form, producing oxides with formulae $\text{W}_n\text{O}_{3n-2}$, with n between approximately 16 and 25.

IR-11.6.4 Unit cell twinning or chemical twinning

This is a structure-building component in which two constituent parts of the structure are twin-related across the interface. The twin plane changes the composition of the host crystal by a definite amount (which may be zero). Ordered, closely spaced arrays of twin planes will lead to homologous series of phases. Disordered twin planes will lead to non-stoichiometric phases in which the twin planes serve as the defects. There is a close parallel between chemical twinning and crystallographic shear (see Section IR-11.6.3).

Example:

1. $(\text{Bi,Pb})_n\text{S}_{n-4}$

The parent structure is PbS which has the *cF8* (*NaCl* type) structure. The twin planes are (311) with respect to the PbS unit cell. Two members of the homologous series are known, $\text{Bi}_8\text{Pb}_{24}\text{S}_{36}$ and $\text{Bi}_8\text{Pb}_{12}\text{S}_{24}$, but other members are found in the quaternary Ag-Bi-Pb-S system. The difference between compounds lies in the separation of the twin planes; each structure is built from slabs of PbS of varying thickness, alternate slabs being twinned across (311) with respect to the parent structure.

IR-11.6.5 **Infinitely adaptive structures**

In some systems it would appear that any composition can yield a fully ordered crystal structure over certain temperature and composition ranges. As the composition changes, so the structure changes to meet this need. The term *infinitely adaptive structures* has been applied to this group of substances.⁸

Examples:

1. Compounds in the Cr_2O_3 - TiO_2 system between the composition ranges $(\text{Cr,Ti})\text{O}_{2.93}$ and $(\text{Cr,Ti})\text{O}_{2.90}$.
2. Compounds in the Nb_2O_5 - WO_3 system with block-type structure between the composition limits Nb_2O_5 and $8\text{WO}_3 \cdot 9\text{Nb}_2\text{O}_5$ ($\text{Nb}_{18}\text{W}_8\text{O}_{69}$).

IR-11.6.6 **Intercalation compounds**

There are several materials in which a guest species is inserted into a host matrix. The process is called intercalation, and the product is called an *intercalation compound*. Common examples of intercalated materials are found in the clay silicates, layered dichalcogenides and electrode materials for lithium batteries; graphite intercalation is considered in detail in Ref. 9. Intercalated materials can be designated by conventional chemical formulae such as Li_xTaS_2 ($0 < x < 1$) or by host-guest designations, such as $\text{TaS}_2 \cdot x\text{Li}$ ($0 < x < 1$). If the stoichiometry is definite, ordinary compound designations may be used, e.g. $3\text{TaS}_2 \cdot 4\text{N}_2\text{H}_4$, $\text{C}_5\text{H}_5\text{N} \cdot 2\text{TiSe}_2$ and KC_8 .

Many intercalation compounds are layered structures and intercalation is a two-dimensional reaction. The term *insertion* is sometimes used for three-dimensional examples, as in the tungsten bronzes, e.g. Na_xWO_3 , and the spinels, e.g. $\text{Li}_x\text{Mn}_2\text{O}_4$, and also as a general term for a reaction involving the transfer of a guest atom, ion or molecule into a host crystal lattice⁴ instead of intercalation. More specifically, *intercalation* is used for an insertion reaction that does not cause a major structural modification of the host.⁴ If the structure of the host is modified significantly, for example by breaking of bonds, then the insertion can be referred to as *topochemical* or *topotactic*.⁴

IR-11.7 **POLYMORPHISM**

IR-11.7.1 **Introduction**

A number of chemical compounds and elements change their crystal structure with external conditions such as temperature and pressure. These various structures are termed polymorphic

forms or modifications, and in the past have been designated using a number of labelling systems, including Greek letters and Roman numerals; the use of such non-systematic labels is not acceptable. A rational system based upon crystal structure should be used wherever possible (*cf.* Sections IR-3.4.4 and IR-4.2.5).

Polytypes and polytypoids can be regarded as a special form of polymorphism and are treated in more detail in Ref. 10.

IR-11.7.2 Use of crystal systems

Polymorphs are indicated by adding an italicized symbol denoting the crystal system after the name or formula. The symbols used are given in Table IR-3.1. For example, ZnS(*c*) corresponds to the zinc blende structure or sphalerite, and ZnS(*h*) to the wurtzite structure. Slightly distorted lattices may be indicated by using the *circa* sign $\sim$. Thus, a slightly distorted cubic lattice would be expressed as ($\sim c$). In order to give more information, simple well-known structures should be designated by giving the type compound in parentheses whenever possible. For example, AuCd above 343 K should be designated AuCd (*CsCl* type) rather than AuCd(*c*).

Properties which strongly depend on lattice and point symmetries may require the addition of the space group to the crystal system abbreviation. For more details see Ref. 11.

IR-11.8 FINAL REMARKS

This Chapter deals with some basic notation and nomenclature of solid-state chemistry. In some areas, such as amorphous systems and glasses, the nomenclature needs further development. The reader is also referred to the work of the International Union of Crystallography.

IR-11.9 REFERENCES

1. Nomenclature of Inorganic Structure Types, J. Lima-de-Faria, E. Hellner, F. Liebau, E. Makovicky and E. Parthé, *Acta Crystallogr., Sect. A*, **46**, 1–11 (1990).
2. M.L. McGlashan, *Chemical Thermodynamics*, Academic Press, London, 1979, pp. 35–36.
3. *Quantities, Units and Symbols in Physical Chemistry*, Second Edn., eds. I. Mills, T. Cvitas, K. Homann, N. Kallay and K. Kuchitsu, Blackwell Scientific Publications, Oxford, 1993, p. 53. (The Green Book. The third edition is planned for publication in 2006.)
4. *Compendium of Chemical Terminology, IUPAC Recommendations*, Second Edn., eds. A.D. McNaught and A. Wilkinson, Blackwell Scientific Publications, Oxford, 1997. (The Gold Book.)
5. Chemical Nomenclature and Formulation of Compositions of Synthetic and Natural Zeolites, R.M. Barrer, *Pure Appl. Chem.*, **51**, 1091–1100 (1979).
6. F.A. Kröger and H.J. Vink, *Solid State Phys.*, **3**, 307–435 (1956).
7. W.B. Pearson, *A Handbook of Lattice Spacings and Structures of Metals and Alloys*, Vol. 2, Pergamon Press, Oxford, 1967, pp. 1–2. For tabulated lattice parameters and

- data on elemental metals and semi-metals, see pp. 79–91. See also, P. Villars and L.D. Calvert, *Pearson's Handbook of Crystallographic Data for Intermetallic Phases*, Vols. 1–3, American Society for Metals, Metals Park, Ohio, USA, 1985.
8. J.S. Anderson. *J. Chem. Soc., Dalton Trans.*, 1107–1115 (1973).
 9. Graphite Intercalation Compounds, Chapter II-6 in *Nomenclature of Inorganic Chemistry II, IUPAC Recommendations 2000*, eds. J.A. McCleverty and N.G. Connelly, Royal Society of Chemistry, 2001.
 10. Nomenclature of Polytype Structures, A. Guinier, G.B. Bokij, K. Boll-Dornberger, J.M. Cowley, S. Durovic, H. Jagodzinski, P. Krishna, P.M. de Wolff, B.B. Zvyagin, D.E. Cox, P. Goodman, Th. Hahn, K. Kuchitsu and S.C. Abrahams, *Acta Crystallogr., Sect. A*, **40**, 399–404 (1984). See also, S.W. Bailey, V.A. Frank-Kamenetskii, S. Goldsztaub, A. Kato, A. Pabst, H. Schulz, H.F.W. Taylor, M. Fleischer and A.J.C. Wilson, *Acta Crystallogr., Sect. A*, **33**, 681–684 (1977).
 11. Structural Phase Transition Nomenclature, J.-C. Tolédano, A.M. Glazer, Th. Hahn, E. Parthé, R.S. Roth, R.S. Berry, R. Metselaar and S.C. Abrahams, *Acta Crystallogr., Sect. A*, **54**, 1028–1033 (1998). Nomenclature of magnetic, incommensurate, composition-changed morphotropic, polytype, transient-structural and quasicrystalline phases undergoing phase transitions, J.-C. Tolédano, R.S. Berry, P.J. Brown, A.M. Glazer, R. Metselaar, D. Pandey, J.M. Perez-Mato, R.S. Roth and S.C. Abrahams, *Acta Crystallogr., Sect. A*, **57**, 614–626 (2001), and erratum in *Acta Crystallogr., Sect. A*, **58**, 79 (2002).

TABLES

Table I Names, symbols and atomic numbers of the elements (see also Section IR-3.1)

<i>Name</i>	<i>Symbol</i>	<i>Atomic number</i>	<i>Name</i>	<i>Symbol</i>	<i>Atomic number</i>
actinium	Ac	89	indium	In	49
aluminium ^a	Al	13	iodine	I	53
americium	Am	95	iridium	Ir	77
antimony	Sb ^b	51	iron	Fe ^g	26
argon	Ar	18	krypton	Kr	36
arsenic	As	33	lanthanum	La	57
astatine	At	85	lawrencium	Lr	103
barium	Ba	56	lead	Pb ^h	82
berkelium	Bk	97	lithium	Li	3
beryllium	Be	4	lutetium	Lu	71
bismuth	Bi	83	magnesium	Mg	12
bohrium	Bh	107	manganese	Mn	25
boron	B	5	meitnerium	Mt	109
bromine	Br	35	mendelevium	Md	101
cadmium	Cd	48	mercury	Hg ⁱ	80
caesium ^c	Cs	55	molybdenum	Mo	42
calcium	Ca	20	neodymium	Nd	60
californium	Cf	98	neon	Ne	10
carbon	C	6	neptunium	Np	93
cerium	Ce	58	nickel	Ni	28
chlorine	Cl	17	niobium	Nb	41
chromium	Cr	24	nitrogen ^j	N	7
cobalt	Co	27	nobelium	No	102
copper	Cu ^d	29	osmium	Os	76
curium	Cm	96	oxygen	O	8
darmstadtium	Ds	110	palladium	Pd	46
dubnium	Db	105	phosphorus	P	15
dysprosium	Dy	66	platinum	Pt	78
einsteinium	Es	99	plutonium	Pu	94
erbium	Er	68	polonium	Po	84
europium	Eu	63	potassium	K ^k	19
fermium	Fm	100	praseodymium	Pr	59
fluorine	F	9	promethium	Pm	61
francium	Fr	87	protactinium	Pa	91
gadolinium	Gd	64	radium	Ra	88
gallium	Ga	31	radon	Rn	86
germanium	Ge	32	rhenium	Re	75
gold	Au ^e	79	rhodium	Rh	45
hafnium	Hf	72	roentgenium	Rg	111
hassium	Hs	108	rubidium	Rb	37
helium	He	2	ruthenium	Ru	44
holmium	Ho	67	rutherfordium	Rf	104
hydrogen	H ^f	1	samarium	Sm	62

Table I *Continued*

<i>Name</i>	<i>Symbol</i>	<i>Atomic number</i>	<i>Name</i>	<i>Symbol</i>	<i>Atomic number</i>
scandium	Sc	21	thorium	Th	90
seaborgium	Sg	106	thulium	Tm	69
selenium	Se	34	tin	Sn ^o	50
silicon	Si	14	titanium	Ti	22
silver	Ag ^l	47	tungsten	W ^p	74
sodium	Na ^m	11	uranium	U	92
strontium	Sr	38	vanadium	V	23
sulfur ⁿ	S	16	xenon	Xe	54
tantalum	Ta	73	ytterbium	Yb	70
technetium	Tc	43	yttrium	Y	39
tellurium	Te	52	zinc	Zn	30
terbium	Tb	65	zirconium	Zr	40
thallium	Tl	81			

^a The alternative spelling ‘aluminum’ is commonly used.

^b The element symbol Sb derives from the name stibium.

^c The alternative spelling ‘cesium’ is commonly used.

^d The element symbol Cu derives from the name cuprum.

^e The element symbol Au derives from the name aurum.

^f The hydrogen isotopes ²H and ³H are named deuterium and tritium, respectively, for which the symbols D and T may be used. However, ²H and ³H are preferred (see Section IR-3.3.2).

^g The element symbol Fe derives from the name ferrum.

^h The element symbol Pb derives from the name plumbum.

ⁱ The element symbol Hg derives from the name hydrargyrum.

^j The name azote provides the root ‘az’ for nitrogen.

^k The element symbol derives K from the name kalium.

^l The element symbol Ag derives from the name argentum.

^m The element symbol Na derives from the name natrium.

ⁿ The name theion provides the root ‘thi’ for sulfur.

^o The element symbol Sn derives from the name stannum.

^p The element symbol W derives from the name wolfram.

Table II *Temporary names and symbols for elements of atomic number greater than 111^a*

<i>Atomic number</i>	<i>Name^b</i>	<i>Symbol</i>
112	ununbium	Uub
113	ununtrium	Uut
114	ununquadium	Uuq
115	ununpentium	Uup
116	ununhexium	Uuh
117	ununseptium	Uus
118	ununoctium	Uuo
119	ununennium	Uue
120	unbinilium	Ubn
121	unbiunium	Ubu
130	untrinilium	Utn
140	unquadnilium	Uqn
150	unpentnilium	Upn
160	unhexnilium	Uhn
170	unseptnilium	Usn
180	unoctnilium	Uon
190	unennilium	Uen
200	binilnilium	Bnn
201	binilunium	Bnu
202	binilbium	Bnb
300	trinilnilium	Tnn
400	quadnilnilium	Qnn
500	pentnilnilium	Pnn
900	ennilnilium	Enn

^a These names are used only when the permanent name has not yet been assigned by IUPAC (see Section IR-3.1.1).
^b One may also write, for example, ‘element 112’.

Table III *Suffixes and endings*^a

a	Terminal vowel of prefixes indicating replacement of: carbon atoms by atoms of other elements in skeletal replacement nomenclature (Section IR-6.2.4.1) and Hantzsch–Widman nomenclature (Section IR-6.2.4.3), <i>e.g.</i> ‘oxa’, ‘aza’; boron atoms by atoms of other elements in boron hydride-based nomenclature (Section IR-6.2.4.4), <i>e.g.</i> ‘carba’, ‘thia’; heteroatoms by carbon atoms in natural product nomenclature (prefix ‘carba’). See Table X for ‘a’ prefixes for all elements.
ane	Ending of names of neutral saturated parent hydrides of elements of Groups 13–17, <i>e.g.</i> thallane, cubane, cyclohexane, cyclohexasilane, diphosphane, tellane, λ^4-tellane. <i>Cf.</i> Section IR-6.2.2 and Table IR-6.1. Last part of endings of a number of parent names of saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>i.e.</i> of ‘irane’, ‘etane’, ‘olane’, ‘ane’, ‘inane’, ‘epane’, ‘ocane’, ‘onane’ and ‘ecane’ (see Section IR-6.2.4.3).
anide	Combined ending of names of anions resulting from the removal of a hydron from a parent hydride with an ‘ane’ name, formed by adding the suffix ‘ide’, <i>e.g.</i> methanide, CH_3^- . <i>Cf.</i> Section IR-6.4.4.
anium	Combined ending of names of cations resulting from the addition of a hydron to a parent structure with an ‘ane’ name, formed by adding the suffix ‘ium’, <i>e.g.</i> phosphanium, PH_4^+ . <i>Cf.</i> Section IR-6.4.1.
ano	Ending resulting from the change of the ‘ane’ ending in names of parent hydrides to form prefixes denoting bridging divalent substituent groups, <i>e.g.</i> diazano, $-\text{HNNH}-$.
ate	General ending of additive names of anions, <i>e.g.</i> tetrahydridoaluminate(1–), $[\text{AlH}_4]^-$. <i>Cf.</i> Section IR-7.1.4 and Table X. Ending of names of anions and esters of inorganic oxoacids having the ‘ic’ ending in the acid name, <i>e.g.</i> nitrate, phosphonate, trimethyl phosphate, and of anions and esters of organic acids, <i>e.g.</i> acetate, methyl acetate, thiocyanate. See Tables IR-8.1 and IR-8.2 and Table IX for more examples of ‘ate’ anion names. See also ‘inate’, ‘onate’.
ato	Ending of name of any anion with an ‘ate’ name (see above) acting as a ligand, <i>e.g.</i> tetrahydridoaluminato(1–), nitrato, acetato. <i>Cf.</i> Sections IR-7.1.3 and IR-9.2.2.3 and Table IX. See also ‘inato’, ‘onato’. Ending of prefixes for certain anionic substituent groups, <i>e.g.</i> carboxylato, $-\text{C}(=\text{O})\text{O}^-$; phosphato, $-\text{O}-\text{P}(=\text{O})(\text{O}^-)_2$. See also ‘onato’.
diene	See ‘ene’.
diide	See ‘ide’.
diido	See ‘ido’.
dium	See ‘ium’.
diyl	Combined suffix composed of the suffix ‘yl’ and the multiplicative prefix ‘di’, indicating the loss of two hydrogen atoms from a parent hydride resulting in a diradical, or a substituent group with two single bonds, if necessary accompanied by locants, <i>e.g.</i> hydrazine-1,2-diyl, $\bullet\text{HNNH}\bullet$ or $-\text{HNNH}-$; phosphanediyli, $\text{HP}<$. See also ‘ylidene’.
diylum	See ‘ylum’.
ecane	Ending of parent names of ten-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.

Table III *Continued*

ecine	Ending of parent names of ten-membered heteromonocycles with the maximum number of non-cumulative double bonds in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
ene	Ending of systematic names of acyclic and cyclic parent structures with double-bond unsaturation, replacing ‘ane’ in the name of the corresponding saturated parent hydride, and if necessary accompanied by locants and multiplicative prefixes specifying the locations and number of double bonds, <i>e.g.</i> diazene, triazene, pentasil-1-ene, cyclopenta-1,3-diene. <i>Cf.</i> Sections IR-6.2.2.3 and IR- 6.2.2.4. Ending of certain acceptable non-systematic names of unsaturated cyclic parent hydrides, <i>e.g.</i> benzene, azulene. See also ‘irene’, ‘ocene’.
enide	Combined ending of names of anions resulting from the removal of a hydron from a parent hydride with an ‘ene’ name, formed by adding the suffix ‘ide’, <i>e.g.</i> diazenide, $\text{HN}=\text{N}^-$. <i>Cf.</i> Section IR-6.4.4.
enium	Combined ending of names of cations resulting from the addition of a hydron to a parent structure with an ‘ene’ name, formed by adding the suffix ‘ium’, <i>e.g.</i> diazenium. <i>Cf.</i> Section IR-6.4.1. Combined ending resulting from the addition of the suffix ‘ium’ to a metallocene name. This leads to ambiguous names, see Section IR-10.2.6.
eno	Ending resulting from the change of the ‘ene’ ending in names of cyclic mancude ring systems to ‘eno’ to form prefixes in fusion nomenclature. (See Section P-25.3 of the Blue Book ^b .) Ending resulting from the change of the ‘ene’ ending in names of parent hydrides to form prefixes denoting bridging divalent substituent groups, <i>e.g.</i> diazeno, $-\text{N}=\text{N}-$.
epane	Ending of parent names of seven-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
epine	Ending of parent names of seven-membered heteromonocycles with the maximum number of non-cumulative double bonds in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
etane	General ending of parent names of four-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3. See also ‘etidine’.
ete	Ending of parent names of four-membered heteromonocycles with the maximum number of non-cumulative double bonds in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
etidine	Ending of parent names of four-membered nitrogen-containing saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
ic	Ending of names of many acids, both inorganic and organic, <i>e.g.</i> sulfuric acid, acetic acid, benzoic acid. For more examples, particularly of inorganic ‘ic’ acid names, see Tables IR-8.1 and IR-8.2, and Table IX. See also ‘inic’ and ‘onic’. Ending formerly added to stems of element names to indicate a higher oxidation state, <i>e.g.</i> ferric chloride, cupric oxide, ceric sulfate. Such names are no longer acceptable.
ide	Ending of names of monoatomic and homopolyatomic anions, <i>e.g.</i> chloride, sulfide, disulfide(2–), triiodide(1–). <i>Cf.</i> Sections IR-5.3.3.2 and IR-5.3.3.3 and Table IX. Ending of names of formally electronegative homoatomic constituents in compositional names, <i>e.g.</i> disulfur dichloride. <i>Cf.</i> Section IR-5.4.

Table III Continued

	Ending of some acceptable non-systematic names of heteropolyatomic anions: cyanide, hydroxide.
	Suffix for names of anions formed by removal of one or more hydrons from a parent hydride, accompanied by locants and multiplicative prefixes as appropriate, <i>e.g.</i> hydrazinide, H_2NNH^- ; hydrazine-1,2-diide, $^-\text{HNNH}^-$; disulfanediide, S_2^{2-} ; methanide, CH_3^- .
ido	Ending of name of any anion with an 'ide' name (see above) acting as a ligand, <i>e.g.</i> chlorido, disulfido(2-) or disulfanediido, hydrazinido, hydrazine-1,2-diido, methanido. <i>Cf.</i> Sections IR-7.1.3 and IR-9.2.2.3 and Table IX.
	Ending of certain prefixes for anionic substituent groups, <i>e.g.</i> oxido for $-\text{O}^-$.
inane	Ending of parent names of six-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
inate	Ending of names of anions and esters of 'inic' oxoacids, <i>e.g.</i> borinate, phosphinate.
inato	Modification of the 'inate' ending of an anion name (see above) used when the anion acts as a ligand.
ine	Ending of the non-systematic, but still acceptable, parent name hydrazine (N_2H_4) and of the now obsolete names of other Group 15 hydrides, <i>e.g.</i> phosphine (PH_3). Ending of names of large heteromonocycles (more than 10 ring atoms) with the maximum number of non-cumulative double bonds for use in fusion nomenclature, <i>e.g.</i> 2 <i>H</i> -1-oxa-4,8,11-triazacyclotetradecine. Last part of endings of a number of parent names in Hantzsch–Widman nomenclature of heteromonocycles, <i>i.e.</i> of 'irine', 'iridine', 'etidine', 'olidine', 'ine', 'inine', 'epine', 'ocine', 'onine' and 'ecine' (see Section IR-6.2.4.3). Ending of a number of parent names of nitrogenous heterocyclic parent hydrides, <i>e.g.</i> pyridine, acridine.
inic	Ending of the parent names of acids of the types $\text{H}_2\text{X}(=\text{O})(\text{OH})$ ($\text{X}=\text{N}, \text{P}, \text{As}, \text{Sb}$), <i>e.g.</i> stibinic acid; $\text{HX}(=\text{O})(\text{OH})$ ($\text{X}=\text{S}, \text{Se}, \text{Te}$), <i>e.g.</i> sulfinic acid; and of borinic acid, H_2BOH .
inide	Combined ending of names of anions resulting from the removal of a hydron from a parent hydride with an 'ine' name, formed by adding the suffix 'ide', <i>e.g.</i> hydrazinide, H_2NNH^- . <i>Cf.</i> Section IR-6.4.4.
inine	Ending of parent names of six-membered heteromonocycles with the maximum number of non-cumulative double bonds in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
inite	Ending of names of anions and esters of oxoacids with an 'inous' name, <i>e.g.</i> phosphinite, H_2PO^- , from phosphinous acid.
inito	Modification of the 'inite' ending of an anion name (see above) used when the anion acts as a ligand.
inium	Combined ending of names of cations resulting from the addition of a hydron to a parent structure with an 'ine' name, formed by adding the suffix 'ium', <i>e.g.</i> hydrazinium, pyridinium. <i>Cf.</i> Section IR-6.4.1.
ino	Ending of some non-systematic substituent group prefixes, <i>e.g.</i> amino, NH_2- ; hydrazino, $\text{H}_2\text{NNH}-$. Ending resulting from the change of the 'ine' ending in names of cyclic mancude ring systems to 'ino' to form prefixes in fusion nomenclature. (See Section P-25.3 of the Blue Book ^b .)

Table III *Continued*

inous	Ending of the parent names of acids of the types $H_2X(OH)$ ($X = N, P, As, Sb$), <i>e.g.</i> stibinous acid. See Table IR-8.1 for other such names.
inoyl	Ending of prefixes for substituent groups formed by removing all hydroxy groups from 'inic' acids (see above), <i>e.g.</i> phosphinoyl, $H_2P(O)-$; seleninoyl, $HSe(O)-$. (See Table IR-8.1 for phosphinic and seleninic acids.)
inyl	Ending of prefixes for the divalent substituent groups $>X=O$ (sulfinyl, seleninyl and tellurinyl for $X = S, Se$ and Te , respectively).
io	Ending of acceptable alternative prefixes for certain cationic substituent groups, <i>e.g.</i> ammonio for azaniumyl, pyridinio for pyridiniumyl (<i>cf.</i> Section IR-6.4.9). Now abandoned ending of prefixes for substituent groups consisting of a single atom, <i>e.g.</i> mercurio, $-Hg-$.
irane	General ending of parent names of three-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3. See also 'iridine'.
irene	General ending of parent names of three-membered heteromonocycles with the maximum number of non-cumulative double bonds (<i>i.e.</i> one double bond) in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3. See also 'irine'.
iridine	Ending of parent names of three-membered nitrogen-containing saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
irine	Ending of parent names of three-membered heteromonocycles with the maximum number of non-cumulative double bonds (<i>i.e.</i> one double bond) and N as the only heteroatom(s) in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
ite	Ending of names of anions and esters of oxoacids having the 'ous' or the 'orous' ending in the acid name, <i>e.g.</i> hypochlorite (from hypochlorous acid), methyl sulfite (from sulfurous acid). <i>Cf.</i> Table IR-8.1. See also 'inite', 'onite'.
ito	Ending of name of any anion with an 'ite' name (see above) acting as a ligand, <i>e.g.</i> nitrito, sulfito. <i>Cf.</i> Sections IR-7.1.3 and IR-9.2.2.3 and Table IX. See also 'inito', 'onito'.
ium	Ending of names of many elements and their cations, <i>e.g.</i> helium, seaborgium, thallium(1+), and of the name of any new element (<i>cf.</i> Ref. 1 of Chapter IR-3). Suffix to indicate addition of hydrons to a parent hydride or other parent structure (see 'anium', 'enium', 'inium', 'onium', 'ynium'), accompanied by multiplying prefixes and locants as appropriate, <i>e.g.</i> hydrazinium, $H_2NNH_3^+$; hydrazine-1,2-dium, $^+H_3NNH_3^+$.
o	Terminal vowel indicating a negatively charged ligand; see 'ato', 'ido', 'ito'. Terminal vowel of prefixes for many inorganic and organic substituent groups, <i>e.g.</i> amino, chloro, oxido, sulfo, thiolato. Terminal vowel of prefixes for fusion components. (See Section P-25.3 of the Blue Book ^b .) See also 'eno', 'ino'. Terminal vowel of infixes used in functional replacement nomenclature (Section IR-8.6) to indicate replacement of oxygen atoms and/or hydroxy groups, <i>e.g.</i> 'amido', 'nitrido', 'thio'.
ocane	Ending of parent names of eight-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
ocene	Ending of the names of certain bis(cyclopentadienyl)metal compounds, <i>e.g.</i> ferrocene. <i>Cf.</i> Section IR-10.2.6.

Table III *Continued*

ocene	Ending of parent names of eight-membered heteromonocycles with the maximum number of non-cumulative double bonds in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
ol	Suffix specifying substitution of a hydrogen atom in a parent hydride for the group –OH, accompanied by locants and multiplicative prefixes if appropriate, <i>e.g.</i> silanol, SiH ₃ OH; trisilane-1,3-diol, SiH ₂ (OH)SiH ₂ SiH ₂ OH. Ending of corresponding suffixes ‘thiol’, ‘selenol’, ‘tellurol’ for –SH, –SeH and –TeH, respectively.
olane	General ending of parent names of five-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> IR-6.2.4.3. See also ‘olidine’.
olate	Suffix specifying substitution of a hydrogen atom in a parent hydride for the substituent –O [–] , accompanied by locants and multiplicative prefixes if appropriate, <i>e.g.</i> silanolate, SiH ₃ O [–] ; trisilane-1,3-diolate, SiH ₂ (O [–])SiH ₂ SiH ₂ O [–] . Ending of corresponding suffixes ‘thiolate’, ‘selenolate’, ‘tellurolate’ for –S [–] , –Se [–] and –Te [–] , respectively.
olato	Modification of the suffix ‘olate’ used when the anion in question acts as a ligand.
ole	Ending of parent names of five-membered heteromonocycles with the maximum number of non-cumulative double bonds in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
olidine	Ending of parent names of five-membered nitrogen-containing saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
onane	Ending of parent names of nine-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
onate	Ending of names of anions and esters of ‘onic’ oxoacids, <i>e.g.</i> boronate, phosphonate, tetrathionate.
onato	Modification of the ‘onate’ ending of an anion used when the anion acts as a ligand. Ending of prefixes of certain anionic substituent groups, <i>e.g.</i> phosphonato, –P(=O)(O [–]) ₂ ; sulfonato, –S(=O) ₂ (O [–]).
one	Suffix specifying the substitution of two hydrogen atoms on the same skeletal atom in a parent hydride for the substituent =O, accompanied by locants and multiplicative prefixes as appropriate, <i>e.g.</i> phosphanone, HP=O; pentane-2,4-dione, CH ₃ C(=O)CH ₂ C(=O)CH ₃ . Ending of corresponding suffixes ‘thione’, ‘selenone’, ‘tellurone’ for =S, =Se and =Te, respectively.
onic	Ending of the parent names of acids of the types HXO(OH) ₂ (X=N, P, As, Sb), <i>e.g.</i> stibonic acid; HXO ₂ (OH) (X=S, Se, Te), <i>e.g.</i> sulfonic acid; and of boronic acid, HB(OH) ₂ . See Table IR-8.1. Ending of the parent names dithionic, trithionic, <i>etc.</i> , acids (see Table IR-8.1).
online	Ending of parent names of nine-membered heteromonocycles with the maximum number of non-cumulative double bonds in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
onite	Ending of names of anions and esters of ‘onous’ oxoacids, <i>e.g.</i> phosphonite, tetrathionite.
onito	Modification of the ‘onite’ ending of an anion name used when the anion acts as a ligand.

Table III *Continued*

onium	Ending of still acceptable non-systematic names of cations formed by hydron addition to a mononuclear parent hydride: ammonium, oxonium (see Section IR-6.4.1).
ono	Ending of prefixes for substituent groups formed from ‘onic’ acids by removal of a hydrogen atom, <i>e.g.</i> phosphono for —P(=O)(OH)_2 . <i>Exception:</i> note that $\text{—S(=O)}_2\text{OH}$ is ‘sulfo’ rather than ‘sulfono’.
onous	Ending of the parent names of acids of the types HX(OH)_2 ($\text{X} = \text{N, P, As, Sb}$), <i>e.g.</i> stibonous acid. Ending of the parent names dithionous, trithionous, <i>etc.</i> , acids (see Table IR-8.1).
onoyl	Ending of prefixes for substituent groups formed by removing all hydroxy groups from ‘onic’ acids, <i>e.g.</i> phosphonoyl, HP(O)< ; selenonoyl, $\text{HSe(O)}_2\text{—}$. (See Table IR-8.1 for phosphonic and selenonic acids.)
onyl	Ending of prefixes for the divalent substituent groups >X(=O)_2 (sulfonyl, selenonyl and telluronyl for $\text{X} = \text{S, Se and Te}$, respectively).
orane	Ending of the acceptable alternative names phosphorane for λ^5 -phosphane (PH_5), arsorane for λ^5 -arsane (AsH_5) and stiborane for λ^5 -stibane (SbH_5).
oryl	Ending of prefixes for substituent groups formed by removing all hydroxy groups from ‘oric’ acids, <i>e.g.</i> phosphoryl, P(O)< , from phosphoric acid.
ous	Ending of parent names of certain inorganic oxoacids, <i>e.g.</i> arsorous acid, seleninous acid. For more examples of ‘ous’ acid names, see Tables IR-8.1 and IR-8.2. See also ‘inous’, ‘onous’. Ending formerly added to stems of element names to indicate a lower oxidation state, <i>e.g.</i> ferrous chloride, cuprous oxide, cerous hydroxide. Such names are no longer acceptable.
triene	See ‘ene’.
triide	See ‘ide’.
trium	See ‘ium’.
triyl	Combined suffix composed of the suffix ‘yl’ and the multiplying prefix ‘tri’, indicating the loss of three hydrogen atoms from a parent hydride resulting in a triradical or a substituent group forming three single bonds, <i>e.g.</i> the substituent groups boranetriyl, —B< ; trisilane-1,2,3-triyl, $\text{—SiH}_2\text{SiHSiH}_2\text{—}$; λ^5 -phosphanetriyl, $\text{H}_2\text{P<}$. (See also ‘ylidyne’ and ‘ylylidene’.)
uide	Suffix specifying the addition of hydride to a parent structure, accompanied by locants and multiplicative prefixes if appropriate, <i>e.g.</i> tellanuide, TeH_3^- .
uido	Modification of the ‘uide’ suffix in an anion name used when the anion acts as a ligand.
y	Terminal vowel of prefixes for some substituent groups, <i>e.g.</i> carboxy, —COOH ; hydroxy, —OH ; oxy, —O— . Terminal vowel in prefixes used in specifying chain and ring atoms in additive nomenclature for inorganic chains and rings, <i>cf.</i> Section IR-7.4. These prefixes are given for all elements in Table X.
yl	Suffix to indicate removal of hydrogen atoms from a parent hydride to form radicals or substituent groups, accompanied by multiplicative prefixes and locants as appropriate, <i>e.g.</i> hydrazinyl, $\text{H}_2\text{NNH}^\bullet$ or $\text{H}_2\text{NNH—}$; hydrazine-1,2-diyl, $\bullet\text{HNNH}^\bullet$ or —HNNH— . (See also ‘diyl’, ‘ylene’, ‘ylylidene’, ‘triyl’, ‘ylylidene’, ‘ylidyne’.)

Table III *Continued*

	Ending of certain non-systematic names of oxidometal cations, <i>e.g.</i> vanadyl for oxidovanadium(2+). These names are no longer acceptable.
ylene	Ending of a few still acceptable names for divalent substituent groups, meaning the same as 'diyl': methylene for methanediyl, $-\text{CH}_2-$; phenylene for benzenediyl, $-\text{C}_6\text{H}_4-$; (1,2-phenylene for benzene-1,2-diyl <i>etc.</i>).
ylidene	Suffix for names of divalent substituent groups formed by the loss of two hydrogen atoms from the same atom of a parent hydride and forming a double bond, <i>e.g.</i> azanylidene, $\text{HN}=\text{}$, and for names of corresponding diradicals. (See also 'diyl'.)
ylidyne	Suffix for names of trivalent substituent groups formed by the loss of three hydrogen atoms from the same atom of a parent hydride and forming a triple bond, <i>e.g.</i> phosphanylidyne, $\text{P}\equiv$. (See also 'ylylidene' and 'triyl'.)
ylum	Suffix for names of cations formed by the loss of hydride ions from parent hydrides, accompanied by locants and multiplicative prefixes as appropriate, <i>e.g.</i> azanylium, NH_2^+ ; disilane-1,2-dylium, $^+\text{H}_2\text{SiSiH}_2^+$.
ylylidene	Combined suffix ('yl' plus 'ylidene') for names of trivalent substituent groups formed by the loss of three hydrogen atoms from the same atom, forming a single bond and a double bond, <i>e.g.</i> azanylylidene, $-\text{N}=\text{}$. (See also 'ylidyne' and 'triyl'.)
yne	Ending of systematic names of acyclic and cyclic parent structures with triple-bond unsaturation, replacing 'ane' in the name of the corresponding saturated parent hydride, and if necessary accompanied by locants and multiplicative prefixes specifying the locations and number of triple bonds, <i>e.g.</i> diazyne (see 'ynium' for an application of this name), ethyne, penta-1,4-diyne.
ynide	Combined ending of names of anions resulting from the removal of a hydron from a parent hydride with an 'yne' name, formed by adding the suffix 'ide', <i>e.g.</i> ethynide, $\text{CH}\equiv\text{C}^-$. <i>Cf.</i> Section IR-6.4.4.
ynium	Combined ending of names of cations resulting from the addition of a hydron to a parent structure with an 'yne' name, formed by adding the suffix 'ium', <i>e.g.</i> diazynium ($\text{N}\equiv\text{NH}^+$). <i>Cf.</i> Section IR-6.4.1.

^a The term 'suffix' is understood here to mean a name part added to a parent name in order to specify a modification of that parent, *e.g.* substitution of a hydrogen atom in a parent hydride by a characteristic group (suffixes such as 'carboxylic acid', 'thiol', *etc.*) or formation of a radical or substituent group by removal of one or more hydrogen atoms (suffixes such as 'yl', 'ylidene', *etc.*). The term 'ending' is used in a broader sense, but also to designate specifically the common last part (last syllable or last few syllables) of systematic names for members of classes of compounds (such as 'ane', 'ene', 'diene', 'yne', *etc.*, for parent hydrides, and 'onic acid', 'inic acid', *etc.*, for inorganic oxoacids).

^b *Nomenclature of Organic Chemistry, IUPAC Recommendations*, eds. W.H. Powell and H. Favre, Royal Society of Chemistry, in preparation. (The Blue Book.)

Table IV *Multiplicative prefixes*

1	mono	21	henicosa
2	di ^a (bis ^b)	22	docosa
3	tri (tris)	23	tricos
4	tetra (tetrakis)	30	triaconta
5	penta (pentakis)	31	hentriaconta
6	hexa (hexakis)	35	pentatriaconta
7	hepta (heptakis)	40	tetraconta
8	octa (octakis)	48	octatetraconta
9	nona (nonakis)	50	pentaconta
10	deca (decakis)	52	dopentaconta
11	undeca	60	hexaconta
12	dodeca	70	heptaconta
13	trideca	80	octaconta
14	tetradeca	90	nonaconta
15	pentadeca	100	hecta
16	hexadeca	200	dicta
17	heptadeca	500	pentacta
18	octadeca	1000	kilia
19	nonadeca	2000	dilia
20	icosa		

^a In the case of a ligand using two donor atoms, the term 'bidentate' rather than 'didentate' is recommended because of prevailing usage.

^b The prefixes bis, tris, *etc.* (examples are given for 1–10 but continue throughout) are used with composite ligand names or in order to avoid ambiguity.

Table V *Geometrical and structural affixes*

Except for those denoted by Greek letters, geometrical and structural affixes are italicized. All are separated from the rest of the name by hyphens.

<i>antiprisma</i>	eight atoms bound into a regular antiprism
<i>arachno</i>	a boron structure intermediate between <i>nido</i> and <i>hypho</i> in degree of openness
<i>asym</i>	asymmetrical
<i>catena</i>	a chain structure; often used to designate linear polymeric substances
<i>cis</i>	two groups occupying adjacent positions in a coordination sphere
<i>closo</i>	a cage or closed structure, especially a boron skeleton that is a polyhedron having all faces triangular
<i>cyclo</i>	a ring structure. (Here, <i>cyclo</i> is used as a modifier indicating structure and hence is italicized. In organic nomenclature, 'cyclo' is considered to be part of the parent name since it changes the molecular formula. It is therefore not italicized.)
δ (delta)	denotes the absolute configuration of chelate ring conformations
Δ (delta)	a structural descriptor to designate deltahedra, or shows absolute configuration
<i>dodecahedro</i>	eight atoms bound into a dodecahedron with triangular faces
η (eta)	specifies the bonding of contiguous atoms of a ligand to a central atom
<i>fac</i>	three groups occupying the corners of the same face of an octahedron
<i>hexahedro</i>	eight atoms bound into a hexahedron (<i>e.g.</i> cube)
<i>hexaprisma</i>	twelve atoms bound into a hexagonal prism
<i>hypho</i>	an open structure, especially a boron skeleton, more closed than a <i>klado</i> structure but more open than an <i>arachno</i> structure
<i>icosahedro</i>	twelve atoms bound into an icosahedron with triangular faces
κ (kappa)	specifies the donor atoms in a ligand
<i>klado</i>	a very open polyboron structure
λ (lambda)	signifies, with its superscript, the bonding number, <i>i.e.</i> the sum of the number of skeletal bonds and the number of hydrogen atoms associated with an atom in a parent compound; denotes the absolute configuration of chelate ring conformations
Λ (lambda)	shows absolute configuration
<i>mer</i>	meridional; three groups occupying vertices of an octahedron so that one is <i>cis</i> to the other two which are themselves mutually <i>trans</i>
μ (mu)	signifies that a group so designated bridges two or more coordination centres
<i>nido</i>	a nest-like structure, especially a boron skeleton that is almost closed
<i>octahedro</i>	six atoms bound into an octahedron
<i>pentaprisma</i>	ten atoms bound into a pentagonal prism
<i>quadro</i>	four atoms bound into a quadrangle (<i>e.g.</i> square)
<i>sym</i>	symmetrical
<i>tetrahedro</i>	four atoms bound into a tetrahedron
<i>trans</i>	two groups occupying positions in a coordination sphere directly opposite each other
<i>triangulo</i>	three atoms bound into a triangle
<i>triprisma</i>	six atoms bound into a triangular prism

Table VI *Element sequence*

→ He	Ne	Ar	Kr	Xe	Rn	→ Li	Na	K	Rb	Cs	Fr	→ Be	Mg	Ca	Sr	Ba	Ra	→ Sc	Y	La	Lu	→ Ti	Zr	Hf	Rf	→ V	Nb	Ta	Db	→ Cr	Mo	W	Sg	→ Mn	Tc	Re	Bh	→ Fe	Ru	Os	Hs	→ Co	Rh	Ir	Mt	→ Ni	Pd	Pt	Ds	→ Cu	Ag	Au	Rg	→ Zn	Cd	Hg	→ B	Al	Ga	In	Tl	→ C	Si	Ge	Sn	Pb	→ N	P	As	Sb	Bi	→ O	S	Se	Te	Po	→ F	Cl	Br	I	At
------	----	----	----	----	----	------	----	---	----	----	----	------	----	----	----	----	----	------	---	----	----	------	----	----	----	-----	----	----	----	------	----	---	----	------	----	----	----	------	----	----	----	------	----	----	----	------	----	----	----	------	----	----	----	------	----	----	-----	----	----	----	----	-----	----	----	----	----	-----	---	----	----	----	-----	---	----	----	----	-----	----	----	---	----

Table VII Ligand abbreviations

Guidelines for the construction and use of ligand abbreviations are given in Section IR-4.4.4 and their use in the formulae of coordination complexes is described in Section IR-9.2.3.4. Abbreviations are listed in alphabetical order but those beginning with a numeral are listed by the first letter of the abbreviation (e.g. 2,3,2-tet appears under the letter ‘t’).

Structural formulae of selected ligands are shown in Table VIII (numbered according to the present Table).

<i>Number and abbreviation</i>	<i>Systematic name</i>	<i>Other name (from which abbreviation derived)^a</i>
1. 4-abu	4-aminobutanoato	
2. Ac	acetyl	
3. acac	2,4-dioxopentan-3-ido	acetylacetonato
4. acacen	2,2'-[ethane-1,2-diylbis(azanylylidene)]bis(4-oxopentan-3-ido)	bis(acetylacetonato)ethylenediamine
5. ade	9H-purin-6-amine	adenine
6. ado	9-β-D-ribofuranosyl-9H-purin-6-amine	adenosine
7. adp	adenosine 5'-diphosphato(3-)	
8. aet	2-aminoethanethiolato	
9. ala	2-aminopropanoato	alaninato
10. ama	2-aminopropanedioato	aminomalonato
11. amp	adenosine 5'-phosphato(2-)	adenosine monophosphato
12. [9]aneN ₃ (also taen)	1,4,7-triazonane	
13. [12]aneN ₄ (also cyclen)	1,4,7,10-tetraazacyclododecane	
14. [14]aneN ₄ (also cyclam)	1,4,8,11-tetraazacyclotetradecane	
15. [18]aneP ₄ O ₂	1,10-dioxo-4,7,13,16-tetraphosphacyclooctadecane	
16. [9]aneS ₃	1,4,7-trithionane	
17. [12]aneS ₄	1,4,7,10-tetrathiaacyclododecane	
18. arg	2-amino-5-carbamimidamidopentanoato	argininato
19. asn	2,4-diamino-4-oxobutanoato	asparaginato
20. asp	2-aminobutanedioato	aspartato
21. atmp	[nitrilotris(methylene)]tris(phosphonato)	aminotris(methylenephosphonato)
22. atp	adenosine 5'-triphosphato(4-)	
23. 2,3-bdta	2,2',2'',2'''-(butane-2,3-diylidinitrilo)tetraacetato	

Table VII Continued

Number and abbreviation	Systematic name	Other name (from which abbreviation derived) ^a
24. benzo-15-crown-5	2,3,5,6,8,9,11,12-octahydro-1,4,7,10,13-benzopentaoxacyclopentadecine	
25. big	bis(carbamimidoyl)azanido	biguanid-3-ido
26. biim	2,2'-bi(1 <i>H</i> -imidazole)-1,1'-diido	2,2'-biimidazolato
27. binap	1,1'-binaphthalene-2,2'-diylbis(diphenylphosphane)	
28. bn	butane-2,3-diamine	
29. bpy	2,2'-bipyridine	
30. 4,4'-bpy	4,4'-bipyridine	
31. Bu	butyl	
32. bzac	1,3-dioxo-1-phenylbutan-2-ido	benzoylacetonoato
33. bzim	1 <i>H</i> -benzimidazol-1-ido	
34. Bz ^b	benzyl	
35. bztz	1,3-benzothiazole	
36. cat	benzene-1,2-diolato	catecholato
37. cbdea	cyclobutane-1,1-dicarboxylato	
38. cdta	2,2',2'',2'''-(cyclohexane-1,2-diylidinitrilo)tetraacetato	
39. C ₅ H ₄ Me	methylcyclopentadienyl	
40. chxn (also dach)	cyclohexane-1,2-diamine	
41. cit	2-hydroxypropane-1,2,3-tricarboxylato	cittrato
42. C ₅ Me ₅ ^c	pentamethylcyclopentadienyl	
43. cod	cycloocta-1,5-diene	
44. cot	cycloocta-1,3,5,7-tetraene	
45. Cp	cyclopentadienyl	
46. cptn	cyclopentane-1,2-diamine	
47. 18-crown-6	1,4,7,10,13,16-hexaoxacyclooctadecane	
48. crypt-211	4,7,13,18-tetraoxa-1,10-diazabicyclo[8.5.5]icosane	cryptand 211
49. crypt-222	4,7,13,16,21,24-hexaoxa-1,10-diazabicyclo[8.8.8]hexacosane	cryptand 222
50. Cy	cyclohexyl	

cyclam (see [14]aneN ₄ , No. 14)		
cyclen (see [12]aneN ₄ , No. 13)		
51. cys	2-amino-3-sulfanypropanoato	cysteinato
52. cyt	4-aminopyrimidin-2(1 <i>H</i>)-one	cytosine
53. dabco	1,4-diazabicyclo[2.2.2]octane	
dach (see chxn, No. 40)		
54. dbm	1,3-dioxo-1,3-diphenylpropan-2-ido	diaminocyclohexane
55. dea	2,2'-azanediyldi(ethan-1-olato)	dibenzoylmethanato
56. depe	ethane-1,2-diylbis(diethylphosphane)	diethanolaminato
57. diars	benzene-1,2-diylbis(dimethylarsane)	1,2-bis(diethylphosphino)ethane
58. dien	<i>N</i> -(2-aminoethyl)ethane-1,2-diamine	diethylenetriamine
59. [14]1,3-dieneN ₄	1,4,8,11-tetraazacyclotetradeca-1,3-diene	
60. diop	[(2,2-dimethyl-1,3-dioxolane-4,5-diyl)bis(methylene)]bis(diphenylphosphane)	
61. diox	1,4-dioxane	
62. dipamp	ethane-1,2-diylbis[(2-methoxyphenyl)phenylphosphane]	'dimer of phenylanisylmethylphosphine'
63. dma	<i>N,N</i> -dimethylacetamide	dimethylacetamide
64. dme	1,2-dimethoxyethane	
65. dmf	<i>N,N</i> -dimethylformamide	
66. dmg	butane-2,3-diylidenebis(azanolato)	dimethylglyoximate
67. dmpe	ethane-1,2-diylbis(dimethylphosphane)	1,2-bis(dimethylphosphino)ethane
68. dmpm	methylenebis(dimethylphosphane)	bis(dimethylphosphino)methane
69. dmso	(methanesulfinyl)methane	dimethyl sulfoxide
70. dpm	2,2,6,6-tetramethyl-3,5-dioxoheptan-4-ido	dipivaloylmethanato
71. dppe	ethane-1,2-diylbis(diphenylphosphane)	1,2-bis(diphenylphosphino)ethane
72. dppl	1,1'-bis(diphenylphosphanyl)ferrocene	
73. dppm	methylenebis(diphenylphosphane)	bis(diphenylphosphino)methane
74. dppp	propane-1,3-diylbis(diphenylphosphane)	1,3-bis(diphenylphosphino)propane
75. dtmpe	(phosphonomethyl)azanediylbis[ethane-2,1-diyl]nitriolobis(methylene)]tetrakis(phosphonato)	diethylenetriaminepentakis- C (methylene)phosphonato) ^d
76. dtpe	2,2',2''/2'''-(carboxylatomethyl)azanediylbis[ethane-2,1-diyl]nitriolotetraacetato	diethylenetriaminepentaacetato
77. ea	2-amino(ethan-1-olato)	ethanolaminato

Table VII Continued

Number and abbreviation	Systematic name	Other name (from which abbreviation derived) ^a
78. edda	2,2'-(ethane-1,2-diylbis(azanediyl))diacetato	ethylenediaminediacetato
79. edta	2,2',2'',2'''-(ethane-1,2-diyl(dinitrilo)tetraacetato	ethylenediaminetetraacetato
80. edtmpa	ethane-1,2-diylbis[nitrilobis(methylene)]tetrakis(phosphonato)	ethylenediaminetetrakis(methylenephosphonato) ^d
81. egta	2,2',2'',2'''-(ethane-1,2-diylbis(oxyethane-2,1-diyl(nitrilo))tetraacetato	ethylene glycol-bis(2-aminoethyl)- <i>N,N,N',N'</i> -tetraacetic acid
82. en	ethane-1,2-diamine	
83. Et	ethyl	
84. Et ₂ dic	<i>N,N</i> -diethylcarbomodithioato	<i>N,N</i> -diethyldithiocarbamato
85. fod	6,6,7,7,8,8,8-heptafluoro-2,2-dimethyl-3,5-dioxooctan-4-ido	
86. fta	1,1,1-trifluoro-2,4-dioxopentan-3-ido	trifluoroacetylacetonato
87. gln	2,5-diamino-5-oxopentanoato	glutaminato
88. glu	2-aminopentanedioato	glutamato
89. gly	aminoacetato	glycinato
90. gua	2-amino-9 <i>H</i> -purin-6(1 <i>H</i>)-one	guanine
91. guo	2-amino-9-β-D-ribofuranosyl-9 <i>H</i> -purin-6(1 <i>H</i>)-one	guanosine
92. hdtmpa	hexane-1,6-diylbis[nitrilobis(methylene)]tetrakis(phosphonato)	hexamethylenediaminetetrakis(methylenephosphonato) ^d
93. hedp	1-hydroxyethane-1,1-diylbis(phosphonato)	1-hydroxyethane-1,1-diphosphonato
94. hfa	1,1,1,5,5,5-hexafluoropentane-2,4-dioxopentan-3-ido	hexafluoroacetylacetonato
95. his	2-amino-3-(imidazol-4-yl)propanoato	histidinato
96. hmpa	hexamethylphosphoric triamide	
97. hmta	1,3,5,7-tetraazatricyclo[3.3.1.1 ^{3,7}]decane	hexamethylenetetramine
98. ida	2,2'-azanediylidiacetato	iminodiacetato
99. ile	2-amino-3-methylpentanoato	isoleucinato
100. im	1 <i>H</i> -imidazol-1-ido	
101. isn	pyridine-4-carboxamide	isonicotinamide

102. leu	2-amino-4-methylpentanoato	leucinato
103. lut	2,6-dimethylpyridine	lutidine
104. lys	2,6-diaminohexanoato	lysinato
105. mal	2-hydroxybutanedioato	malato
106. male	(Z)-butenedioato	maleato
107. malo	propanedioato	malonato
108. Me	methyl	
109. 2-Mepy	2-methylpyridine	
110. met	2-amino-4-(methylsulfonyl)butanoato	methioninato
111. mnt	1,2-dicyanoethene-1,2-dithiolato	maleonitriledithiolato
112. napy	1,8-naphthyridine	
113. nbd	bicyclo[2.2.1]hepta-2,5-diene	norbornadiene
114. nia	pyridine-3-carboxamide	nicotinamide
115. nmp	N-methylpyrrolidine	
116. nta	2,2',2''-nitrioltriacetato	
117. oep	2,3,7,8,12,13,17,18-octaethylporphyrin-21,23-diido	
118. ox	ethanedioato	oxalato
119. pc	phthalocyanine-29,31-diido	
120. 1,2-pdta	2,2',2'',2'''-(propane-1,2-diyl(dinitrilo))tetraacetato	1,2-propylenediaminetetraacetato
121. 1,3-pdta	2,2',2'',2'''-(propane-1,3-diyl(dinitrilo))tetraacetato	1,3-propylenediaminetetraacetato
122. Ph	phenyl	
123. phe	2-amino-3-phenylpropanoato	phenylalaninato
124. phen	1,10-phenanthroline	
125. pip	piperidine	
126. pmdien	2,2'-(methylazanediyl)bis(N,N-dimethylethan-1-amine)	N,N,N',N'',N'''-pentamethyl- C -diethylenetriamine ^d
127. pn	propane-1,2-diamine	
128. pplX	2,18-bis(2-carboxyethyl)-3,7,12,17-tetramethyl-8,13-divinylporphyrin-21,23-diido	protoporphyrinato IX
129. pro	pyrrolidine-2-carboxylato	prolinato
130. ptn	pentane-2,4-diamine	
131. py	pyridine	

Table VII Continued

Number and abbreviation	Systematic name	Other name (from which abbreviation derived) ^a
132. pyz	pyrazine	
133. pz	1 <i>H</i> -pyrazol-1-ido	
134. qdt	quinoxaline-2,3-dithiolato	
135. quin	quinolin-8-olato	
136. sal	2-hydroxybenzoato	salicylato
137. salan	2-[(phenylimino)methyl]phenolato	salicylideneanilinato
138. saldien	2,2'-[azanediy]bis(ethane-2,1-diylazanylylidenemethanylylidene)]diphenolato	bis(salicylidene)diethylenetriaminato
139. salen	2,2'-[ethane-1,2-diylbis(azanylylidenemethanylylidene)]diphenolato	bis(salicylidene)ethylenediaminato
140. salgly	<i>N</i> -(2-oxidobenzylidene)glycinato	salicylideneglycinato
141. salpn	2,2'-[propane-1,2-diylbis(azanylylidenemethanylylidene)]diphenolato	bis(salicylidene)propylenediaminato
142. saltn	2,2'-[propane-1,3-diylbis(azanylylidenemethanylylidene)]diphenolato	bis(salicylidene)trimethylenediaminato
143. sdta ^e	2,2',2'',2'''-[(1,2-diphenylethane-1,2-diyl)dinitrilo]tetraacetato	stilbenediaminetetraacetato
144. sepf ^f	1,3,6,8,10,13,16,19-octaazabicyclo[6.6.6]icosane	
145. ser	2-amino-3-hydroxypropanoato	serinato
146. stien ^e	1,2-diphenylethane-1,2-diamine	stilbenediamine
tacn (see [9]aneN ₃ , No. 12)		1,4,7-triazacyclononane
147. tap	propane-1,2,3-triamine	1,2,3-triaminopropane
148. tart	2,3-dihydroxybutanedioato	tartrato
149. tcne	ethenetetracarbonitrile	tetracyanoethylene
150. tenq	2,2'-(cyclohexa-2,5-diene-1,4-diylidene)di(propanedinitrile)	tetracyanoquinodimethane
151. tdt	4-methylbenzene-1,2-dithiolato	
152. tea	2,2',2''-nitrioltri(ethan-1-olato)	triethanolaminato
153. terpy	2,2':6':2''-terpyridine	terpyridine
154. 2,3,2-tet	<i>N,N'</i> -bis(2-aminoethyl)propane-1,3-diamine	1,4,8,11-tetrazaundecane
155. 3,3,3-tet	<i>N,N'</i> -bis(3-aminopropyl)propane-1,3-diamine	1,5,9,13-tetraza-tridecane
156. tetren	<i>N,N'</i> -(azanediy)diethane-2,1-diyl)di(ethane-1,2-diamine)	tetraethylenepentamine

157. tfa	trifluoroacetato		
158. thf	oxolane		tetrahydrofuran
159. thiox	1,4-oxathiane		thioxane
160. thr	2-amino-3-hydroxybutanoato		threoninato
161. tht	thiolane		tetrahydrothiophene
162. thy	5-methylpyrimidine-2,4(1 <i>H</i> ,3 <i>H</i>)-dione		thymine
163. tmen	<i>N,N,N',N'</i> -tetramethylethane-1,2-diamine		
164. tmp	5,10,15,20-tetrakis(2,4,6-trimethylphenyl)porphyrin-21,23-diido		5,10,15,20-tetramesitylporphyrin-21,23-diido
165. tn	propane-1,3-diamine		trimethylenediamine
166. Tol (<i>o</i> -, <i>m</i> - or <i>p</i> -)	2-, 3- or 4-methylphenyl		tolyl (<i>o</i> -, <i>m</i> - or <i>p</i> -)
167. Tp	hydridotris(pyrazolido- <i>N</i>)borato(1-), or tris(1 <i>H</i> -pyrazol-1-yl)boranuido		hydrottris(pyrazolyl)borato
168. Tp ^g	tris(3,5-dimethylpyrazolido- <i>N</i>)hydridoborato(1-)		hydrottris(3,5-dimethylpyrazolyl)borato
169. tpp	5,10,15,20-tetraphenylporphyrin-21,23-diido		
170. tren	<i>N,N</i> -bis(2-aminoethyl)ethane-1,2-diamine		tris(2-aminoethyl)amine
171. trien	<i>N,N'</i> -bis(2-aminoethyl)ethane-1,2-diamine		triethylenetetramine
172. triphos ^h	[(phenylphosphanediy)bis(ethane-2,1-diyl)]bis(diphenylphosphane)		
173. tris	2-amino-2-(hydroxymethyl)propane-1,3-diol		
174. ttp	2-amino-3-(1 <i>H</i> -indol-3-yl)propanoato		tris(hydroxymethyl)aminomethane
175. tsalen	2,2'-[ethane-1,2-diyl]bis(azanylylidenemethanylylidene)]dibenzeneethiolato		tryptophanato
176. tffa	4,4,4-trifluoro-1,3-dioxo-1-(2-thienyl)butan-2-ido		bis(thiosalicylidene)ethylenediaminato
177. ttha	2,2',2'',2'''-(ethane-1,2-diyl)bis{[(carboxylatomethyl)azanediy]ethane-2,1-diylinitrilo}tetraacetato		thenoyltrifluoroacetato
178. ttp	5,10,15,20-tetrakis(4-methylphenyl)porphyrin-21,23-diido		triethylenetetraminehexaacetato
179. tu	thiourea		5,10,15,20-tetra- <i>p</i> -tolylporphyrin-21,23-diido
180. tyr	2-amino-3-(4-hydroxyphenyl)propanoato		tyrosinato
181. tz	1,3-thiazole		thiazole
182. ura	pyrimidine-2,4(1 <i>H</i> ,3 <i>H</i>)-dione		uracil
183. val	2-amino-3-methylbutanoato		valinato

^a Many of these names are no longer acceptable.

^b The abbreviation Bz has often been used previously for 'benzoyl', and Bzl has been used for 'benzyl'. Use of the alternatives, PhCO and PhCH₂, is therefore preferable.

^c The use of the abbreviation Cp* for pentamethylcyclopentadienyl is discouraged. It can lead to confusion because the asterisk, *, is also used to represent an excited state, an optically active substance, *etc.*

- ^d The symbol ‘ $\frown$ ’ is used to divide the name, necessitated by the line break. In the absence of the line break this symbol is omitted. Note that all *hyphens* are true parts of the name.
- ^e The abbreviation derives from the non-systematic name stilbenediamine which incorrectly implies the presence of a C=C double bond in the ligand.
- ^f The abbreviation derives from the non-systematic name sepulchrate which incorrectly implies that the ligand is anionic.
- ^g The use of Tp’ is preferred to Tp* for the reasons given in footnote c. A general procedure for abbreviating substituted hydridotris(pyrzolido-*N*)borate ligands has been proposed [see S. Trofimenko, *Chem. Rev.*, **93**, 943–980 (1993)]. For example, Tp’ becomes Tp^{Me}, the superscript denoting the methyl groups at the 3- and 5-positions of the pyrazole rings.
- ^h The abbreviation triphos should not be used for the four-phosphorus ligand PhP(CH₂PPh₂)₃.

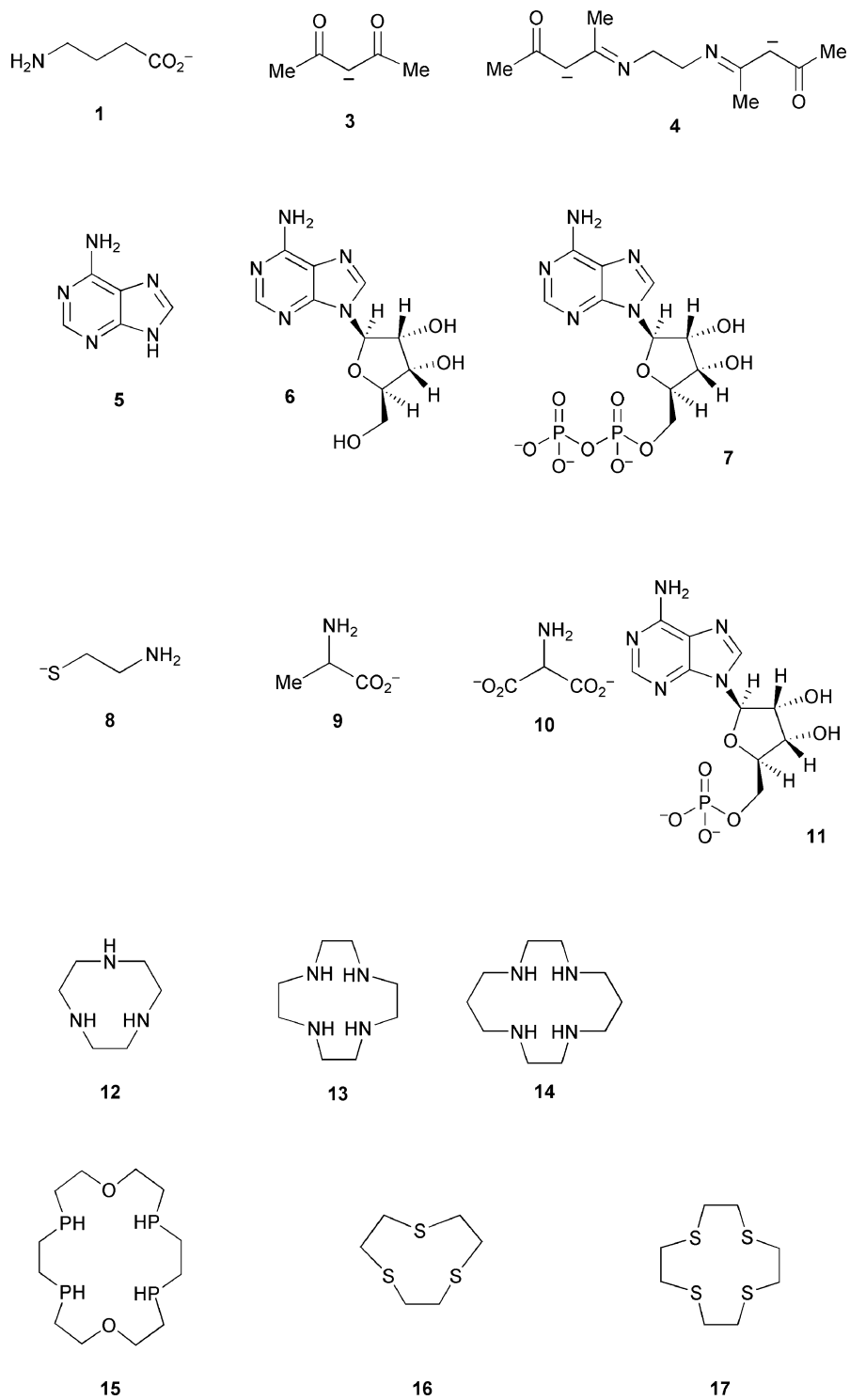
Table VIII Structural formulae of selected ligands (numbered according to Table VII)

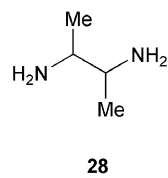
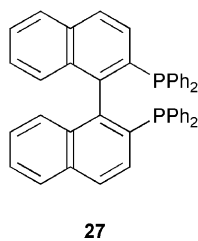
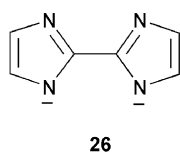
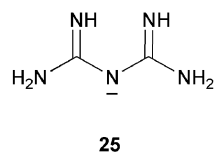
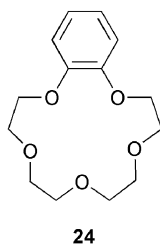
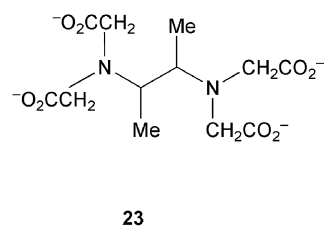
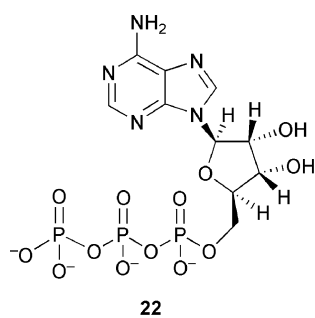
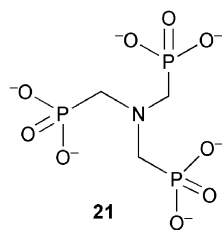
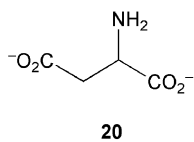
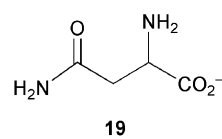
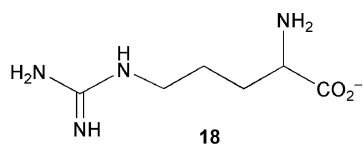
Table VIII *Continued*

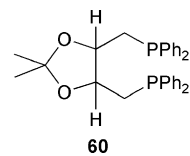
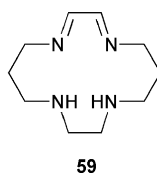
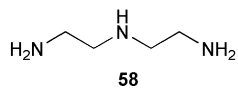
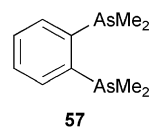
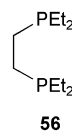
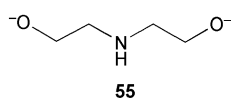
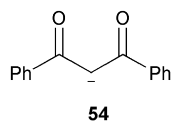
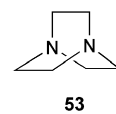
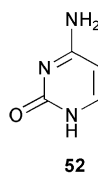
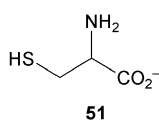
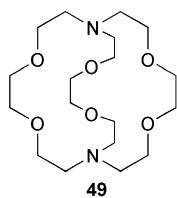
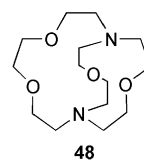
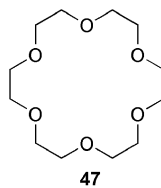
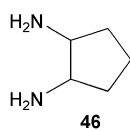
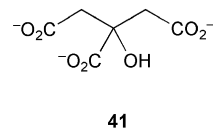
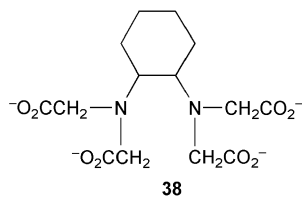
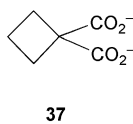
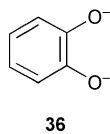
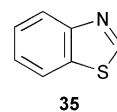
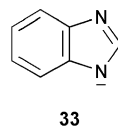
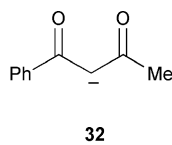
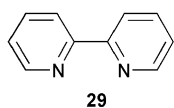
Table VIII *Continued*

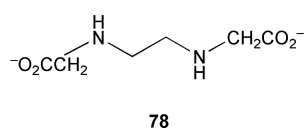
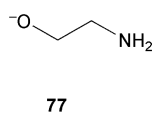
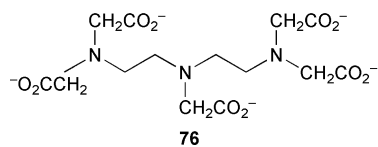
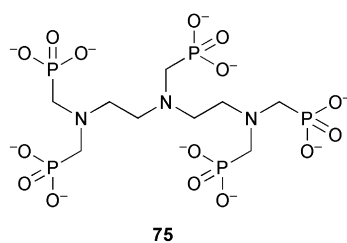
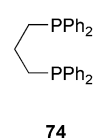
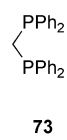
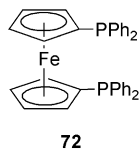
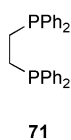
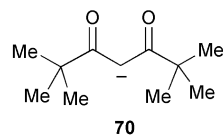
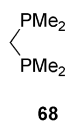
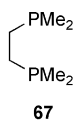
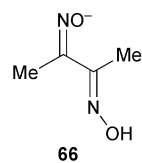
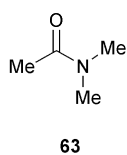
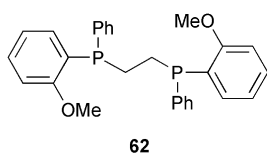
Table VIII *Continued*

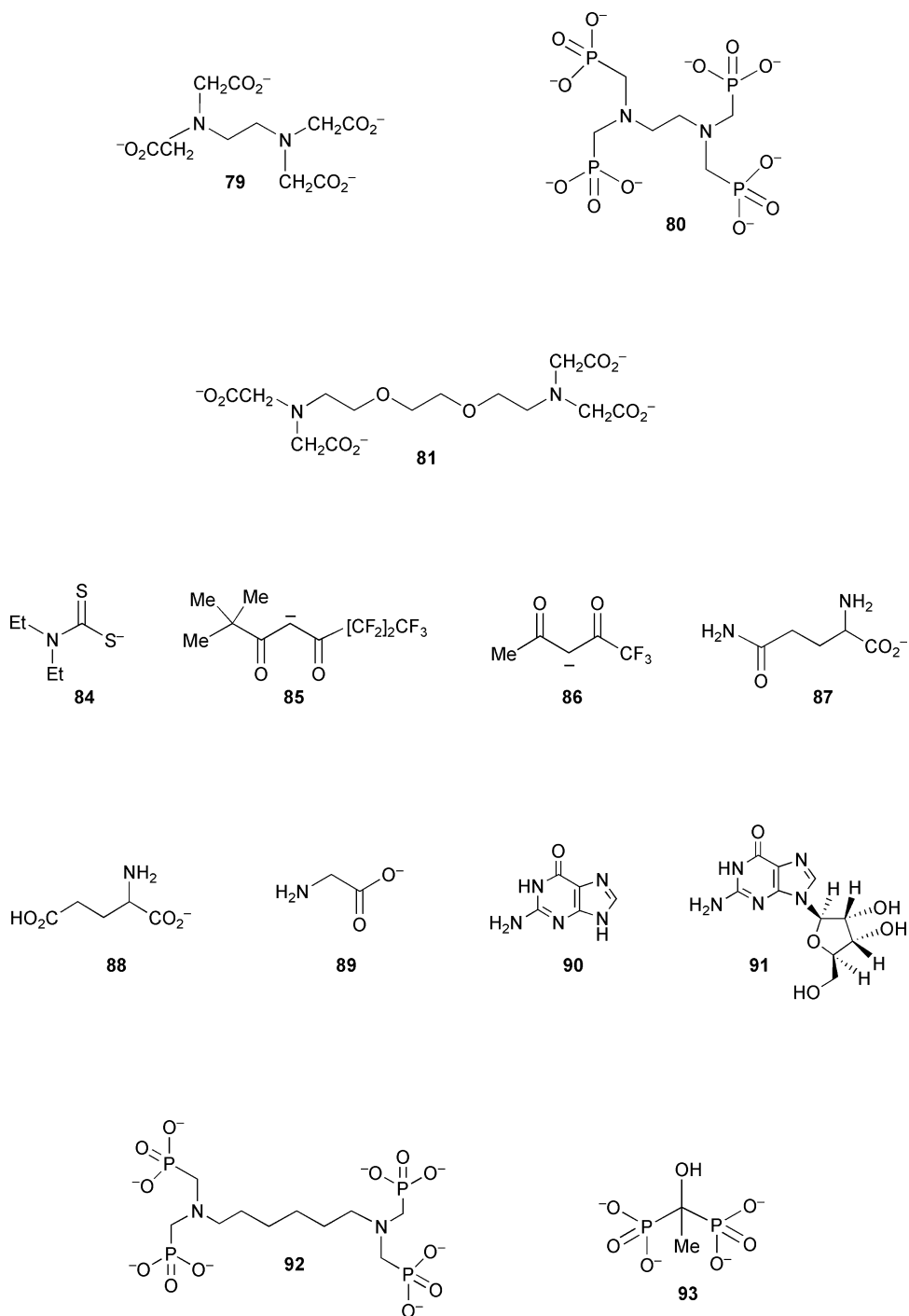
Table VIII *Continued*

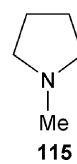
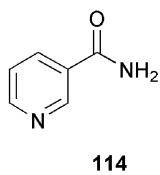
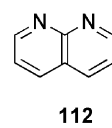
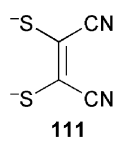
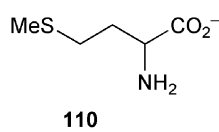
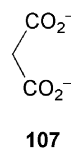
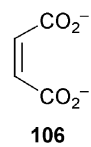
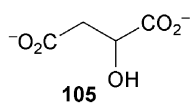
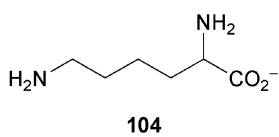
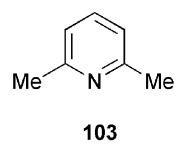
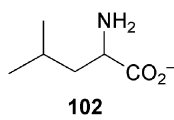
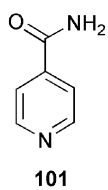
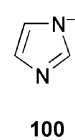
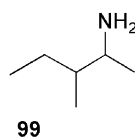
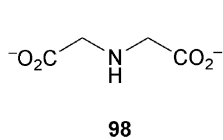
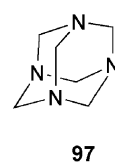
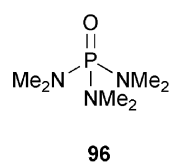
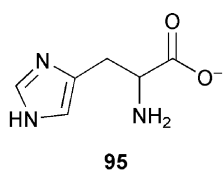
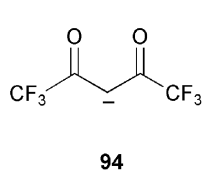
Table VIII *Continued*

Table VIII Continued

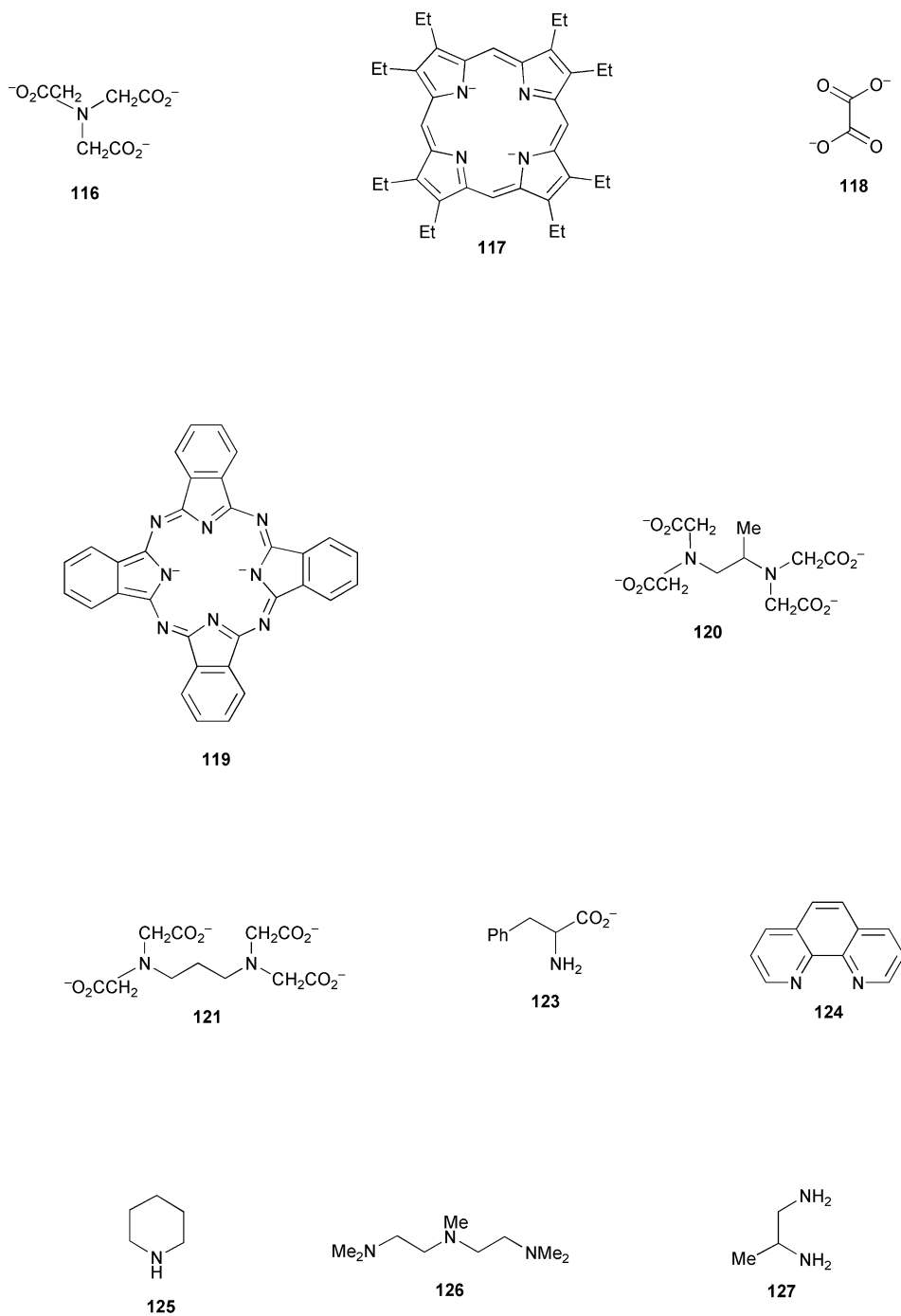


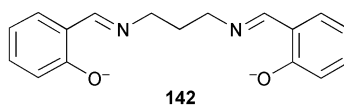
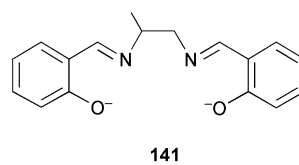
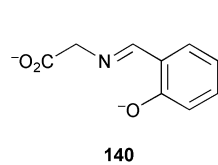
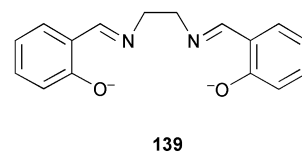
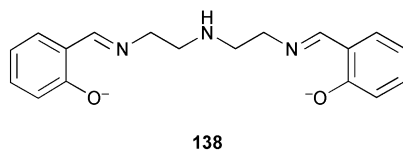
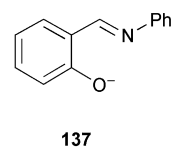
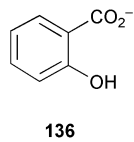
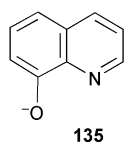
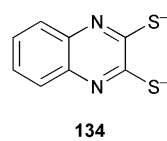
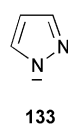
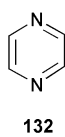
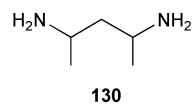
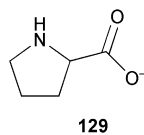
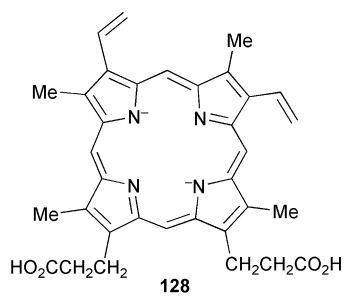
Table VIII *Continued*

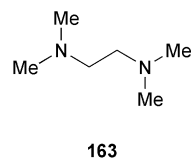
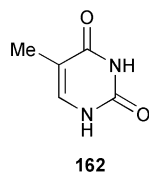
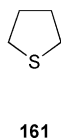
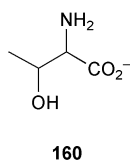
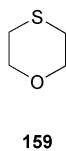
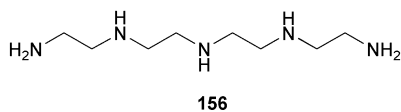
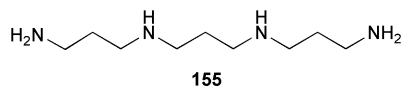
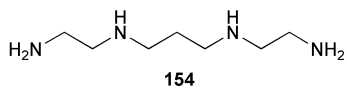
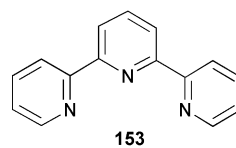
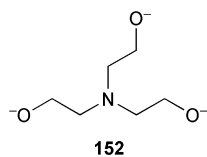
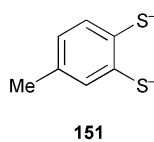
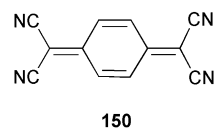
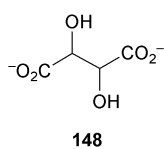
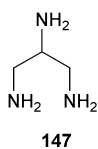
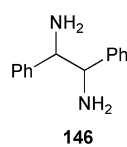
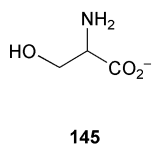
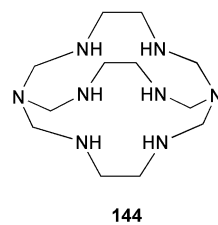
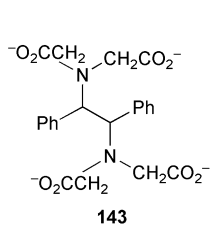
Table VIII *Continued*

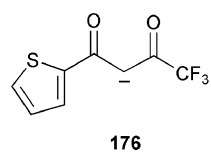
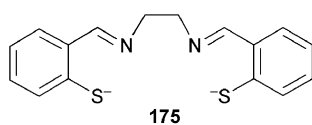
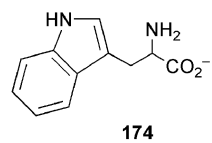
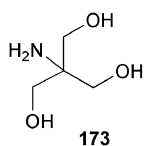
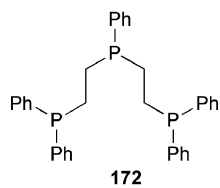
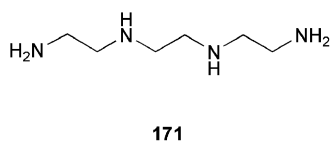
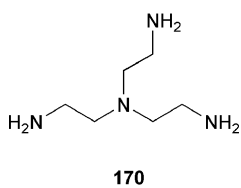
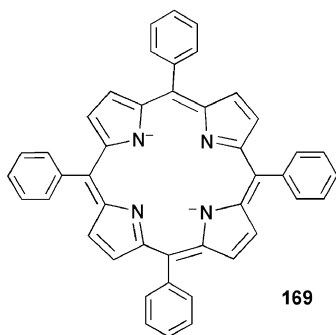
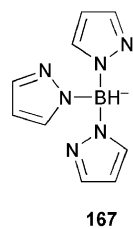
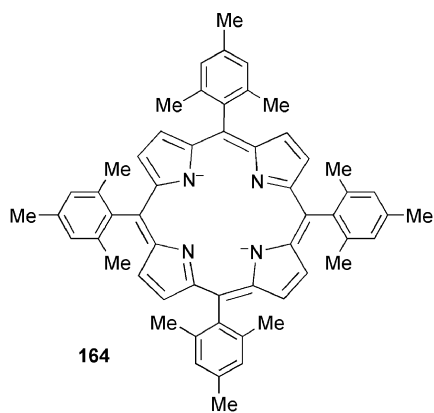
Table VIII *Continued*

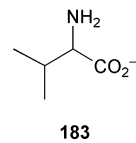
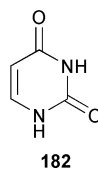
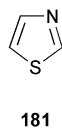
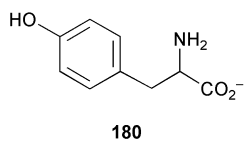
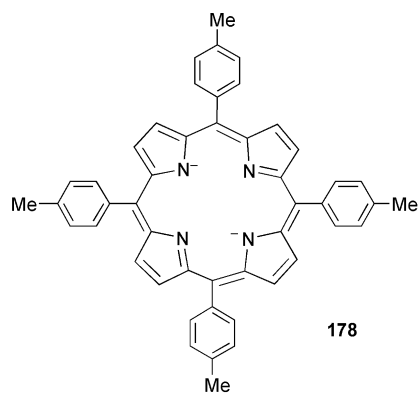
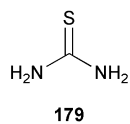
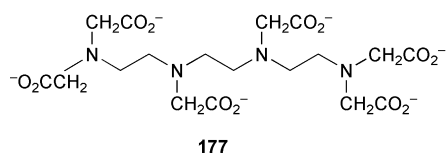
Table VIII *Continued*

Table IX Names of homoatomic, binary and certain other simple molecules, ions, compounds, radicals and substituent groups

This Table names a large number of homoatomic and binary compounds and species, and some heteropolynuclear entities, and thus may be used as a reference for names of simple compounds and a source of examples to guide in the naming of further compounds. It may be necessary to browse the Table to find (families of) compounds that match those of interest. For example, all the oxides of potassium are named; corresponding compounds of the other alkali metals, not included here, are named analogously. Several silicon and germanium hydride species are named; names for corresponding tin and lead species are not necessarily included.

Some inorganic acids and their corresponding bases are included in this Table, but more acid names are given in Tables IR-8.1 and IR-8.2. Only a few simple carbon-containing compounds and substituent groups are included. In particular, organic ligands belonging to the general classes alcohols, thiolates, phenolates, carboxylates, amines, phosphanes and arsanes as well as (partially) dehydrated amines, phosphanes and arsanes are generally *not* included. Their naming is described and exemplified in Section IR-9.2.2.3.

Entries in the first column are ordered alphabetically according to the formulae as they appear here. Formulae for *binary* species are written in this column according to the position of the two elements in Table VI (Section IR-4.4), *e.g.* ammonia is found under 'NH₃', but selane under 'H₂Se' and AlLi under 'LiAl'. In case of doubt, cross references should aid in finding the correct entry in the Table. However, in the first column, formulae of *ternary* and *quaternary* compounds are written strictly alphabetically, *e.g.* 'ClSCN^{•-}' is found under the entry 'CClNS', and 'HPO₄⁻' under 'HO₄P', and these formulae are ordered as described in Section IR-4.4.2.2. In the columns to the right of the first column, special formats may be used for formulae in order to stress a particular structure, *e.g.* under the entry 'BrHO₃' one finds 'HOBrO₂' rather than 'HBrO₃' or '[BrO₂(OH)]', the two formats presented in Table IR-8.2.

The symbol 'C' is used for dividing names when this is made necessary by a line break. When the name is reconstructed from the name given in the Table, this symbol should be omitted. Thus, all *hyphens* in the Table are true parts of the names. The symbols '>' and '<' placed next to an element symbol both denote two single bonds connecting the atom in question to two other atoms.

For a given compound, the various systematic names, if applicable, are given in the order: stoichiometric names (Chapter IR-5), substitutive names (Chapter IR-6), additive names (Chapter IR-7) and hydrogen names (Section IR-8.4). Acceptable names which are not entirely systematic (or not formed according to any of the systems mentioned above) are given at the end after a semicolon. No order of preference is implied by the order in which formulae and names are listed, but in practice it may be useful to select particular formulae and names for particular uses. Thus, for sodium chloride the formula [NaCl] and the additive name 'chloridosodium' may be used specifically for the molecular compound, which can be regarded as a coordination compound, whereas 'sodium chloride' may be used, and traditionally *is* used, for the compound in general and for the solid with the composition NaCl. Corresponding remarks apply to a number of hydrides for which a stoichiometric name may be applied to the compound with the stoichiometry in question without further structural implications (such as 'aluminium trihydride' for AlH₃ or 'dihydrogen disulfide' for H₂S₂), whereas a parent hydride name (needed anyway for naming certain derivatives) or an additive name may be used to denote specifically the molecular compound or entity (such as 'alumane' or 'trihydroaluminium' for the molecular entity [AlH₃]) and 'disulfane' for HSSH).

Note from the examples above that in order to stress the distinctions discussed, the square brackets are sometimes used in the Table to enclose formulae for molecular entities that are otherwise often written with no enclosing marks. When a formula with square brackets is shown, there will also be a coordination-type additive name.

Formula for uncharged atom or group	Name			
	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
Ac	actinium	actinium	actinide ^d	actinido
Ag	silver	silver	argentide	argentido
Al	aluminium	aluminium (general) Al ⁺ , aluminium(1+) Al ³⁺ , aluminium(3+)	aluminide (general) Al ⁻ , aluminide(1-)	aluminido (general) Al ⁻ , aluminido(1-)
AlCl	AlCl, aluminium monochloride [AlCl], chloridoaluminium	AlCl ⁺ , chloridoaluminium(1+)		
AlCl ₃ (see also Al ₂ Cl ₆)	AlCl ₃ , aluminium trichloride [AlCl ₃], trichloroalumane, trichloridoaluminium			
AlCl ₄			AlCl ₄ ⁻ , tetrachloroalumanuide, tetrachloridoaluminate(1-)	AlCl ₄ ⁻ , tetrachloroalumanuido, tetrachloridoaluminate(1-)
AlH	AlH, aluminium monohydride [AlH], λ ¹ -alumane (parent hydride name), hydridoaluminium	AlH ⁺ , hydridoaluminium(1+)		
AlH ₂	-AlH ₂ , alumanyl			
AlH ₃	AlH ₃ , aluminium trihydride [AlH ₃], alumane (parent hydride name), trihydridoaluminium	AlH ₃ ^{•+} , alumaniumyl, trihydridoaluminium(•1+)	AlH ₃ ^{•-} , alumanidyl, trihydridoaluminate(•1-) ^e	
AlH ₄			AlH ₄ ⁻ , alumanuide, tetrahydridoaluminate(1-)	AlH ₄ ⁻ , alumanuido, tetrahydridoaluminate(1-)
AlO	AlO, aluminium mon(o)oxide [AlO], oxidoaluminium	AlO ⁺ , oxidoaluminium(1+)	AlO ⁻ , oxidoaluminate(1-)	
AlSi	AlSi, aluminium monosilicide [AlSi], silicidoaluminium			

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
Al ₂	Al ₂ , dialuminium		Al ₂ ⁻ , dialuminide(1-)	
Al ₂ Cl ₆	[Cl ₂ Al(μ-Cl) ₂ AlCl ₂], di-μ-chlorido-bis(dichlorido- $\ominus$ aluminium)			
Al ₄			Al ₄ ²⁻ , tetraaluminide(2-)	
Am	americium	americium	americide	americido
Ar	argon	argon (general) Ar ⁺ , argon(1+)	argonide	argonido
ArBe		ArBe ⁺ , beryllidoargon(1+)		
ArF	ArF, argon monofluoride [ArF], fluoroargon	ArF ⁺ , fluoroargon(1+)		
ArHe		ArHe ⁺ , helidoargon(1+)		
ArLi		ArLi ⁺ , lithidoargon(1+)		
Ar ₂	Ar ₂ , diargon	Ar ₂ ⁺ , diargon(1+)		
As	arsenic >As-, arsanytriyl	arsenic	arsenide (general) As ³⁻ , arsenide(3-), arsanytriide; arsenide	arsenido (general) As ³⁻ , arsanytriido; arsenido
AsH	AsH, arsenic monohydride AsH ^{2•} , arsanylidene, hydridoarsenic(2•) >AsH, arsanydiyl =AsH, arsanylidene	AsH ²⁺ , arsanebis(ylum), hydridoarsenic(1+)	AsH ²⁻ , arsanydiide, hydridoarsenate(2-)	AsH ²⁻ , arsanydiido, hydridoarsenate(2-)

AsHO	>AsH(O), oxo- λ^5 -arsanedyl; arsonoyl = AsH(O), oxo- λ^5 -arsanylidene; arsonoylidene			
AsHO ₂	>AsO(OH), hydroxy(oxo)- λ^5 -arsanedyl; hydroxyarsoryl = AsO(OH), hydroxy(oxo)- λ^5 - arsanylidene; hydroxyarsorylidene	AsHO ₂ ²⁻ , hydridodioxidoarsenate(2-); arsonite	AsHO ₂ ²⁻ , hydridodioxidoarsenate(2-); arsonito	
AsHO ₃		AsHO ₃ ²⁻ , hydridotrioxidoarsenate(2-); arsonate	AsHO ₃ ²⁻ , hydridotrioxidoarsenate(2-); arsonato	
AsH ₂	AsH ₂ , arsenic dihydride AsH ₂ [•] , arsanyl, dihydridoarsenic(•) -AsH ₂ , arsanyl	AsH ₂ [•] , arsane, dihydridoarsenate(1-)	AsH ₂ [•] , arsano, dihydridoarsenate(1-)	
AsH ₂ O	-AsH ₂ O, oxo- λ^5 -arsanyl; arsinoyl	AsH ₂ O ⁻ , dihydridoarsenate(1-); arsinite	AsH ₂ O ⁻ , dihydridoarsenate(1-); arsinito	
AsH ₂ O ₂		AsH ₂ O ₂ ⁻ , dihydridodioxidoarsenate(1-); arsinate	AsH ₂ O ₂ ⁻ , dihydridodioxidoarsenate(1-); arsinato	
AsH ₂ O ₃	-As(O)(OH) ₂ , dihydroxyoxo- λ^5 - arsanyl; dihydroxyarsoryl, arsono	AsO(OH) ₂ ⁻ , dihydroxydioxidoarsenate(1-)	AsO(OH) ₂ ⁻ , dihydroxydioxidoarsenate(1-)	
AsH ₃	AsH ₃ , arsenic trihydride [AsH ₃], arsane (parent hydride name), trihydridoarsenic	AsH ₃ ^{•+} , arsaniumyl, trihydridoarsenic(•1+) -AsH ₃ ⁺ , arsaniumyl	AsH ₃ ^{•+} , arsanuidyl, trihydridoarsenate(•1-) ^e	
AsH ₄	-AsH ₄ , λ^5 -arsanyl	AsH ₄ ⁺ , arsanium, tetrahydridoarsenic(1+)		
AsH ₅	AsH ₅ , arsenic pentahydride [AsH ₅], λ^5 -arsane (parent hydride name), pentahydridoarsenic			

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
AsO	>As(O)–, oxo- λ^5 -arsanetriyl; arsoryl =As(O)–, oxo- λ^5 -arsanylidene; arsorylidene $\equiv$ As(O), oxo- λ^5 -arsanyldiyl; arsoryldiyl			
AsO ₃			AsO ₃ ^{3–} , trioxidoarsenate(3–); arsenite, arsorite –As(=O)(O [–]) ₂ , dioxidooxo- λ^5 -arsanyl; arsonato	AsO ₃ ^{3–} , trioxidoarsenato(3–); arsenito, arsorito
AsO ₄			AsO ₄ ^{3–} , tetraoxidoarsenate(3–); arsenate, arsorate	AsO ₄ ^{3–} , tetraoxidoarsenato(3–); arsenato, arsorato
AsS ₄			AsS ₄ ^{3–} , tetrasulfidoarsenate(3–)	AsS ₄ ^{3–} , tetrasulfidoarsenato(3–)
As ₂ H			HAs=As [–] , diarsenide HAsAs ^{3–} , diarsanetriide	HAs=As [–] , diarsenido HAsAs ^{3–} , diarsanetriido
As ₂ H ₂	HAs=AsH, diarsene		H ₂ AsAs ^{2–} , diarsane-1,1-diide HAsAsH ^{2–} , diarsane-1,2-diide	HAs=AsH, diarsene H ₂ AsAs ^{2–} , diarsane-1,1-diido HAsAsH ^{2–} , diarsane-1,2-diido
As ₂ H ₄	H ₂ AsAsH ₂ , diarsane			H ₂ AsAsH ₂ , diarsane
As ₄	As ₄ , tetraarsenic			As ₄ , tetraarsenic
At	astatine (general) At [•] , astatine(•), monoastatine	astatine	At [–] , astatide(1–); astatide	astatido (general) At [–] , astatido(1–); astatido
AtH, see HAt				

At ₂	At ₂ , diastatine		auride		aurido
Au	gold	gold (general) Au ⁺ , gold(1+) Au ³⁺ , gold(3+)			aurido
B	boron >B-, boranetriyl ≡B, boranylidene	boron (general) B ⁺ , boron(1+) B ³⁺ , boron(3+)	boride (general) B ⁻ , boride(1-) B ³⁻ , boride(3-); boride		borido (general) B ⁻ , borido(1-) B ³⁻ , borido(3-); borido
BH	>BH, boranediyl =BH, boranylidene	BH ²⁺ , boranebis(ylum), hydridoboron(2+)	BH ²⁻ , boranediide, hydridoborate(2-)		BH ²⁻ , boranediido, hydridoborato(2-)
BHO ₃			BO ₂ (OH) ²⁻ , hydroxidodioxidoborate(2-); hydrogenborate		BO ₂ (OH) ²⁻ , hydroxidodioxidoborato(2-); hydrogenborato
BH ₂	-BH ₂ , boranyl	BH ₂ ⁺ , boranylium, dihydridoboron(1+)	BH ₂ ⁻ , boranide, dihydridoborate(2-)		BH ₂ ⁻ , boranido, dihydridoborato(2-)
BH ₂ O	-BH(OH), hydroxyboranyl				
BH ₂ O ₂	-B(OH) ₂ , dihydroxyboranyl; borono				
BH ₃	BH ₃ , boron trihydride [BH ₃], borane (parent hydride name), trihydridoboron	BH ₃ ^{•+} , boraniumyl, trihydridoboron(•1+)	BH ₃ ^{•-} boranuidyl, trihydridoborate(•1-) ^e -BH ₃ ⁻ , boranuidyl		BH ₃ ^{•-} , trihydridoborato(•1-)
BH ₄		BH ₄ ⁺ , boranium, tetrahydridoboron(1+)	BH ₄ ⁻ , boranuide, tetrahydridoborate(1-)		BH ₄ ⁻ , boranuideo, tetrahydridoborato(1-)
BO	BO, boron mon(o)oxide [BO], oxidoboron	BO ⁺ , oxidoboron(1+)	BO ⁻ , oxidoborate(1-)		BO ⁻ , oxidoborato(1-)
BO ₂			(BO ₂ ⁻) _n = (OBO) _n ⁿ⁻ , <i>catena</i> -poly[(oxidoborate-μ- oxido)(1-)]; metaborate		
BO ₃			BO ₃ ³⁻ , trioxidoborate(3-); borate		BO ₃ ³⁻ , trioxidoborato(3-); borato
Ba	barium	barium	baride		barido
BaO	barium oxide				

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
BaO ₂	Ba ²⁺ O ₂ ²⁻ , barium dioxide(2-); barium peroxide			
Be	beryllium	beryllium (general) Be ⁺ , beryllium(1+) Be ²⁺ , beryllium(2+)	beryllide	beryllido
BeH	BeH, beryllium monohydride [BeH], hydridoberyllium	BeH ⁺ , hydridoberyllium(1+)	BeH ⁻ , hydridoberyllate(1-)	BeH ⁻ , hydridoberyllato(1-)
Bh	bohrium	bohrium	bohride	bohrido
Bi	bismuth	bismuth	bismuthide (general) Bi ³⁻ , bismuthide(3-), bismuthanetriide; bismuthide	bismuthido (general) Bi ³⁻ , bismuthido(3-), bismuthanetriido; bismuthido
BiH	>BiH, bismuthanediyl =BiH, bismuthanylidene BiH ^{2•} , bismuthanylidene, hydridobismuth(2•)	BiH ²⁺ , bismuthanebis(ylum), hydridobismuth(2+)	BiH ²⁻ , bismuthanediide, hydridobismuthate(2-)	BiH ²⁻ , bismuthanediido, hydridobismuthato(2-)
BiH ₂	-BiH ₂ , bismuthanyl BiH ₂ [•] , bismuthanyl, dihydridobismuth(•)	BiH ₂ ⁺ , bismuthanylium, dihydridobismuth(1+)	BiH ₂ ⁻ , bismuthanide, dihydridobismuthate(1-)	BiH ₂ ⁻ , bismuthanido, dihydridobismuthato(1-)
BiH ₃	BiH ₃ , bismuth trihydride [BiH ₃], bismuthane (parent hydride name), trihydridobismuth =BiH ₃ , λ ⁵ -bismuthanylidene	BiH ₃ ^{•+} , bismuthaniumyl, trihydridobismuth(•1+)	BiH ₃ ^{•-} , bismuthanuidyl, trihydridobismuthate(•1-) ^e	

BiH ₄		BiH ₄ ⁺ , bismuthanium, tetrahydridobismuth(1+)		
Bi ₅		Bi ₅ ⁴⁺ , pentabismuth(4+)		
Bk	berkelium	berkelium	berkelide	berkelido
Br	bromine (general) Br [•] , bromine(•), monobromine -Br, bromo	bromine (general) Br ⁺ , bromine(1+)	bromide (general) Br ⁻ , bromide(1-); bromide	bromido (general) Br ⁻ , bromido(1-); bromido
BrCN	BrCN, cyanobromane, bromidonitridocarbon			
BrH, see HBr				
BrHO	HOBr, bromanol, hydrodibromine ^f ; hypobromous acid			
BrHO ₂	HOBrO, hydroxy-λ ³ -bromanone, hydrodioxidobromine; bromous acid			
BrHO ₃	HOBrO ₂ , hydroxy-λ ⁵ -bromanedione, hydrodioxidobromine; bromic acid			
BrHO ₄	HOBrO ₃ , hydroxy-λ ⁷ -bromanetrione, hydrodioxidobromine; perbromic acid			
Br ₂	Br ₂ , dibromine	Br ₂ ^{•+} , dibromine(•1+)	Br ₂ ^{•-} , dibromide(•1-)	Br ₂ , dibromine
Br ₃	Br ₃ , tribromine		Br ₃ ⁻ , tribromide(1-); tribromide	Br ₃ ⁻ , tribromido(1-); tribromido
C	carbon (general) C, monocarbon >C<, methanetetrayl =C=, methanediyldiene	carbon (general) C ⁺ , carbon(1+)	carbide (general) C ⁻ , carbide(1-) C ⁴⁻ , carbide(4-), methanetetraide; carbide	carbido (general) C ⁻ , carbido(1-) C ⁴⁻ , carbido(4-), methanetetrayl, methanetetraido
CClNS			ClSCN ^{•-} , (chloridosulfato)nitrido⇌ carbonate(•1-)	

Table IX Continued

Formula for uncharged atom or group	Name			
	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
CH	CH [•] , hydridocarbon(•) CH ^{3•} , methylidyne, hydridocarbon(3•), carbyne ≡CH, methyldyne -CH=, methanylylidene -CH<, methanetriyl	CH ⁺ , λ ² -methanylium, hydridocarbon(1+)	CH ⁻ , λ ² -methanide, hydridocarbonate(1-) CH ³⁻ , methanetriide, hydridocarbonate(3-)	CH ⁻ , λ ² -methanido, hydridocarbonato(1-) CH ³⁻ , methanetriyl, methanetriido, hydridocarbonato(3-)
CHN	HCN, hydrogen cyanide HCN = [CHN], methanenitrile, hydridonitridocarbon; formonitrile >C=NH, carbonimidoyl =C=NH, iminomethylidene, carbonimidoylidene			
CHNO	HCNO = [N(CH)O], formonitrile oxide, (hydridocarbonato)oxidonitrogen HNCO = [C(NH)O], (hydridonitrato)oxidocarbon; isocyanic acid HOCN = [C(OH)N], hydroxidonitridocarbon; cyanic acid HONC = [NC(OH)], λ ² -methylidenehydroxylamine, carbidohydroxidonitrogen		HNCO ^{•-} , (hydridonitrato)oxido⊖ carbonato(•1-) HOCN ^{•-} , hydroxidonitridocarbonato(•1-)	HNCO ^{•-} , (hydridonitrato)oxido⊖ carbonato(•1-) HOCN ^{•-} , hydroxidonitridocarbonato(•1-)

CHNOS			$\text{HONCS}^{\bullet-}$, (hydroxidonitrato)sulfido $\oslash$ carbonato($\bullet 1-$) $\text{HOSCN}^{\bullet-}$, (hydroxidosulfato)nitrido $\oslash$ carbonato($\bullet 1-$)	$\text{HONCS}^{\bullet-}$, (hydroxidonitrato)sulfido $\oslash$ carbonato($\bullet 1-$) $\text{HOSCN}^{\bullet-}$, (hydroxidosulfato)nitrido $\oslash$ carbonato($\bullet 1-$)
CHNO ₂			$\text{HOOCN}^{\bullet-}$, (dioxidanido)nitridocarbonate($\bullet 1-$) $\text{HONCO}^{\bullet-}$, (hydroxidonitrato)oxido $\oslash$ carbonato($\bullet 1-$)	$\text{HOOCN}^{\bullet-}$, (dioxidanido)nitrido $\oslash$ carbonato($\bullet 1-$) $\text{HONCO}^{\bullet-}$, (hydroxidonitrato)oxido $\oslash$ carbonato($\bullet 1-$)
CHNS	$\text{HCNS} = \text{HC}\equiv\text{N}^+\text{S}^-$ $= [\text{N}(\text{CH})\text{S}]$, (methyldyneammonium) $\oslash$ sulfanide, (hydridocarbonato)sulfidonitrogen $\text{HNCS} = [\text{C}(\text{NH})\text{S}]$, (hydridonitrato)sulfidocarbon; isothiocyanic acid $\text{HSCN} = [\text{CN}(\text{SH})]$, nitridosulfanidocarbon; thiocyanic acid $\text{HSNC} = [\text{NC}(\text{SH})]$, λ^2 -methylenethiohydroxylamine, carbidosulfanidonitrogen			

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
CHNSe	HCNSe = HC≡N ⁺ Se ⁻ = [N(CH)Se], (methyldiyneammonium)⊖ selenide, (hydriodocarbonato)selenidonitrogen HNCS _e = [C(NH)Se], (hydridonitrato)selenidocarbon; isoselenocyanic acid HSeCN = [CN(SeH)], nitridoselaniidocarbon; selenocyanic acid HSeNC = [NC(SeH)], λ ² -methylideneseleno⊖ hydroxylamine, carbidoselaniidnitrogen			
CHO	HCO [•] , oxomethyl, hydriidooxidocarbon(•) -CH(O), methanoyl, formyl			
CHOS ₂	HOCS ₂ [•] , hydroxidodisulfidocarbon(•)			
CHO ₂	HOCO [•] , hydroxidooxidocarbon(•)			

CHO ₃	HOCO ₂ [•] , hydroxidodioxidocarbon(•) HOOCO [•] , (dioxidanido)oxidocarbon(•)		HCO ₃ ⁻ , hydroxidodioxidocarbonate(1-); hydrogencarbonate	HCO ₃ ⁻ , hydroxidodioxidocarbonato(1-); hydrogencarbonato
CH ₂	CH ₂ , λ ² -methane CH ₂ ^{2•} , methylenide, dihydridocarbon(2•); carbene >CH ₂ , methanediyl, methylene =CH ₂ , methylenide		CH ₂ ²⁻ , methanediide, dihydridocarbonate(2-) -CH ₂ ⁻ , methanidyl	>CH ₂ , methanediyl, methylene =CH ₂ , methylenide CH ₂ ²⁻ , methanediido, dihydridocarbonato(2-)
CH ₂ N	H ₂ CN [•] , dihydridonitridocarbon(•)			
CH ₂ NO	H ₂ NCO [•] , (dihydridonitrato)oxidocarbon(•) HNCOH [•] , (hydridonitrato)hydroxido carbon(•)			
CH ₃	CH ₃ [•] , methyl -CH ₃ <i>or</i> -Me, methyl	CH ₃ ⁺ , methylum, trihydridocarbon(1+)	CH ₃ ⁻ , methanide, trihydridocarbonate(1-)	CH ₃ ⁻ , methyl, methanido, trihydridocarbonato(1-)
CH ₄	CH ₄ , methane (parent hydride name), tetrahydridocarbon	CH ₄ ^{•+} , methaniumyl, tetrahydridocarbon(•1+)	CH ₄ ^{•-} , methanuidyl, tetrahydridocarbonate(•1-) ^e	
CH ₅		CH ₅ ⁺ , methanium, pentahydridocarbon(1+)		
CN	CN [•] , nitridocarbon(•); cyanyl -CN, cyano -NC, isocyano	CN ⁺ , azanylidynemethylum, nitridocarbon(1+)	CN ⁻ , nitridocarbonate(1-); cyanide	nitridocarbonato (general) CN ⁻ , nitridocarbonato(1-); cyanido = [nitridocarbonato(1-)-κC]
CN ₂			NCN ²⁻ , dinitridocarbonate(2-)	NCN ²⁻ , dinitridocarbonato(2-)
CNO	OCN [•] , nitridooxidocarbon(•) -OCN, cyanato -NCO, isocyanato -ONC, λ ² -methylenideneazanylideneoxy -CNO, (oxo-λ ² -azanylidyne)methyl		OCN ⁻ , nitridooxidocarbonate(1-); cyanate ONC ⁻ , carbidooxidonitrato(1-); fulminate OCN ^{•2-} , nitridooxidocarbonate(•2-)	OCN ⁻ , nitridooxidocarbonato(1-); cyanato ONC ⁻ , carbidooxidonitrato(1-); fulminato

Table IX Continued

Formula for uncharged atom or group	Name			
	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
CNS	SCN [•] , nitridosulfidocarbon(•) –SCN, thiocyanato –NCS, isothiocyanato –SNC, λ^2 -methylidene- $\text{C}\equiv$ azanylylidenesulfanediyyl –CNS, (sulfanylidene- λ^5 - azanylidene)methyl		SCN [–] , nitridosulfidocarbonato(1–); thiocyanato SNC [–] , carbidosulfidonitrato(1–)	SCN [–] , nitridosulfidocarbonato(1–); thiocyanato SNC [–] , carbidosulfidonitrato(1–)
CNSe	SeCN [•] , nitridoselenidocarbon(•) –SeCN, selenocyanato –NCSe, isoselenocyanato –SeNC, λ^2 -methylidene- $\text{C}\equiv$ azanylylideneselenanediyyl –CNSe, (selanylidene- λ^5 - azanylidene)methyl		SeCN [–] , nitridoselenidocarbonato(1–); selenocyanato SeNC [–] , carbidoselenidonitrato(1–)	SeCN [–] , nitridoselenidocarbonato(1–); selenocyanato SeNC [–] , carbidoselenidonitrato(1–)
CO	CO, carbon mon(o)oxide >C=O, carbonyl =C=O, carbonylidene	CO ^{•+} , oxidocarbon(•1+) CO ²⁺ , oxidocarbon(2+)	CO ^{•–} , oxidocarbonato(•1–)	CO, oxidocarbon, oxidocarbonato (general); carbonyl = oxidocarbon- κC (general) CO ^{•+} , oxidocarbon(•1+) CO ^{•–} , oxidocarbonato(•1–)
COS	C(O)S, carbonyl sulfide, oxidosulfidocarbon			
CO ₂	CO ₂ , carbon dioxide, dioxidocarbon		CO ₂ ^{•–} , oxidooxomethyl, dioxidocarbonato(•1–)	CO ₂ , dioxidocarbon CO ₂ ^{•–} , oxidooxomethyl, dioxidocarbonato(•1–)

CO ₃				CO ₃ ^{••-} , trioxidocarbonate(•1-), OCOO ^{•-} , (dioxido)oxidocarbonate(•1-), oxidoperoxidocarbonate(•1-) CO ₃ ²⁻ , trioxidocarbonate(2-); carbonate	CO ₃ ²⁻ , trioxidocarbonato(2-); carbonato
CS	carbon monosulfide >C=S, carbonothioyl; thiocarbonyl =C=S, carbonothioylidene	CS ^{•+} , sulfidocarbon(•1+)		CS ^{••-} , sulfidocarbonate(•1-)	CS, sulfidocarbon, sulfidocarbonato, thiocarbonyl (general); CS ^{•+} , sulfidocarbon(•1+) CS ^{•-} , sulfidocarbonato(•1-)
CS ₂	CS ₂ , disulfidocarbon, carbon disulfide			CS ₂ ^{••-} , sulfidothioxomethyl, disulfidocarbonate(•1-)	CS ₂ , disulfidocarbon CS ₂ ^{•-} , sulfidothioxomethyl, disulfidocarbonato(•1-)
CS ₃				CS ₃ ²⁻ , trisulfidocarbonate(2-)	CS ₃ ²⁻ , trisulfidocarbonato(2-)
C ₂	C ₂ , dicarbon	C ₂ ⁺ , dicarbon(1+)		C ₂ ⁻ , dicarbide(1-) C ₂ ²⁻ , dicarbide(2-), ethynediide, acetylenediide; acetylde	dicarbido (general) C ₂ ²⁻ , dicarbido(2-), ethynediido, ethyne-1,2-diyl
C ₂ H	HCC [•] , ethynyl, hydridodicarbon(•)				
C ₂ N ₂	NCCN, ethanedinitrile, bis(nitridocarbon)(C-C); oxalonitrile			NCCN ^{•-} , bis(nitridocarbonato)(C-C)(•1-)	
C ₂ N ₂ O ₂	NCOOCN, dioxidanedicarbonitrile, bis[cyanidooxygen](O-O)			NCOOCN ^{•-} , bis[cyanidooxygenato](O-O)(•1-) ^e OCNNCO ^{•-} , bis(carbonylnitrate)(N-N)(•1-) ^e	NCOOCN ^{•-} , bis[cyanidooxygenato](O-O)(•1-) OCNNCO ^{•-} , bis(carbonylnitrate)(N-N)(•1-)
C ₂ N ₂ S ₂	NCSSCN, disulfanedicarbonitrile, bis[cyanidosulfur](S-S)			NCSSCN ^{•-} , bis[cyanidosulfate](S-S)(•1-) ^e	NCSSCN ^{•-} , bis[cyanidosulfato](S-S)(•1-)
C ₃ O ₂	C ₃ O ₂ , tricarbon dioxide O=C=C=C=O, propa-1,2-diene-1,3-dione				
C ₁₂ O ₉	C ₁₂ O ₉ , dodecacarbon nonaoxide				

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
Ca	calcium	calcium (general) Ca ²⁺ , calcium(2+)	calcide	calcido
Cd	cadmium	cadmium (general) Cd ²⁺ , cadmium(2+)	cadmide	cadmido
Ce	cerium	cerium (general) Ce ³⁺ , cerium(3+) Ce ⁴⁺ , cerium(4+)	ceride	cerido
Cf	californium	californium	californide	californido
Cl	chlorine (general) Cl [•] , chlorine(•), monochlorine -Cl, chloro	chlorine (general) Cl ⁺ , chlorine(1+)	chloride (general) Cl ⁻ , chloride(1-); chloride	chlorido (general) Cl ⁻ , chlorido(1-); chlorido
ClF	ClF, fluoridochlorine, chlorine monofluoride	ClF ⁺ , fluoridochlorine(1+)		
ClF ₂			ClF ₂ ⁻ , difluoridochlorate(1-)	ClF ₂ ⁻ , difluoridochlorato(1-)
ClF ₄		ClF ₄ ⁺ , tetrafluoridochlorine(1+)	ClF ₄ ⁻ , tetrafluoridochlorate(1-)	ClF ₄ ⁻ , tetrafluoridochlorato(1-)
ClH, see HCl				
ClHN			NHCl ⁻ , chloroazanide, chloridohydridonitrate(1-)	NHCl ⁻ , chloroazanido, chloridohydridonitrate(1-)
ClHO	HOCl, chloranol, hydroxidochlorine ^f ; hypochlorous acid		HOCl ^{•-} , hydroxidochlorate(•1-)	
ClHO ₂	HOClO, hydroxy-λ ³ -chloranone, hydroxidooxidochlorine; chlorous acid			

ClHO ₃	HOClO ₂ , hydroxy-λ ⁵ -chloranediene, hydroxidodioxichlorine; chloric acid				
ClHO ₄	HOClO ₃ , hydroxy-λ ⁷ -chloranetrione, hydroxidotrioxichlorine; perchloric acid				
Cl ₂	Cl ₂ , dichlorine	Cl ₂ ^{•+} , dichlorine(•1+)	Cl ₂ ^{•-} , dichloride(•1-)	Cl ₂ , dichlorine Cl ₂ ^{•-} , dichlorido(•1-)	
Cl ₂ OP	-PCl ₂ (O), dichlorooxo-λ ⁵ -phosphanyl, phosphorodichloridoyl				
Cl ₄		Cl ₄ ⁺ , tetrachlorine(1+)			
Cm	curium	curium	curide	curido	
Co	cobalt	cobalt (general) Co ²⁺ , cobalt(2+) Co ³⁺ , cobalt(3+)	cobaltide	cobaltido	
Cr	chromium	chromium (general) Cr ²⁺ , chromium(2+) Cr ³⁺ , chromium(3+)	chromide	chromido	
CrO	CrO, chromium mon(o)oxide, chromium(II) oxide				
CrO ₂	CrO ₂ , chromium dioxide, chromium(IV) oxide				
CrO ₃	CrO ₃ , chromium trioxide, chromium(VI) oxide				
CrO ₄	[Cr(O ₂) ₂], dioxodichromium		CrO ₄ ²⁻ , tetraoxodichromate(2-); chromate CrO ₄ ³⁻ , tetraoxodichromate(3-) CrO ₄ ⁴⁻ , tetraoxodichromate(4-)	CrO ₄ ²⁻ , tetraoxodichromato(2-); chromato CrO ₄ ³⁻ , tetraoxodichromato(3-) CrO ₄ ⁴⁻ , tetraoxodichromato(4-)	
CrO ₅	[CrO(O ₂) ₂], oxidodioxodichromium				

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
CrO ₆			CrO ₂ (O ₂) ₂ ²⁻ , dioxidodiperoxidochromate(2-)	
CrO ₈			Cr(O ₂) ₄ ²⁻ , tetraperoxidochromate(2-) Cr(O ₂) ₄ ³⁻ , tetraperoxidochromate(3-)	
Cr ₂ O ₃	Cr ₂ O ₃ , dichromium trioxide, chromium(III) oxide			
Cr ₂ O ₇			Cr ₂ O ₇ ²⁻ , heptaoxidodichromate(2-) O ₃ CrOCrO ₃ ²⁻ , μ -oxido-bis(trioxidochromate)(2-); dichromate	Cr ₂ O ₇ ²⁻ , heptaoxidodichromato(2-) O ₃ CrOCrO ₃ ²⁻ , μ -oxido-bis(trioxidochromato)(2-); dichromato
Cs	caesium	caesium	caeside	caesido
Cu	copper	copper (general) Cu ⁺ , copper(1+) Cu ²⁺ , copper(2+)	cupride	cuprido
D, see H				
D ₂ , see H ₂				
D ₂ O, see H ₂ O				
Db	dubnium	dubnium	dubnide	dubnido
Ds	darmstadtium	darmstadtium	darmstadtide	darmstadtido
Dy	dysprosium	dysprosium	dysproside	dysprosido
Er	erbium	erbium	erbide	erbido

Es	einsteinium	einsteinium	einsteinide	einsteinido	
Eu	euprium	euprium	eupride	euprido	
F	fluorine F [•] ; fluorine(•), monofluorine -F, fluoro	fluorine (general) F ⁺ , fluorine(1+)	fluoride (general) F ⁻ , fluoride(1-); fluoride	F ⁻ , fluorido(1-); fluorido	
FH, see HF					
FHO	HOF, fluoranol, fluoridohydridoxygen				
FNS	NSF, fluoridonitridosulfur				
FN ₃	FNNN, fluorido-1κF-trinitrogen(2 N-N)				
FO, see OF					
F ₂	F ₂ , difluorine	F ₂ ⁺ , difluorine(•1+)	F ₂ ⁻ , difluoride(•1-)	F ₂ , difluorine	
F ₂ N ₂	FN=NF, difluorido-1κF,2κF- dinitrogen(N-N), difluorodiazene				
Fe	iron	iron (general) Fe ²⁺ , iron(2+) Fe ³⁺ , iron(3+)	ferride	ferrido	
Fm	fermium	fermium	fermide	fermido	
Fr	francium	francium	francide	francido	
Ga	gallium	gallium	gallide	gallido	
GaH ₂	-GaH ₂ , gallanyl				
GaH ₃	GaH ₃ , gallium trihydride [GaH ₃], gallane (parent hydride name), trihydridogallium				
Gd	gadolinium	gadolinium	gadolinide	gadolinido	
Ge	germanium >Ge<, germanetetrayl =Ge=, germanediylidene >GeH-, germanetriyl =GeH-, germanylylidene ≡GeH, germlyidyne	germanium (general) Ge ²⁺ , germanium(2+) Ge ⁴⁺ , germanium(4+)	germide (general) Ge ⁴⁻ , germide(4-); germide	germido (general) Ge ⁴⁻ , germido(4-); germido	
GeH					

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
GeH ₂	>GeH ₂ , germandiyl =GeH ₂ , germylidene			
GeH ₃	–GeH ₃ , germyl	GeH ₃ ⁺ , germylum, trihydridogermanium(1+)	GeH ₃ [–] , germanide, trihydridogermanate(1–)	GeH ₃ [–] , germanido, trihydridogermanato(1–)
GeH ₄	GeH ₄ , germane (parent hydride name), tetrahydridogermanium			
Ge ₄			Ge ₄ ^{4–} , tetragermide(4–)	
H	hydrogen H [•] , hydrogen(•), monohydrogen (natural or unspecified isotopic composition) ¹ H [•] , protium(•), monoprotrium ² H [•] = D [•] , deuterium(•), monodeuterium ³ H [•] = T [•] , tritium(•), monotritium	hydrogen (general) H ⁺ , hydrogen(1+), hydron (natural or unspecified isotopic composition) ¹ H ⁺ , protium(1+), proton ² H ⁺ = D ⁺ , deuterium(1+), deuteron ³ H ⁺ = T ⁺ , tritium(1+), triton	hydride (general) H [–] , hydride (natural or unspecified isotopic composition) ¹ H [–] , protide ² H [–] = D [–] , deuteride ³ H [–] = T [–] , tritide	hydrido protido deuterido tritido
HAt	HAt, hydrogen astatide [HAt], astatidohydrogen			
HBr	HBr, hydrogen bromide [HBr], bromane (parent hydride name), bromidohydrogen			
HCO, see CHO				

HCl	HCl, hydrogen chloride [HCl], chlorane (parent hydride name), chloridohydrogen	HCl ⁺ , chloraniumyl, chloridohydrogen(•1+)	
HF	HF, hydrogen fluoride [HF], fluorane (parent hydride name), fluoridohydrogen	HF ⁺ , fluoraniumyl, fluoridohydrogen(•1+)	
HF ₂			FHF ⁻ , fluoro fluoranuide, μ -hydridodifluorate(1-), difluoridohydrogenate(1-)
HI	HI, hydrogen iodide [HI], iodane (parent hydride name), iodidohydrogen		
HIO	HOI, iodanol, hydroxidoiodine ¹ ; hypiodous acid		
HIO ₂	HOIO, hydroxy- λ^3 -iodanone, hydroxidoiodine; iodous acid		
HIO ₃	HOIO ₂ , hydroxy- λ^5 -iodanedione, hydroxidoiodine; iodic acid		HOIO ₂ ^{•-} , hydroxidoiodine(•1-)
HIO ₄	HOIO ₃ , hydroxy- λ^7 -iodanetrione, hydroxidoiodine; periodic acid		
H _n N _m , see N _m H _n			
HMnO ₄	HMnO ₄ = [MnO ₃ (OH)], hydroxidotrioxomanganese		HMnO ₄ ⁻ = [MnO ₃ (OH)] ⁻ , hydroxidotrioxomanganate(1-)

Table IX Continued

Formula for uncharged atom or group	Name			
	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
HNO	HNO = [NH(O)], azanone, hydridoxygenitrogen HON ^{2•} , hydroxidonitrogen(2•) >NH(O), oxo-λ ⁵ -azanediyl; azonoyl =NH(O), oxo-λ ⁵ -azanylidene; azonoylidene >N-OH, hydroxyazanediyl =N-OH, hydroxyazanylidene; hydroxyimino	HNO ^{•+} = [NH(O)] ^{•+} , hydridoxygenitrogen(•1+)	HON ²⁻ , hydroxidonitrate(2-)	HON ²⁻ , hydroxidonitrate(2-)
HNO ₂	HNO ₂ = [NO(OH)], hydrodioxidonitrogen; nitrous acid >N(O)(OH), hydroxyoxo-λ ⁵ -azanediyl; hydroxyazoryl =N(O)(OH), hydroxyoxo-λ ⁵ -azanylidene; hydroxyazorylidene			
HNO ₃	HNO ₃ = [NO ₂ (OH)], hydrodioxididonitrogen; nitric acid HNO(O ₂) = [NO(OOH)], dioxidanidooxygenitrogen, peroxynitrous acid			

HNO ₄	HNO ₄ = [NO ₂ (OOH)], (dioxidanido)dioxidonitrogen; peroxynitric acid			
HNS	>S(=NH), imino-λ ⁴ -sulfanediyl; sulfiminidoyl			
HN ₂ O ₂	–NHNO ₂ , nitrozazanyl, nitroamino		[HON=NO] [–] , 2-hydroxydiazen-1-olate, hydroxido-1κO-oxido-2κO- dinitrate(N–N)(1–)	
HN ₂ O ₃			HN ₂ O ₃ [–] = N(H)(O)NO ₂ [–] , hydrido-1κH-trioxido- 1κO,2κ ² O-dinitrate(N–N)(1–)	
HN ₃ O			HON ₃ ^{•–} , hydroxido-1κO- trinitrate(2 N–N)(•1–)	
HO	HO [•] , oxidanyl, hydridoxygen(•); hydroxyl –OH, oxidanyl; hydroxy	HO ⁺ , oxidanylium, hydridoxygen(1+); hydroxylum	HO [–] , oxidanide, hydridoxygenate(1–); hydroxide	HO [–] , oxidanido; hydroxido
HOP	HPO = [P(H)O], phosphanone, hydridoxygenphosphorus >PH(O), oxo-λ ⁵ -phosphanediyl; phosphonoyl = PH(O), oxo-λ ⁵ -phosphanylidene; phosphonoylidene = P–OH, hydroxyphosphanylidene			
HOS	–SH(O), oxo-λ ⁴ -sulfanyl –SOH, hydroxysulfanyl –OSH, sulfanyloxy		HSO [–] , sulfanolate, hydridoxygenate(1–)	HSO [–] , sulfanolato, hydridoxygenate(1–)
HOSe	–SeH(O), oxo-λ ⁴ -selanyl –SeOH, hydroxyselanyl –OSeH, selanyloxy			

Table IX Continued

Formula for uncharged atom or group	Name			
	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
HO ₂	HO ₂ •, dioxidanyl, hydridodioxygen(•) –OOH, dioxidanyl; hydroperoxy	HO ₂ ⁺ , dioxidanylium, hydridodioxygen(1+)	HO ₂ [–] , dioxidanide, hydrogen(peroxide)(1–)	HO ₂ [–] , dioxidanido, hydrogen(peroxido)(1–)
HO ₂ P	P(O)(OH), hydroxyphosphanone, hydroxidooxidophosphorus >P(O)(OH), hydroxyoxo-λ ⁵ -phosphanedyl; hydroxyphosphoryl =P(O)(OH), hydroxyoxo-λ ⁵ -phosphanylidene; hydroxyphosphorylidene		HOPO ^{•–} , hydroxidooxidophosphate(•1–) HPO ₂ ^{2–} , hydridodioxidophosphate(2–)	HOPO ^{•–} , hydroxidooxidophosphato(•1–) HPO ₂ ^{2–} , hydridodioxidophosphato(2–)
HO ₂ S	HOOS•, hydrido-1κH-sulido-2κS-dioxygen(•) HOSO•, hydroxidooxidofur(•) HSOO•, (hydridosulfato)dioxygen(•) –S(O)(OH), hydroxyoxo-λ ⁴ -sulfanyl; hydroxysulfinyl, sulfinio –S(O) ₂ H, dioxo-λ ⁶ -sulfanyl		HOSO [–] , hydroxysulfanolate, hydroxidooxidosulfate(1–)	HOSO [–] , hydroxysulfanolato, hydroxidooxidosulfato(1–)

HO ₂ Se	–Se(O)(OH), hydroxyoxo-λ ⁴ -selenyl; hydroxyseleninyl, selenino –Se(O) ₂ H, dioxo-λ ⁶ -selenyl			
HO ₃	HO ₃ [•] , hydriotrioxigen(•) HO ₃ [•] , trioxidanyl, hydrido- 1κH-trioxygen(2 O–O)(•) –OOOH, trioxidanyl			
HO ₃ P	P(O) ₂ (OH), hydroxy-λ ⁵ -phosphanedione, hydroxidodioxidophosphorus	HOPO ₂ ^{•–} , hydroxidodioxidophosphate(•1–) PHO ₃ ^{2–} , hydriotrioxidophosphate(2–); phosphonate HPO ₃ ^{2–} = PO ₂ (OH) ^{2–} , hydroxidodioxidophosphate(2–); hydrogenphosphite	HOPO ₂ ^{•–} , hydroxidodioxidophosphato(•1–) PHO ₃ ^{2–} , hydriotrioxidophosphato(2–); phosphonate HPO ₃ ^{2–} = PO ₂ (OH) ^{2–} , hydroxidodioxidophosphato(2–); hydrogenphosphito	
HO ₃ S	–S(O) ₂ (OH), hydroxydioxo-λ ⁶ -sulfanyl, hydroxysulfonyl; sulfo	HSO ₃ [–] , hydroxidodioxidosulfate(1–), hydrogensulfite	HSO ₃ [–] , hydroxidodioxidosulfato(1–), hydrogensulfito	
HO ₃ Se	HOSeO ₂ [•] , hydroxidodioxidoselenium(•) –Se(O) ₂ (OH), hydroxydioxo-λ ⁶ -selenyl, hydroxyselenonyl; selenono	HSO ₃ [–] , hydroxidodioxidoselenate(1–)	HSO ₃ [–] , hydroxidodioxidoselenato(1–)	
HO ₄ P		HOPO ₃ ^{•–} = PO ₃ (OH) ^{•–} , hydroxidotrioxidophosphate(•1–) HPO ₄ ^{2–} = PO ₃ (OH) ^{2–} , hydroxidotrioxidophosphate(2–); hydrogenphosphate	HOPO ₃ ^{•–} = PO ₃ (OH) ^{•–} , hydroxidotrioxidophosphato(•1–) HPO ₄ ^{2–} , hydroxidotrioxidophosphato(2–); hydrogenphosphato	
HO ₄ S	HOSeO ₃ [•] , hydro- xidotrioxidosulfur(•) –OS(O) ₂ (OH), hydroxysulfonyloxy; sulfooxy	HSO ₄ [–] , hydroxidotrioxidosulfate(1–); hydrogensulfate	HSO ₄ [–] , hydroxidotrioxidosulfato(1–); hydrogensulfato	

Table IX Continued

Formula for uncharged atom or group		Name		
	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
HO ₄ Se			HSeO ₄ [−] , hydroxidotrioxoselenate(1−)	HSeO ₄ [−] , hydroxidotrioxoselenato(1−)
HO ₅ P			HOPO ₄ ^{•−} = PO ₂ (OH)(OO) ^{•−} , (dioxido)hydroxidodioxido phosphate(•1−)	PO ₂ (OH)(OO) ^{•−} , (dioxido)hydroxidodioxido phosphato(•1−)
HO ₅ S	HOSO ₄ [•] = [SO ₂ (OH)(OO)] [•] , (dioxido)hydroxidodioxidosulfur(•)			
HS	−SH, sulfanyl	HS ⁺ , sulfanylium, hydridosulfur(1+)	HS [−] , sulfanide, hydrogen(sulfide)(1−)	HS [−] , sulfanido, hydrogen(sulfido)(1−)
HS ₂	HS [•] , sulfanyl, hydridosulfur(•)		HSS [−] , disulfanide	HSS [−] , disulfanido
HS ₃	−SSH, disulfanyl		HSSS [−] , trisulfanide	HSSS [−] , trisulfanido
HS ₄	−SSSH, trisulfanyl		HSSSS [−] , tetrasulfanide	HSSSS [−] , tetrasulfanido
HS ₅	−SSSSH, pentasulfanyl		HSSSSS [−] , pentasulfanide	HSSSSS [−] , pentasulfanido
HSe	HSe [•] , selanyl, hydridoselenium(•)	HSe ⁺ , selanylium, hydridoselenium(1+)	HSe [−] , selanide, hydrogen(selenide)(1−)	HSe [−] , selanido, hydrogen(selenido)(1−)
−SeH, selanyl			HSeSe [−] , diselanide	HSeSe [−] , diselanido
−SeSeH, diselanyl			HTe [−] , tellanide, hydrogen(tellanide)(1−)	
HTe	HTe [•] , tellanyl, hydridotellurium(•)	HTe ⁺ , tellanylium, hydridotellurium(1+)		
−TeH, tellanyl				
HTe ₂	−TeTeH, ditellanyl		HTeTe [−] , ditellanide	HTeTe [−] , ditellanido
H ₂	H ₂ , dihydrogen	H ₂ ^{•+} , dihydrogen(•1+)		
D ₂ , dideuterium		¹ H ₂ ^{•+} , diprotium(•1+)		
T ₂ , ditritium		D ₂ ^{•+} , dideuterium(•1+)		
		T ₂ ^{•+} , ditritium(•1+)		

H_2Br	$H_2Br^\bullet$, λ^3 -bromanyl, dihydrdobromine($\bullet$)	H_2Br^+ , bromanium, dihydrdobromine($1+$)		
H_2Cl	$H_2Cl^\bullet$, λ^3 -chloranyl, dihydrdochlorine($\bullet$)	H_2Cl^+ , chloranium, dihydrdochlorine($1+$)		
H_2F	$H_2F^\bullet$, λ^3 -fluoranyl, dihydrdofluorine($\bullet$)	H_2F^+ , fluoranium, dihydrdofluorine($1+$)		
H_2I	$H_2I^\bullet$, λ^3 -iodanyl, dihydrdoiodine($\bullet$)	H_2I^+ , iodanium, dihydrdoiodine($1+$)		
H_2IO_2	$-I(OH)_2$, dihydroxy- λ^3 -iodanyl			
H_2MnO_4	$H_2MnO_4 = [MnO_2(OH)_2]$, dihydroxidodioxidomanganese			
H_2N_m , see N_mH_2				
H_2NO	$H_2NO^\bullet$, aminooxidanyl, dihydrdooxidonitrogen($\bullet$); aminoxyl $HONH^\bullet$, hydroxyazanyl, hydridohydroxididonitrogen($\bullet$) $-NH(OH)$, hydroxyazanyl, hydroxyamino $-ONH_2$, aminooxy $-NH_2(O)$, oxo- λ^5 -azanyl; azinoyl	$HONH^-$, hydroxyazanide, hydridohydroxididonitrate($1-$) H_2NO^- , azanolate, aminooxidamide, dihydrdooxidonitrate($1-$)	$NHOH^-$, hydroxyazanido, hydridohydroxididonitrate($1-$) H_2NO^- , azanolate, aminooxidamide, dihydrdooxidonitrate($1-$)	
H_2NOS	$-S(O)NH_2$, azanyloxo- λ^4 -sulfanyl; aminosulfanyl			
H_2NO_2S	$-S(O)_2NH_2$, azanyldioxo- λ^6 -sulfanyl; aminosulfonyl; sulfamoyl			
H_2NO_3		$[NO(OH)_2]^+$, dihydroxidooxididonitrogen($1+$)		
H_2NS	$-SNH_2$, azanyl/sulfanyl; aminosulfanyl $-NH_2(S)$, sulfanylidene- λ^5 -azanyl; azinothioyl			

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
H_2N_m , see N_mH_2				
H_2O	H_2O , dihydrogen oxide; water $\text{H}_2\text{O} = [\text{OH}_2]$, oxidane (parent hydride name), dihydrodooxygen $^1\text{H}_2\text{O}$, diprotium oxide; $(^1\text{H}_2)\text{water}$ $\text{D}_2\text{O} = ^2\text{H}_2\text{O}$, dideuterium oxide; $(^2\text{H}_2)\text{water}$ $\text{T}_2\text{O} = ^3\text{H}_2\text{O}$, ditritium oxide; $(^3\text{H}_2)\text{water}$			H_2O , aqua
H_2OP	$-\text{PH}_2\text{O}$, oxo- λ^5 -phosphanyl; phosphinoyl		PH_2O^- dihydrodooxidophosphate(1-); phosphinite	PH_2O^- , dihydrodooxidophosphato(1-); phosphinito
H_2OSb	$-\text{SbH}_2\text{O}$, oxo- λ^5 -stibanyl, stibinoyl			
H_2O_2	H_2O_2 , dihydrogen peroxide; hydrogen peroxide HOOH , dioxidane (parent hydride name), bis(hydrodooxygen)($O-O$)	$\text{HOOH}^{\bullet+}$, dioxidaniumyl, bis(hydrodooxygen)($O-O$)($\bullet 1^+$)		HOOH , dioxidane
$\text{H}_2\text{O}_2\text{P}$	$-\text{P}(\text{OH})_2$, dihydroxyphosphanyl $-\text{PH}(\text{O})(\text{OH})$, hydroxyoxo- λ^5 -phosphanyl		PH_2O_2^- , dihydrodooxidophosphate(1-); phosphinate	PH_2O_2^- , dihydrodooxidophosphato(1-); phosphinato

H_2O_3B				$H_2BO_3^- = [BO(OH)_2]^-$, dihydroxidooxidoborate(1-); dihydrogenborate	$H_2BO_3^- = [BO(OH)_2]^-$, dihydroxidooxidoborate(1-); dihydrogenborato
H_2O_3P	$-P(O)(OH)_2$, dihydroxyoxo- λ^5 -phosphanyl; dihydroxyphosphoryl, phosphono			$[PHO_2(OH)]^-$, hydrihydroxidodioxido- phosphate(1-); hydrogenphosphonate $[PO(OH)_2]^-$, dihydroxidooxidophosphate(1-); dihydrogenphosphite	$[PHO_2(OH)]^-$, hydrihydroxidodioxido- phosphato(1-); hydrogenphosphonato $[PO(OH)_2]^-$, dihydroxidooxidophosphato(1-); dihydrogenphosphito
H_2O_4P	$(HO)_2PO_2^\bullet$, (dihydroxido)dioxidophosphorus($\bullet$)			$H_2PO_4^-$, dihydroxidodioxidophosphate(1-); dihydrogenphosphate	$H_2PO_4^-$, dihydroxidodioxidophosphato(1-); dihydrogenphosphato
$H_2O_5P_2$				$P_2H_2O_5^{2-} = [PH(O)_2OPH(O)_2]^{2-}$, μ -oxido- bis(hydriodioxidophosphate)(2-); diphosphonate	$P_2H_2O_5^{2-} = [PH(O)_2OPH(O)_2]^{2-}$, μ -oxido- bis(hydriodioxidophosphato)(2-); diphosphonato
H_2PS	$-PH_2(S)$, sulfanylidene- λ^5 -phosphanyl; phosphinothioyl				
H_2Po	H_2Po , dihydrogen polonide $H_2Po = [PoH_2]$, polane (parent hydride name), dihydriodopolonium				
H_2S	H_2S , dihydrogen sulfide; hydrogen sulfide $H_2S = [SH_2]$, sulfane (parent hydride name), dihydridosulfur	$H_2S^{\bullet+}$, sulfaniumyl, dihydridosulfur($\bullet 1+$) $-SH_2^+$, sulfaniumyl		$H_2S^{\bullet-}$, sulfaniumyl, dihydridosulfate($\bullet 1-$) ^e	H_2S , sulfane
H_2S_2	H_2S_2 , dihydrogen disulfide HSSH, disulfane (parent hydride name), bis(hydridosulfur)($S-S$)	$HSSH^{\bullet+}$, disulfaniumyl, bis(hydridosulfur)($S-S$)($\bullet 1+$)		$HSSH^{\bullet-}$, disulfaniumyl, bis(hydridosulfate)($S-S$)($\bullet 1-$) ^e	HSSH, disulfane
H_2S_3	H_2S_3 , dihydrogen trisulfide HSSSH, trisulfane (parent hydride name)				HSSSH, trisulfane

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
H ₂ S ₄	H ₂ S ₄ , dihydrogen tetrasulfide HSSSSH, tetrasulfane (parent hydride name)			HSSSSH, tetrasulfane
H ₂ S ₅	H ₂ S ₅ , dihydrogen pentasulfide HSSSSSH, pentasulfane (parent hydride name)			HSSSSSH, pentasulfane
H ₂ Se	H ₂ Se, dihydrogen selenide; hydrogen selenide H ₂ Se = [SeH ₂], selane (parent hydride name), dihydridoselenium	H ₂ Se ^{•+} , seleniumyl, dihydridoselenium(•1+) -SeH ₂ ⁺ , seleniumyl	H ₂ Se ^{•-} , selenuidyl, dihydridoselenate(•1-) ^e	H ₂ Se, selane
H ₂ Se ₂	H ₂ Se ₂ , dihydrogen diselenide HSeSeH, diselane (parent hydride name), bis(hydridoselenium)(Se-Se)	HSeSeH ^{•+} , diselaniumyl, bis(hydridoselenium)(Se-Se)(•1+)	HSeSeH ^{•-} , diselanuidyl, bis(hydridoselenate)(Se-Se)(•1-) ^e	HSeSeH, diselane
H ₂ Te	H ₂ Te, dihydrogen tellanide; hydrogen tellanide H ₂ Te = [TeH ₂], tellane (parent hydride name), dihydridotellurium	H ₂ Te ^{•+} , tellaniumyl, dihydridotellurium(•1+) -TeH ₂ ⁺ , tellaniumyl	H ₂ Te ^{•-} , tellanuidyl, dihydridotellurate(•1-) ^e	H ₂ Te, tellane
H ₃		H ₃ ⁺ , trihydrogen(1+)		
H ₃ N _m , see N _m H ₃				

H_3NO	HONH_2 , azanol, dihydrodihydroxidodinitrogen; hydroxylamine (parent name for organic derivatives)	$\text{HONH}_2^{\bullet+}$, hydroxyazanumyl, dihydrodihydroxidodinitrogen($\bullet 1+$)		HONH_2 , azanol, dihydrodihydroxidodinitrogen; hydroxylamine
H_3NP	$-\text{PH}_2(=\text{NH})$, imino- λ^5 -phosphanyl; phosphinimidoyl			
H_3O		H_3O^+ , oxidanium, trihydrodooxygen($1+$), aquahydrogen($1+$); oxonium (<i>not</i> hydronium)		
H_3OS	$\text{H}_3\text{OS}^+ = [\text{SH}_3(\text{O})]^+$, oxo- λ^5 -sulfanylium, trihydrodooxidosulfur($1+$)			
H_3OSi	$-\text{OSiH}_3$, silyloxy			
$\text{H}_3\text{O}_4\text{S}$		$[\text{SO}(\text{OH})_3]^+ = \text{H}_3\text{SO}_4^+$, trihydrodooxidosulfur($1+$), trihydrogen(tetraoxidosulfate)($1+$)		
$\text{H}_3\text{O}_5\text{P}$	$[\text{PO}(\text{OH})_2(\text{OOH})]$, (dioxidanido)dihydroxido $\ominus$ oxidophosphorus; peroxyphosphoric acid, phosphoroperoxoic acid			
H_3S	$\text{H}_3\text{S}^\bullet$, λ^4 -sulfanyl, trihydrodisulfur($\bullet$)	H_3S^+ , sulfanium, trihydrodisulfur($1+$)	H_3S^- , sulfanuide, trihydrodisulfate($1-$)	
H_3Se	$\text{H}_3\text{Se}^\bullet$, λ^4 -selanyl, trihydrodiselenium($\bullet$)	H_3Se^+ , selanium, trihydrodiselenium($1+$)	H_3Se^- , selanuide, trihydrodiselenate($1-$)	
H_3Te	$\text{H}_3\text{Te}^\bullet$, λ^4 -tellanyl, trihydrodote tellurium($\bullet$)	H_3Te^+ , tellanium, trihydrodote tellurium($1+$)	H_3Te^- , tellanuide, trihydrodote tellurate($1-$)	
H_4N_m , see N_mH_4				
H_4NO		NH_2OH_2^+ , aminooxidanium, aquadihydroxidodinitrogen($1+$) NH_3OH^+ , hydroxyazanum, trihydrodihydroxidodinitrogen($1+$); hydroxyammonium		

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
H ₄ O		H ₄ O ²⁺ , oxidanadium, tetrahydridoxygen(2+)		
H ₅ IO ₆	IO(OH) ₅ , pentahydroxy-λ ⁷ -iodanone, pentahydroxidoiodine; orthoperiodic acid			
H ₅ N ₂ , see N ₂ H ₅				
H ₅ O ₂		[H(H ₂ O) ₂] ⁺ , μ-hydrido- bis(dihydrooxygen)(1+), diaquahydrogen(1+)		
H ₆ N ₂ , see N ₂ H ₆				
H _n N _m , see N _m H _n				
He	helium	helium (general) He ^{•+} , helium(•1+)	helide	helido
HeH		HeH ⁺ , hydridohelium(1+)		
He ₂		He ₂ ⁺ , dihelium(1+) He ₂ ²⁺ , dihelium(2+)		
Hf	hafnium	hafnium	hafnide	hafnido
Hg	mercury	mercury (general) Hg ²⁺ , mercury(2+)	mercuride	mercurido
Hg ₂		Hg ₂ ²⁺ , dimercury(2+)		
Ho	holmium	holmium	holmide	holmido

Hs	hassium	hassium	hassium	hasside	hassido
I	iodine (general) I [•] , iodine(•), monoiodine –I, iodo	iodine (general) I ⁺ , iodine(1+)	iodide (general) I [–] , iodide(1 [–]); iodide		I [–] , iodido(1 [–]); iodido
ICl ₂	ICl ₂ [•] , dichloroiodine(•) –ICl ₂ , dichloro-λ ³ -iodanyl	ICl ₂ ⁺ , dichloroiodanium, dichloroiodine(1+)			
IF	IF, iodine fluoride [IF], fluoroiodine				
IF ₄		IF ₄ ⁺ , tetrafluoro-λ ³ -iodanium, tetrafluoroiodine(1+)	IF ₄ [–] , tetrafluoro-λ ³ -iodanuide, tetrafluoroiodate(1 [–])		IF ₄ [–] , tetrafluoro-λ ³ -iodanuïdo, tetrafluoroiodato(1 [–])
IF ₆			IF ₆ [–] , hexafluoro-λ ⁵ -iodanuide, hexafluoroiodate(1 [–])		IF ₆ [–] , hexafluoro-λ ⁵ -iodanuïdo, hexafluoroiodato(1 [–])
IH, see HI					
I ₂	I ₂ , diiodine	I ₂ ^{•+} , diiodine(•1+)	I ₂ ^{•–} , diiodide(•1 [–])		I ₂ , diiodine
I ₃	I ₃ , triiodine		I ₃ [–] , triiodide(1 [–]); triiodide		I ₃ [–] , triiodido(1 [–]); triiodido
In	indium	indium	indide		indido
InH ₂	–InH ₂ , indiganyl				
InH ₃	InH ₃ , indium trihydride [InH ₃], indigane (parent hydride name), trihydridoindium				
Ir	iridium	iridium	iridide		iridido
K	potassium	potassium	potasside		potassido
KO ₂	KO ₂ , potassium dioxide(1 [–]); potassium superoxide				
KO ₃	KO ₃ , potassium trioxide(1 [–]); potassium ozonide				
K ₂ O	K ₂ O, dipotassium oxide				
K ₂ O ₂	K ₂ O ₂ , dipotassium dioxide(2 [–]); potassium peroxide				
Kr	krypton	krypton	kryptonide		kryptonido
La	lanthanum	lanthanum	lanthanide ^d		lanthanido
Li	lithium	lithium (general) Li ⁺ , lithium(1+)	lithide (general) Li [–] , lithide(1 [–]); lithide		lithido Li [–] , lithido(1 [–]); lithido

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
LiAl	[LiAl], aluminidolithium			
LiBe	[LiBe], beryllidolithium			
LiCl	LiCl, lithium chloride [LiCl], chloridolithium	LiCl ⁺ , chloridolithium(1+)	LiCl ⁻ , chloridolithate(1-)	LiCl ⁻ , chloridolithato(1-)
LiH	LiH, lithium hydride [LiH], hydridolithium	LiH ⁺ , hydridolithium(1+)	LiH ⁻ , hydridolithate(1-)	LiH ⁻ , hydridolithato(1-)
LiMg	LiMg, lithium monomagneside	LiMg ⁺ , magnesidolithium(1+)		
Li ₂	Li ₂ , dilithium	Li ₂ ⁺ , dilithium(•1+)	Li ₂ ⁻ , dilithide(•1-)	Li ₂ ⁻ , dilithido(•1-)
Lr	lawrencium	lawrencium	lawrencide	lawrencido
Lu	lutetium	lutetium	lutetide	lutetido
Md	mendelevium	mendelevium	mendelevide	mendelevido
Mg	magnesium	magnesium (general) Mg ⁺ , magnesium(1+) Mg ²⁺ , magnesium(2+)	magneside (general) Mg ⁻ , magneside(1-)	magnesido Mg ⁻ , magnesido(1-)
Mn	manganese	manganese (general) Mn ²⁺ , manganese(2+) Mn ³⁺ , manganese(3+)	manganide	manganido
MnO	MnO, manganese mon(o)oxide, manganese(II) oxide			
MnO ₂	MnO ₂ , manganese dioxide, manganese(IV) oxide			
MnO ₃		MnO ₃ ⁺ , trioxidomanganese(1+)		

MnO ₄			MnO ₄ ⁻ , tetraoxidomanganate(1-), permanganate MnO ₄ ²⁻ , tetraoxidomanganate(2-), manganate(VI) MnO ₄ ³⁻ , tetraoxidomanganate(3-), manganate(V)	MnO ₄ ⁻ , tetraoxidomanganato(1-), permanganato MnO ₄ ²⁻ , tetraoxidomanganato(2-), manganato(VI) MnO ₄ ³⁻ , tetraoxidomanganato(3-), manganato(V)
Mn ₂ O ₃	Mn ₂ O ₃ , dimanganese trioxide, manganese(III) oxide			
Mn ₂ O ₇	Mn ₂ O ₇ , dimanganese heptaoxide, manganese(VII) oxide [O ₃ MnOMnO ₃], μ-oxido-bis(trioxidomanganese)			
Mn ₃ O ₄	Mn ₃ O ₄ , trimanganese tetraoxide Mn ^{II} Mn ^{III} ₂ O ₄ , manganese(II,III) tetraoxide			
Mo	molybdenum	molybdenum	molybdenide	molybdenido
Mt	meitnerium	meitnerium	meitneride	meitnerido
Mu	Mu [•] = μ ⁺ e ⁻ , muonium	Mu ⁺ = μ ⁺ , muon	Mu ⁻ = μ ⁺ (e ⁻) ² , muonide	
N	nitrogen N [•] , nitrogen(•), mononitrogen -N<, azanetriyl; nitrilo -N=, azanylylidene ≡N, azanylidyne	nitrogen (general) N ⁺ , nitrogen(1+)	nitride (general) N ³⁻ , nitride(3-), azanetriide; nitride =N-, azanidylydene; amidylidene -N ²⁻ , azanediidyl	N ³⁻ , nitrido(3-), azanetriido
NCO, see CNO				
NCS, see CNS				
NCl ₂			NCl ₂ ⁻ , dichloroazanide, dichloridonitrate(1-)	NCl ₂ ⁻ , dichloroazanido, dichloridonitrato(1-)
NF			NF ²⁻ , fluoroazanediide, fluoridonitrate(2-)	NF ²⁻ , fluoroazanediido, fluoridonitrato(2-)

Table IX Continued

Formula for uncharged atom or group	Name			
	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
NF ₃	NF ₃ , nitrogen trifluoride [NF ₃], trifluoroazane, trifluoridonitrogen			NF ₃ , trifluoroazane, trifluoridonitrogen
NF ₄		NF ₄ ⁺ , tetrafluoroammonium, tetrafluoroazanium, tetrafluoridonitrogen(1+)		
NH	NH ^{2•} , azanylidene, hydridonitrogen(2•); nitrene >NH, azanediyl =NH, azanylidene; imino	NH ⁺ , azanyliumdiyl, hydridonitrogen(1+) NH ²⁺ , azanebis(ylium), hydridonitrogen(2+)	NH [−] , azanidyl, hydridonitrate(1−) NH ^{2−} , azanediide, hydridonitrate(2−); imide −NH [−] , azanidyl; amidyl	NH ^{2−} , azanediido, hydridonitrate(2−); imido
NH ₂	NH ₂ [•] , azanyl, dihydridonitrogen(•); aminyl −NH ₂ , azanyl; amino	NH ₂ ⁺ , azanylium, dihydridonitrogen(1+)	NH ₂ [−] , azanide, dihydridonitrate(1−); amide	NH ₂ [−] , azanido, dihydridonitrate(1−), amido
NH ₃	NH ₃ , azane (parent hydride name), amine (parent name for certain organic derivatives), trihydridonitrogen; ammonia	NH ₃ ^{•+} , azaniumyl, trihydridonitrogen(•1+) −NH ₃ ⁺ , azaniumyl; ammonio	NH ₃ ^{•−} , azanuidyl, trihydridonitrate(•1−) ^e	NH ₃ , ammine
NH ₄	NH ₄ [•] , λ ⁵ -azanyl, tetrahydridonitrogen(•)	NH ₄ ⁺ , azanium; ammonium		

NO	NO, nitrogen mon(o)oxide (<i>not</i> nitric oxide) NO [•] , oxoazanyl, oxidonitrogen(•); nitrosyl -N=O, oxoazanyl; nitroso >N(O)-, oxo- λ^3 -azanetriyl; azoryl =N(O)-, oxo- λ^5 -azanylidene; azorylidene $\equiv$ N(O), oxo- λ^5 -azanylidyne; azorylidyne -O ⁺ =N ⁻ , azanidyldeneoxoazaniumyl	NO ⁺ , oxidonitrogen(1+) (<i>not</i> nitrosyl) NO ^{•2+} , oxidonitrogen(•2+)	NO ⁻ , oxidonitrate(1-) NO ^(2•-) , oxidonitrate(•2-1-)	NO, oxidonitrogen (general); nitrosyl = oxidonitrogen-κN (general) NO ⁺ , oxidonitrogen(1+) NO ⁻ , oxidonitrate(1-)
NO ₂	NO ₂ , nitrogen dioxide NO ₂ [•] = ONO [•] , nitrosooxidanyl, dioxidonitrogen(•); nitryl -NO ₂ , nitro -ONO, nitrosoxy	NO ₂ ⁺ , dioxidonitrogen(1+) (<i>not</i> nitryl)	NO ₂ ⁻ , dioxidonitrate(1-); nitrite NO ₂ ^{•2-} , dioxidonitrate(•2-)	NO ₂ ⁻ , dioxidonitrate(1-); nitrite NO ₂ ^{•2-} , dioxidonitrate(•2-)
NO ₃	NO ₃ , nitrogen trioxide NO ₃ [•] = O ₂ NO [•] , nitrooxidanyl, trioxidonitrogen(•) ONOO [•] , nitrosodioxidanyl, oxidoperoxidonitrogen(•) -ONO ₂ , nitrooxy		NO ₃ ⁻ , trioxidonitrate(1-); nitrate NO ₃ ^{•2-} , trioxidonitrate(•2-) [NO(OO)] ⁻ , oxidoperoxidonitrate(1-); peroxynitrite	NO ₃ ⁻ , trioxidonitrate(1-); nitrate NO ₃ ^{•2-} , trioxidonitrate(•2-) [NO(OO)] ⁻ , oxidoperoxidonitrate(1-); peroxynitrite
NO ₄			NO ₂ (O ₂) ⁻ , dioxidoperoxidonitrate(1-); peroxynitrate	NO ₂ (O ₂) ⁻ , dioxidoperoxidonitrate(1-); peroxynitrate
NS	NS, nitrogen monosulfide NS [•] , sulfidonitrogen(•) -N=S, sulfanylideneazanyl; thionitroso	NS ⁺ , sulfidonitrogen(1+) (<i>not</i> thionitrosyl)	NS ⁻ , sulfidonitrate(1-)	NS, sulfidonitrogen, sulfidonitrate, thionitrosyl (general) NS ⁺ , sulfidonitrogen(1+) NS ⁻ , sulfidonitrate(1-)

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
N ₂	N ₂ , dinitrogen =N ⁺ =N ⁻ , (azanidylidene)azanumylidene; diazo =NN=, diazene-1,2-diylidene; hydrazinediylidene -N=N-, diazene-1,2-diyl; azo	N ₂ ^{•+} , dinitrogen(•1+) N ₂ ²⁺ , dinitrogen(2+) -N ⁺ ≡N, diazyn-1-ium-1-yl	N ₂ ²⁻ , dinitride(2-) N ₂ ⁴⁻ , dinitride(4-), diazanetetraide; hydrazinetetraide	N ₂ , dinitrogen N ₂ ²⁻ , dinitride(2-) N ₂ ⁴⁻ , dinitride(4-), diazanetetraide; hydrazinetetraide
N ₂ H		N≡NH ⁺ , diazynium	N=NH ⁻ , diazenide NNH ³⁻ , diazanetriide, hydrazinetriide	N=NH ⁻ , diazenido NNH ³⁻ , diazanetriido, hydrazinetriido
N ₂ H ₂	HN=NH, diazene N=NH ₂ ⁺ , diazen-2-ium-1-ide H ₂ NN ^{2•} , diazanylidene, hydrazinylidene =NNH ₂ , diazanylidene; hydrazinylidene •HNNH [•] , diazane-1,2-diyl; hydrazine-1,2-diyl -HNNH-, diazane-1,2-diyl; hydrazine-1,2-diyl	HNNH ²⁺ , diazynylium	HNNH ²⁻ , diazane-1,2-diide, hydrazine-1,2-diide H ₂ NN ²⁻ , diazane-1,1-diide, hydrazine-1,1-diide	HN=NH, diazene N=NH ₂ ⁺ , diazen-2-ium-1-ido HNNH ²⁻ , diazane-1,2-diido, hydrazine-1,2-diido H ₂ NN ²⁻ , diazane-1,1-diido, hydrazine-1,1-diido
N ₂ H ₃	H ₂ NNH [•] , diazanyl, trihydridodinitrogen(N—N)(•); hydrazinyl -NHNH ₂ , diazanyl; hydrazinyl 2-NNH ₃ ⁺ , diazan-2-ium-1,1-diide	HN=NH ₂ ⁺ , diazenium	H ₂ NNH ⁻ , diazanide, hydrazinide	2-NNH ₃ ⁺ , diazan-2-ium-1,1-diido H ₂ NNH ⁻ , diazanido, hydrazinido

N_2H_4	H_2NNH_2 , diazane (parent hydride name), hydrazine (parent name for organic derivatives) $\text{—}NHNH_3^+$, diazan-2-ium-1-ido	$H_2NNH_2^{\bullet+}$, diazaniumyl, bis(dihydridonitrogen)($N\text{—}N$)($\bullet 1+$); hydraziniumyl $H_2N=NH_2^{2+}$, diazenedidium $H_2NNH_3^+$, diazanium, hydrazinium	H_2NNH_2 , diazane, hydrazine $\text{—}NHNH_3^+$, diazan-2-ium-1-ido
N_2H_5			
N_2H_6		$H_3NNH_3^{2+}$, diazanedidium, hydrazinedidium	
N_2O	N_2O , dinitrogen oxide (<i>not</i> nitrous oxide) NNO, oxidodinitrogen($N\text{—}N$) $\text{—}N(O)=N\text{—}$, azoxy	$N_2O^{\bullet-}$, oxidodinitrate($\bullet 1-$)	N_2O , dinitrogen oxide (general) NNO, oxidodinitrogen($N\text{—}N$) $N_2O^{\bullet-}$, oxidodinitrate($\bullet 1-$)
N_2O_2	N_2O_2 , dinitrogen dioxide ONNO, bis(oxidonitrogen)($N\text{—}N$)	$N_2O_2^{2-}$, diazenediolate, bis(oxidonitrate)($N\text{—}N$)(2-)	$N_2O_2^{2-}$, bis(oxidonitrate)($N\text{—}N$)(2-)
N_2O_3	N_2O_3 , dinitrogen trioxide O_2NNO , trioxido-1 $\kappa^2O_2,2\kappa O$ - dinitrogen($N\text{—}N$) $NO^+NO_2^-$, oxidonitrogen(1+) dioxidinitrate(1-) ONONO, dinitrosooxidane, μ -oxido-bis(oxidonitrogen)	$N_2O_3^{2-} = [O_2NNO]^{2-}$, trioxido-1 $\kappa^2O_2,2\kappa O$ - dinitrate($N\text{—}N$)(2-)	
N_2O_4	N_2O_4 , dinitrogen tetraoxide O_2NNO_2 , bis(dioxidonitrogen)($N\text{—}N$) ONONO, 1,2-dinitrosodioxidane, bis(nitrosyloxygen)($O\text{—}O$), 2,5-diazy-1,3,4,6- tetraoxy-[6]catena $NO^+NO_3^-$, oxidonitrogen(1+) trioxidinitrate(1-)		

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
N ₂ O ₅	N ₂ O ₅ , dinitrogen pentaoxide O ₂ NONO ₂ , dinitrooxidane, μ-oxido- bis(dioxidonitrogen)(N—N) NO ₂ ⁺ NO ₃ [−] , dioxidonitrogen(1+) trioxidonitrate(1−)			
N ₃	N ₃ [•] , trinitrogen(•) −N≡N ⁺ =N [−] , azido		N ₃ [−] , trinitride(1−); azide	N ₃ [−] , trinitrido(1−); azido
N ₃ H	N ₃ H, hydrogen trinitride(1−); hydrogen azide [NNNH], hydrido-1κ <i>H</i> -trinitrogen(2 <i>N</i> — <i>N</i>)			
N ₃ H ₂	−NHN=NH, triaz-2-en-1-yl			
N ₃ H ₄	−NHNHNH ₂ , triazan-1-yl			
N ₅		N ₅ ⁺ , pentanitrogen(1+)		
N ₆			N ₆ ^{•−} , hexanitride(•1−)	N ₆ ^{•−} , hexanitrido(•1−)
Na	sodium	sodium (general) Na ⁺ , sodium(1+)	sodide (general) Na [−] , sodide(1−); sodide	sodido Na [−] , sodido(1−); sodido
NaCl	NaCl, sodium chloride [NaCl], chloridosodium	NaCl ¹⁺ , chloridosodium(1+)	NaCl [−] , chloridosodate(1−)	
Na ₂	Na ₂ , disodium	Na ₂ ⁺ , disodium(1+)	Na ₂ [−] , disodide(1−)	Na ₂ [−] , disodido(1−)
Nb	niobium	niobium	niobide	niobido
Nd	neodymium	neodymium	neodymide	neodymido

Ne	neon	neon (general) Ne ⁺ , neon(1+)	neonide	neonido
NeH		NeH ⁺ , hydridoneon(1+)		
NeHe		NeHe ⁺ , helidoneon(1+)		
Ni	nickel	nickel (general) Ni ²⁺ , nickel(2+) Ni ³⁺ , nickel(3+)	nickelide	nickelido
No	nobelium	nobelium	nobelide	nobelido
Np	neptunium	neptunium	neptunide	neptunido
NpO ₂	NpO ₂ , neptunium dioxide	NpO ₂ ⁺ , dioxidoneptunium(1+) [<i>not</i> neptunyl(1+)] NpO ₂ ²⁺ , dioxidoneptunium(2+) [<i>not</i> neptunyl(2+)]		
O	oxygen (general) O, monooxygen O ^{2•} , oxidanylidene, monooxygen(2•) >O, oxy, epoxy (in rings) =O, oxo	oxygen (general) O ^{•+} , oxygen(•1+)	oxide (general) O ^{•-} , oxidanidyl, oxide(•1-) O ²⁻ , oxide(2-); oxide -O ⁻ , oxido	O ²⁻ , oxido
OBr	OBr, oxygen (mono)bromide ^f OBr [•] , bromidooxygen(•) ^f ; bromosyl -BrO, oxo-λ ³ -bromanyl; bromosyl -OBr, bromooxy	OBr ⁺ , bromidooxygen(1+) ^f (<i>not</i> bromosyl)	OBr ⁻ , bromidooxygenate(1-) ^f ; oxidobromate(1-) ^f , hypobromite	OBr ⁻ , bromidooxygenato(1-) ^f ; oxidobromato(1-) ^f , hypobromito
OCN, see CNO				
OCl	OCl, oxygen (mono)chloride ^f OCl [•] , chloridooxygen(•) ^f ; chlorosyl -ClO, oxo-λ ³ -chloranyl; chlorosyl -OCl, chlorooxy		OCl ⁻ , chloridooxygenate(1-) ^f ; oxidochlorate(1-) ^f , hypochlorite	OCl ⁻ , chloridooxygenato(1-) ^f ; oxidochlorato(1-) ^f , hypochlorito
OD ₂ , see H ₂ O				

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
OF	OF, oxygen (mono)fluoride OF [•] , fluoridooxygen(•) -FO, oxo-λ ³ -fluoranyl; fluorosyl	OF ⁺ , fluoridooxygen(1+)	OF ⁻ , fluoridooxygenate(1-)	
OF ₂	OF ₂ , oxygen difluoride [OF ₂], difluoridooxygen			
OH _n , see H _n O (n = 1-4)				
O ¹ H ₂ , see H ₂ O				
OI	OI, oxygen (mono)iodide ^f OI [•] , iodidooxygen(•) ^f ; iodosyl -IO, oxo-λ ³ -iodanyl; iodosyl -OI, iodoxy	OI ⁺ , iodidooxygen(1+) ^f (not iodosyl)	OI ⁻ , iodidooxygenate(1-) ^f ; oxidoiodate(1-) ^f ; hypiodite OI ^{•2-} , iodidooxygenate(•2-) ^f	OI ⁻ , iodidooxygenato(1-) ^f ; oxidoiodato(1-) ^f ; hypiodito
ONC, see CNO				
OT ₂ , see H ₂ O				
O ₂	O ₂ , dioxygen O ₂ ^{2•} , dioxidanediyl, dioxygen(2•) -OO-, dioxidanediyl; peroxy	O ₂ ^{•+} , dioxidanilylumyl, dioxygen(•1+) O ₂ ²⁺ , dioxidanebis(ylium), dioxygen(2+)	O ₂ ^{•-} , dioxidanidyl, dioxide(•1-); superoxide (not hyperoxide) O ₂ ²⁻ , dioxidanediide, dioxide(2-); peroxide	dioxido (general) O ₂ , dioxygen O ₂ ^{•-} , dioxide(•1-); superoxido O ₂ ²⁻ , dioxidanediido, dioxido(2-); peroxido
O ₂ Br	O ₂ Br, dioxygen bromide ^f BrO ₂ [•] , dioxidobromine(•) -BrO ₂ , dioxo-λ ⁵ -bromanyl; bromyl -OBrO, oxo-λ ³ -bromanyloxy	BrO ₂ ⁺ , dioxidobromine(1+) (not bromyl)	BrO ₂ ⁻ , dioxidobromate(1-); bromite	BrO ₂ ⁻ , dioxidobromato(1-); bromito

O_2Cl	O_2Cl , dioxygen chloride ^f $ClO_2^\bullet$, dioxidochlorine($\bullet$) $ClOO^\bullet$, chloridodioxygen($O-O$)($\bullet$) - ClO_2 , dioxo- λ^5 -chloranyl; chloryl - $OCIO$, oxo- λ^3 -chloranyloxy	ClO_2^+ , dioxidochlorine(1+) (<i>not</i> chloryl)	ClO_2^- , dioxidochlorate(1-); chlorite	ClO_2^- , dioxidochlorato(1-); chlorito
O_2Cl_2		$O_2Cl_2^+$, (dioxygen dichloride)(1+) ^f		
O_2F_2	O_2F_2 , dioxygen difluoride FOOF, difluorodioxidane, bis(fluoridooxygen)($O-O$)			
O_2I	O_2I , dioxygen iodide ^f $IO_2^\bullet$, dioxidoiodine($\bullet$) - IO_2 , dioxo- λ^5 -iodanyl; iodyl - OIO , oxo- λ^3 -iodanyloxy	IO_2^+ , dioxidoiodine(1+) (<i>not</i> iodyl)	IO_2^- , dioxidoiodate(1-); iodite	IO_2^- , dioxidoiodato(1-); iodito
O_3	O_3 , trioxxygen; ozone - $OOO-$, trioxidanediyl		$O_3^\bullet$, trioxidanidyl, trioxide($\bullet$ 1-); ozonide	O_3 , trioxxygen; ozone $O_3^\bullet$, trioxido($\bullet$ 1-); ozonido
O_3Br	O_3Br , trioxxygen bromide ^f $BrO_3^\bullet$, trioxidobromine($\bullet$) - BrO_3 , trioxo- λ^7 -bromanyl; perbromyl - $OBRO_2$, dioxo- λ^5 -bromanyloxy	BrO_3^+ , trioxidobromine(1+) (<i>not</i> perbromyl)	BrO_3^- , trioxidobromate(1-); bromate	BrO_3^- , trioxidobromato(1-); bromato
O_3Cl	O_3Cl , trioxxygen chloride ^f $ClO_3^\bullet$, trioxidochlorine($\bullet$) - ClO_3 , trioxo- λ^7 -chloranyl; perchloryl - $OCIO_2$, dioxo- λ^5 -chloranyloxy	ClO_3^+ , trioxidochlorine(1+) (<i>not</i> perchloryl)	ClO_3^- , trioxidochlorate(1-); chlorate	ClO_3^- , trioxidochlorato(1-); chlorato
O_3I	O_3I , trioxxygen iodide ^f $IO_3^\bullet$, trioxidoiodine($\bullet$) - IO_3 , trioxo- λ^7 -iodanyl; periodyl - OIO_2 , dioxo- λ^5 -iodanyloxy	IO_3^+ , trioxidoiodine(1+) (<i>not</i> periodyl)	IO_3^- , trioxidoiodate(1-); iodate	IO_3^- , trioxidoiodato(1-); iodato
O_4Br	O_4Br , tetraoxxygen bromide ^f $BrO_4^\bullet$, tetraoxidobromine($\bullet$) - $OBRO_3$, trioxo- λ^7 -bromanyloxy		BrO_4^- , tetraoxidobromate(1-); perbromate	BrO_4^- , tetraoxidobromato(1-); perbromato

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
O ₄ Cl	O ₄ Cl, tetraoxygen chloride ^f ClO ₄ •, tetraoxidochlorine(•) -OClO ₃ , trioxo-λ ⁷ -chloranyl oxy		ClO ₄ ⁻ , tetraoxidochlorate(1-); perchlorate	ClO ₄ ⁻ , tetraoxidochlorato(1-); perchlorato
O ₄ I	O ₄ I, tetraoxygen iodide ^f IO ₄ •, tetraoxidoiodine(•) -OIO ₃ , trioxo-λ ⁷ -iodanyl oxy		IO ₄ ⁻ , tetraoxidoiodate(1-); periodate	IO ₄ ⁻ , tetraoxidoiodato(1-); periodato
O ₅ I			IO ₅ ³⁻ , pentaaxidoiodate(3-)	IO ₅ ³⁻ , pentaaxidoiodato(3-)
O ₆ I			IO ₆ ⁵⁻ , hexaaxidoiodate(5-); orthoperiodate	IO ₆ ⁵⁻ , hexaaxidoiodato(5-); orthoperiodato
O ₉ I ₂			I ₂ O ₉ ⁴⁻ , nonaaxidodiiodate(4-) [O ₃ I(μ-O) ₃ IO ₃] ⁴⁻ , tri-μ-oxido-bis(trioxidoiodate)(4-)	I ₂ O ₉ ⁴⁻ , nonaaxidodiiodato(4-) [O ₃ I(μ-O) ₃ IO ₃] ⁴⁻ , tri-μ-oxido-bis(trioxidoiodato)(4-)
Os	osmium	osmium	osmide	osmido
P	phosphorus (general) P•, phosphorus(•), monophosphorus >P-, phosphanetriyl	phosphorus (general) P ⁺ , phosphorus(1+)	phosphide (general) P ⁻ , phosphide(1-) P ³⁻ , phosphide(3-), phosphanetriide; phosphide	P ³⁻ , phosphido, phosphanetriido
PF			PF ²⁻ , fluorophosphanediide, fluoridophosphate(2-)	PF ²⁻ , fluorophosphanediido, fluoridophosphato(2-)
PF ₂			PF ₂ ⁻ , difluorophosphanide, difluoridophosphate(1-)	PF ₂ ⁻ , difluorophosphanido, difluoridophosphato(1-)
PF ₃	PF ₃ , phosphorus trifluoride [PF ₃], trifluorophosphane, trifluoridophosphorus			

PF ₄		PF ₄ ⁺ , tetrafluorophosphanium, tetrafluoridophosphorus(1+)	PF ₄ ⁻ , tetrafluorophosphanuide, tetrafluoridophosphate(1-)	PF ₄ ⁻ , tetrafluorophosphanuido, tetrafluoridophosphato(1-)
PF ₅	PF ₅ , phosphorus pentafluoride [PF ₅], pentafluoro-λ ⁵ -phosphane, pentafluoridophosphorus			
PF ₆			PF ₆ ⁻ , hexafluoro-λ ⁵ -phosphanuide, hexafluoridophosphate(1-)	PF ₆ ⁻ , hexafluoro-λ ⁵ -phosphanuido, hexafluoridophosphato(1-)
PH	PH ^{2•} , phosphanylidene, hydridophosphorus(2•) >PH, phosphanediyl =PH, phosphanylidene	PH ^{•+} , phosphanyliumyl, hydridophosphorus(•1+) PH ²⁺ , phosphanebis(ylium), hydridophosphorus(2+)	PH ^{•-} , phosphanidyl, hydridophosphate(•1-) PH ²⁻ , phosphanediide, hydridophosphate(2-)	PH ²⁻ , phosphanediido, hydridophosphato(2-)
PH ₂	PH ₂ [•] , phosphanyl, dihydridophosphorus(•) -PH ₂ , phosphanyl	PH ₂ ⁺ , phosphanylium, dihydridophosphorus(1+)	PH ₂ ⁻ , phosphanide, dihydridophosphate(1-)	PH ₂ ⁻ , phosphanido, dihydridophosphato(1-)
PH ₃	PH ₃ , phosphorus trihydride [PH ₃], phosphane (parent hydride name), trihydridophosphorus -PH ₃ , λ ⁵ -phosphanyl	PH ₃ ^{•+} , phosphaniumyl, trihydridophosphorus(•1+) -PH ₃ ⁺ , phosphaniumyl	PH ₃ ^{•-} , phosphanuidyl, trihydridophosphate(•1-) ^e	PH ₃ , phosphane
PH ₄		PH ₄ ⁺ , phosphanium, tetrahydridophosphorus(1+)	PH ₄ ⁻ , phosphanuide, tetrahydridophosphate(1-)	PH ₄ ⁻ , phosphanuido, tetrahydridophosphato(1-)
PH ₅	PH ₅ , phosphorus pentahydride [PH ₅], λ ⁵ -phosphane (parent hydride name), pentahydridophosphorus			
PN	P≡N, nitridophosphorus >P≡N, azanylidyne-λ ⁵ -phosphanediyl; phosphoronitridoyl			

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
PO	PO [•] , oxophosphanyl, oxidophosphorus(•), phosphorus mon(o)oxide; phosphoryl >P(O)–, oxo-λ ⁵ -phosphanetriyl; phosphoryl =P(O)–, oxo-λ ⁵ -phosphanylidene; phosphorylidene ≡P(O), oxo-λ ⁵ -phosphanylidyne; phosphorylidyne	PO ⁺ , oxidophosphorus(1+) (not phosphoryl)	PO [–] , oxidophosphate(1–)	
PO ₂	–P(O) ₂ , dioxo-λ ⁵ -phosphanyl		PO ₂ [–] , dioxidophosphate(1–)	PO ₂ [–] , dioxidophosphato(1–)
PO ₃			PO ₃ [–] , trioxidophosphate(1–) PO ₃ ^{•2–} , trioxidophosphate(•2–) PO ₃ ^{3–} , trioxidophosphate(3–); phosphite (PO ₃ [–]) _n = {P(O) ₂ O} _n ^{n–} , catena-poly[(dioxidophosphate-μ- oxido)(1–)]; metaphosphate –P(O)(O [–]) ₂ , dioxidooxo-λ ⁵ -phosphanyl; phosphonato	PO ₃ [–] , trioxidophosphato(1–) PO ₃ ^{•2–} , trioxidophosphato(•2–) PO ₃ ^{3–} , trioxidophosphato(3–); phosphito

PO_4			$\text{PO}_4^{\bullet 2-}$, tetraoxidophosphate($\bullet 2-$) PO_4^{3-} , tetraoxidophosphate($3-$); phosphate	PO_4^{3-} , tetraoxidophosphato($3-$); phosphato
PO_5			$\text{PO}_5^{\bullet 2-} = \text{PO}_3(\text{OO})^{\bullet 2-}$, trioxidoperoxidophosphate($\bullet 2-$) $\text{PO}_5^{3-} = \text{PO}_3(\text{OO})^{3-}$, trioxidoperoxidophosphate($3-$); peroxyphosphato, phosphoroperoxoato	$\text{PO}_5^{3-} = \text{PO}_3(\text{OO})^{3-}$, trioxidoperoxidophosphato($3-$); peroxyphosphato, phosphoroperoxoato
PS	$\text{PS}^\bullet$, sulfidophosphorus($\bullet$); -PS, thiophosphoryl	PS^+ , sulfidophosphorus($1+$) (<i>not</i> thiophosphoryl)		
PS_4			PS_4^{3-} , tetrasulfidophosphate($3-$)	PS_4^{3-} , tetrasulfidophosphato($3-$)
P_2	P_2 , diphosphorus	P_2^+ , diphosphorus($1+$)	P_2^- , diphosphide($1-$) P_2^{2-} , diphosphide($2-$)	P_2 , diphosphorus P_2^- , diphosphido($1-$) P_2^{2-} , diphosphido($2-$)
P_2H			$\text{HP} = \text{P}^-$, diphosphenide PPH^{3-} , diphosphanetriide	$\text{HP} = \text{P}^-$, diphosphenido PPH^{3-} , diphosphanetriido
P_2H_2	$\text{HP} = \text{PH}$, diphosphene (parent hydride name) $\text{H}_2\text{PP}^{2\bullet}$, diphosphanylidene $= \text{PPH}_2$, diphosphanylidene $= \text{HPPH}^-$, diphosphane-1,2-diyl		HPPH^{2-} , diphosphane-1,2-diide H_2PP^{2-} , diphosphane-1,1-diide	$\text{HP} = \text{PH}$, diphosphene HPPH^{2-} , diphosphane-1,2-diido H_2PP^{2-} , diphosphane-1,1-diido
P_2H_3	$\text{H}_2\text{PPH}^\bullet$, diphosphanyl, trihydridodiphosphorus($P-P$)($\bullet$) $= \text{HPPH}_2$, diphosphanyl		H_2PPH^- , diphosphanide	H_2PPH^- , diphosphanido
P_2H_4	H_2PPH_2 , diphosphane (parent hydride name)			H_2PPH_2 , diphosphane
P_2O_6			$\text{O}_3\text{PPO}_3^{3-}$, bis(trioxidophosphate)($P-P$)($4-$); hypodiphosphate	$\text{O}_3\text{PPO}_3^{3-}$, bis(trioxidophosphato)($P-P$)($4-$); hypodiphosphato
P_2O_7			$\text{O}_3\text{POPO}_3^{4-}$, μ -oxido- bis(trioxidophosphate)($4-$); diphosphate	$\text{O}_3\text{POPO}_3^{4-}$, μ -oxido- bis(trioxidophosphato)($4-$); diphosphato

Table IX Continued

Formula for uncharged atom or group	Name			
	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
P ₂ O ₈			O ₃ POOPO ₃ ⁴⁻ , μ-peroxido-1κO,2κO'- bis(trioxidophosphate)(4-); peroxydiphosphate	O ₃ POOPO ₃ ⁴⁻ , μ-peroxido-1κO,2κO'- bis(trioxidophosphate)(4-); peroxydiphosphate
P ₄	P ₄ , tetraphosphorus			P ₄ , tetraphosphorus
Pa	protactinium	protactinium	protactinide	protactinido
Pb	lead	lead (general) Pb ²⁺ , lead(2+) Pb ⁴⁺ , lead(4+)	plumbide	plumbido
PbH ₄	PbH ₄ , plumbane (parent hydride name), tetrahydrido-lead, lead tetrahydride			
Pb ₉			Pb ₉ ⁴⁻ , nonaplumbide(4-)	
Pd	palladium	palladium (general) Pd ²⁺ , palladium(2+) Pd ⁴⁺ , palladium(4+)	palladiide	palladiido
Pm	promethium	promethium	promethide	promethido
Po	polonium	polonium	polonide	polonido
PoH ₂ , see H ₂ Po				
Pr	praseodymium	praseodymium	praseodymide	praseodymido
Pt	platinum	platinum (general) Pt ²⁺ , platinum(2+) Pt ⁴⁺ , platinum(4+)	platinide	platinido

Pu	plutonium	plutonium	plutonide	plutonido
PuO ₂	PuO ₂ , plutonium dioxide	PuO ₂ ⁺ , dioxidoplutonium(1+) [not plutonyl(1+)] PuO ₂ ²⁺ , dioxidoplutonium(2+) [not plutonyl(2+)]		
Ra	radium	radium	radide	radio
Rb	rubidium	rubidium	rubidide	rubidido
Re	rhénium	rhénium	rhénide	rhénido
ReO ₄			ReO ₄ ⁻ , tetraoxidorhenate(1-) ReO ₄ ²⁻ , tetraoxidorhenate(2-)	ReO ₄ ⁻ , tetraoxidorhenato(1-) ReO ₄ ²⁻ , tetraoxidorhenato(2-)
Rf	rutherfordium	rutherfordium	rutherfordide	rutherfordido
Rg	roentgenium	roentgenium	roentgenide	roentgenido
Rh	rhodium	rhodium	rhodide	rhodido
Rn	radon	radon	radonide	radonido
Ru	ruthénium	ruthénium	ruthénide	ruthénido
S	sulfur (general) S, monosulfur =S, sulfanylidene; thioxo -S-, sulfanedyl	sulfur (general) S ⁺ , sulfur(1+)	sulfide (general) S ^{•-} , sulfamidyl, sulfide(•1-) S ²⁻ , sulfanedide, sulfide(2-); sulfide -S-, sulfido	sulfido (general) S ^{•-} , sulfamidyl, sulfido(•1-) S ²⁻ , sulfanedido, sulfido(2-)
SCN, see CNS				
SH, see HS				
SH ₂ , see H ₂ S				
SNC, see CNS				
SO	SO, sulfur mon(o)oxide [SO], oxidosulfur > SO, oxo-λ ⁴ -sulfanedyl; sulfinyl	SO ^{•+} , oxidosulfur(•1+) (not sulfinyl or thionyl)	SO ^{•-} , oxidosulfate(•1-)	[SO], oxidosulfur
SO ₂	SO ₂ , sulfur dioxide [SO ₂], dioxidosulfur >SO ₂ , dioxo-λ ⁶ -sulfanedyl; sulfonyl, sulfonyl		SO ₂ ^{•-} , dioxidosulfate(•1-) SO ₂ ²⁻ , dioxidosulfate(2-), sulfanediolate	[SO ₂], dioxidosulfur SO ₂ ²⁻ , dioxidosulfato(2-), sulfanediolato

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
SO ₃	SO ₃ , sulfur trioxide		SO ₃ ^{•-} , trioxidosulfate(•1-) SO ₃ ²⁻ , trioxidosulfate(2-); sulfite -S(O) ₂ (O ⁻), oxidodioxo-λ ⁶ -sulfanyl; sulfonato	SO ₃ ²⁻ , trioxidosulfato(2-); sulfito
SO ₄	-OS(O) ₂ O ⁻ , sulfonylbis(oxy)		SO ₄ ^{•-} , tetraoxidosulfate(•1-) SO ₄ ²⁻ , tetraoxidosulfate(2-); sulfate	SO ₄ ²⁻ , tetraoxidosulfato(2-); sulfato
SO ₅			SO ₅ ^{•-} = SO ₃ (OO) ^{•-} , trioxidoperoxidosulfate(•1-) SO ₅ ²⁻ = SO ₃ (OO) ²⁻ , trioxidoperoxidosulfate(2-); peroxysulfate, sulfuroperoxoate	SO ₅ ²⁻ = SO ₃ (OO) ²⁻ , trioxidoperoxidosulfato(2-); peroxysulfato, sulfuroperoxoato
S ₂	S ₂ , disulfur -SS-, disulfanediyl >S=S, sulfanylidene-λ ⁴ -sulfanediyl; sulfinothioyl	S ₂ ^{•+} , disulfur(•1+)	S ₂ ^{•-} , disulfanidyl, disulfide(•1-) S ₂ ²⁻ , disulfide(2-), disulfanediide -SS-, disulfanidyl	S ₂ ²⁻ , disulfido(2-), disulfanediido
S ₂ O	>S(=O)(=S), oxosulfanylidene-λ ⁶ -sulfanediyl; sulfonothioyl			

S_2O_2			$S_2O_2^{2-} = OSSO_2^{2-}$, disulfanediolate, bis(oxidosulfate)($S-S$)(2-) $S_2O_2^{2-} = SOOS^{2-}$, dioxidanedithiolate, peroxybis(sulfanide), bis(sulfidoxygenate)($O-O$)(2-) $S_2O_2^{2-} = SO_2S^{2-}$, dioxido-1 κ^2O -disulfate($S-S$)(2-); dioxidosulfidosulfate(2-); thiosulfite, sulfurothioite	$S_2O_2^{2-} = OSSO_2^{2-}$, disulfanediolate, bis(oxidosulfate)($S-S$)(2-) $S_2O_2^{2-} = SOOS^{2-}$, dioxidanedithiolate, peroxybis(sulfanide), bis(sulfidoxygenate)($O-O$)(2-) $S_2O_2^{2-} = SO_2S^{2-}$, dioxido-1 κ^2O -disulfate($S-S$)(2-); dioxidosulfidosulfate(2-); thiosulfite, sulfurothioite
S_2O_3			$S_2O_3^{\bullet-} = SO_3S^{\bullet-}$, trioxido-1 κ^3O -disulfate($S-S$)(•1-), trioxidosulfidosulfate(•1-) $S_2O_3^{2-} = SO_3S^{2-}$, trioxido-1 κ^3O -disulfate($S-S$)(2-); trioxidosulfidosulfate(2-); thiosulfate, sulfurothioate	$S_2O_3^{2-} = SO_3S^{2-}$, trioxido-1 κ^3O -disulfate($S-S$)(2-); trioxidosulfidosulfate(2-); thiosulfate, sulfurothioate
S_2O_4			$S_2O_4^{2-} = O_2SSO_2^{2-}$, bis(dioxidosulfate)($S-S$)(2-); dithionite	$S_2O_4^{2-} = O_2SSO_2^{2-}$, bis(dioxidosulfate)($S-S$)(2-); dithionite
S_2O_5			$S_2O_5^{2-} = O_3SSO_2^{2-}$, pentaoxido- 1 κ^3O , 2 κ^2O -disulfate($S-S$)(2-) $S_2O_5^{2-} = O_2SOSO_2^{2-}$, μ -oxido-bis(dioxidosulfate)(2-) $S_2O_6^{2-} = O_3SSO_3^{2-}$, bis(trioxidosulfate)($S-S$)(2-); dithionate	$S_2O_5^{2-} = O_3SSO_2^{2-}$, pentaoxido- 1 κ^3O , 2 κ^2O -disulfate($S-S$)(2-) $S_2O_5^{2-} = O_2SOSO_2^{2-}$, μ -oxido-bis(dioxidosulfate)(2-) $S_2O_6^{2-} = O_3SSO_3^{2-}$, bis(trioxidosulfate)($S-S$)(2-); dithionate
S_2O_6				
S_2O_7			$S_2O_7^{2-} = O_3SOSO_3^{2-}$, μ -oxido-bis(trioxidosulfate)(2-); disulfate	$S_2O_7^{2-} = O_3SOSO_3^{2-}$, μ -oxido-bis(trioxidosulfate)(2-); disulfate

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
S_2O_8			$S_2O_8^{2-} = O_3SOOSO_3^{2-}$, μ -peroxido-1κO,2κO'-bis(trioxidosulfate)(2-); peroxydisulfate	$S_2O_8^{2-} = O_3SOOSO_3^{2-}$, μ -peroxido-1κO,2κO'-bis(trioxidosulfate)(2-); peroxydisulfate
S_3	S_3 , trisulfur -SSS-, trisulfaneyl >S(=S) ₂ , bis(sulfanylidene)-λ ⁶ -sulfaneyl; sulfonodithioyl, dithiosulfonyl	S_3^{2+} , trisulfur(2+)	$S_3^{\bullet-}$, trisulfide(•1-) SSS ^{•-} , trisulfaneyl S_3^{2-} , trisulfide(2-) SSS ²⁻ , trisulfaneylide	$S_3^{\bullet-}$, trisulfido(•1-) SSS ^{•-} , trisulfaneyl S_3^{2-} , trisulfido(2-) SSS ²⁻ , trisulfaneydiido
S_4	S_4 , tetrasulfur -SSSS-, tetrasulfaneyl	S_4^{2+} , tetrasulfur(2+)	S_4^{2-} , tetrasulfide(2-) SSSS ²⁻ , tetrasulfaneydiide	S_4^{2-} , tetrasulfido(2-) SSSS ²⁻ , tetrasulfaneydiido
S_4O_6			$S_4O_6^{2-} = O_3SSSO_3^{2-}$, disulfanedisulfonate, bis[(trioxidosulfato)sulfate]⊖ (S-S)(2-); tetrathionate $S_4O_6^{\bullet 3-} = O_3SSSO_3^{\bullet 3-}$, bis[(trioxidosulfato)sulfate]⊖ (S-S)(•3-)	$S_4O_6^{2-} = O_3SSSO_3^{2-}$, disulfanedisulfonate, bis[(trioxidosulfato)sulfate]⊖ (S-S)(2-); tetrathionate $S_4O_6^{\bullet 3-} = O_3SSSO_3^{\bullet 3-}$, bis[(trioxidosulfato)sulfate]⊖ (S-S)(•3-)
S_5	S_5 , pentasulfur		S_5^{2-} , pentasulfide(2-) SSSSS ²⁻ , pentasulfaneydiide	S_5^{2-} , pentasulfido(2-) SSSSS ²⁻ , pentasulfaneydiido
S_8	S_8 , octasulfur	S_8^{2+} , octasulfur(2+)	S_8^{2-} , octasulfide(2-) S[S] ₆ S ²⁻ , octasulfaneydiide	S_8 , octasulfur S_8^{2-} , octasulfido(2-) S[S] ₆ S ²⁻ , octasulfaneydiido

Sb	antimony >Sb-, stibanetriyl	antimony	antimonide (general) Sb ³⁻ , antimonide(3-), stibanetriide; antimonide	antimonido (general) Sb ³⁻ , antimonido, stibanetriido
SbH	SbH [•] , stibanylidene, hydridoantimony(2•) >SbH, stibanediyl =SbH, stibanylidene	SbH ²⁺ , stibanebis(ylum), hydridoantimony(2+)	SbH ²⁻ , stibanediide, hydridoantimonate(2-)	SbH ²⁻ , stibanediido, hydridoantimonato(2-)
SbH ₂	SbH ₂ [•] , stibanyl, dihydridoantimony(•) -SbH ₂ , stibanyl	SbH ₂ ⁺ , stibanylium, dihydridoantimony(1+)	SbH ₂ ⁻ , stibanide, dihydridoantimonate(1-)	SbH ₂ ⁻ , stibanido, dihydridoantimonato(1-)
SbH ₃	SbH ₃ , antimony trihydride [SbH ₃], stibane (parent hydride name), trihydridoantimony -SbH ₄ , λ ⁵ -stibanyl	SbH ₃ ^{•+} , stibaniumyl, trihydridoantimony(•1+) -SbH ₃ ⁺ , stibaniumyl	SbH ₃ ^{•-} , stibanuidyl, trihydridoantimonate(•1-) ^e	SbH ₃ , stibane
SbH ₄		SbH ₄ ⁺ , stibanium, tetrahydridoantimony(1+)		
SbH ₅	SbH ₅ , antimony pentahydride [SbH ₅], λ ⁵ -stibane (parent hydride name), pentahydridoantimony			
Sc	scandium	scandium	scandide	scandido
Se	Se (general) Se, monoselenium >Se, selanediyl =Se, selanylidene; selenoxo	selenium	selenide (general) Se ^{•-} , selanidyl, selenide(•1-) Se ²⁻ , selanediide, selenide(2-); selenide	selenido (general) Se ^{•-} , selanidyl, selenido(•1-) Se ²⁻ , selanediido, selenido(2-)
SeCN, see CNSe				
SeH, see HSe				
SeH ₂ , see H ₂ Se				
SeO	SeO, selenium mon(o)oxide [SeO], oxidoselenium > SeO, seleninyl			[SeO], oxidoselenium

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
SeO ₂	SeO ₂ , selenium dioxide [SeO ₂], dioxidoselenium >SeO ₂ , selenonyl		SeO ₂ ²⁻ , dioxidoselenate(2-)	[SeO ₂], dioxidoselenium SeO ₂ ²⁻ , dioxidoselenato(2-)
SeO ₃	SeO ₃ , selenium trioxide		SeO ₃ ^{•-} , trioxidoselenate(•1-) SeO ₃ ²⁻ , trioxidoselenate(2-); selenite	SeO ₃ ²⁻ , trioxidoselenato(2-); selenito
SeO ₄			SeO ₄ ²⁻ , tetraoxidoselenate(2-); selenate	SeO ₄ ²⁻ , tetraoxidoselenato(2-); selenato
Sg	seaborgium	seaborgium	seaborgide	seaborgido
Si	silicon >Si<, silanetetrayl =Si=, silanediylidene	silicon (general) Si ⁺ , silicon(•1+) Si ⁴⁺ , silicon(4+)	silicide (general) Si ⁻ , silicide(•1-) Si ⁴⁻ , silicide(4-); silicide	silicido (general) Si ⁻ , silicido(•1-) Si ⁴⁻ , silicido(4-); silicido
SiC	SiC, silicon carbide [SiC], carbidosilicon	SiC ⁺ , carbidosilicon(1+)		
SiH		SiH ⁺ , silanyliumdiyl, hydridosilicon(1+)	SiH ⁻ , silanediyl, hydridosilicate(1-)	
SiH ₂	SiH ₂ ^{2•} , silylidene, dihydridosilicon(2•) >SiH ₂ , silanediyl =SiH ₂ , silylidene			
SiH ₃	SiH ₃ [•] , silyl, trihydridosilicon(•) -SiH ₃ , silyl	SiH ₃ ⁺ , silylium, trihydridosilicon(1+)	SiH ₃ ⁻ , silanide, trihydridosilicate(1-)	SiH ₃ ⁻ , silanido

SiH ₄	SiH ₄ , silicon tetrahydride [SiH ₄], silane (parent hydride name), tetrahydrosilicon			
SiO	SiO, oxidosilicon, silicon mon(o)oxide	SiO ⁺ , oxidosilicon(1+)		
SiO ₂	SiO ₂ , silicon dioxide			
SiO ₃			SiO ₃ ^{•-} , trioxidosilicate(•1-) (SiO ₃ ²⁻) _n = tSi(O) ₂ O ₃ ²ⁿ⁻ , <i>catena</i> -poly[(dioxidosilicate- μ-oxido)(1-)]; metasilicate	SiO ₃ ^{•-} , trioxidosilicato(•1-)
SiO ₄			SiO ₄ ⁴⁻ , tetraoxidosilicate(4-); silicate	SiO ₄ ⁴⁻ , tetraoxidosilicato(4-); silicato
Si ₂	Si ₂ , disilicon	Si ₂ ⁺ , disilicon(1+)	Si ₂ ⁻ , disilicide(1-)	
Si ₂ H ₄	>SiHSiH ₃ , disilane-1,1-diyl -SiH ₂ SiH ₂ ⁻ , disilane-1,2-diyl = SiHSiH ₃ , disilanylidene			
Si ₂ H ₅	Si ₂ H ₅ [•] , disilanyl, pentahydridosilicon(Si—Si)(•) -Si ₂ H ₅ , disilanyl	Si ₂ H ₅ ⁺ , disilanylium	Si ₂ H ₅ ⁻ , disilanide	Si ₂ H ₅ ⁻ , disilanido
Si ₂ H ₆	Si ₂ H ₆ , disilane (parent hydride name)			Si ₂ H ₆ , disilane
Si ₂ O ₇			Si ₂ O ₇ ⁶⁻ , μ-oxido-bis(trioxidosilicate)(6-); disilicate	Si ₂ O ₇ ⁶⁻ , μ-oxido-bis(trioxidosilicato)(6-); disilicato
Si ₄			Si ₄ ⁴⁻ , tetrasilicide(4-)	
Sm	samarium	samarium	samaride	samarido
Sn	tin	tin (general) Sn ²⁺ , tin(2+) Sn ⁴⁺ , tin(4+)	stannide	stannido
SnCl ₃			SnCl ₃ ⁻ , trichloridostannate(1-)	SnCl ₃ ⁻ , trichloridostannato(1-)
SnH ₄	SnH ₄ , tin tetrahydride [SnH ₄], stannane (parent hydride name), tetrahydridotin			

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
Sn ₅			Sn ₅ ²⁻ , pentastannide(2-)	Sn ₅ ²⁻ , pentastannido(2-)
Sr	strontium	strontium	strontide	strontido
T, see H				
T ₂ , see H ₂				
T ₂ O, see H ₂ O				
Ta	tantalum	tantalum	tantalide	tantalido
Tb	terbium	terbium	terbide	terbido
Tc	technetium	technetium	technetide	technetido
TcO ₄			TcO ₄ ⁻ , tetraoxido technetate(1-) TcO ₄ ²⁻ , tetraoxido technetate(2-)	TcO ₄ ⁻ , tetraoxido technetato(1-) TcO ₄ ²⁻ , tetraoxido technetato(2-)
Te	tellurium >Te, tellanediyl =Te, tellanylidene; telluroxo	tellurium	telluride (general) Te ^{•-} , tellanidylyl, telluride(•1-) Te ²⁻ , tellanediide, telluride(2-); telluride	tellurido (general) Te ^{•-} , tellanidylyl, tellurido(•1-) Te ²⁻ , tellanediido, tellurido(2-)
TeH, see HTe				
TeH ₂ , see H ₂ Te				
TeO ₃			TeO ₃ ^{•-} , trioxido tellurate(•1-) TeO ₃ ²⁻ , trioxido tellurate(2-)	TeO ₃ ^{•-} , trioxido tellurato(•1-) TeO ₃ ²⁻ , trioxido tellurato(2-)
TeO ₄			TeO ₄ ²⁻ , tetraoxido tellurate(2-); tellurate	TeO ₄ ²⁻ , tetraoxido tellurato(2-); tellurato
TeO ₆			TeO ₆ ⁶⁻ , hexaoxido tellurate(6-); orthotellurate	TeO ₆ ⁶⁻ , hexaoxido tellurato(6-); orthotellurato
Th	thorium	thorium	thoride	thorido

Ti	titanium	titanium	titanide	titanido
TiO	TiO, titanium(II) oxide	TiO ²⁺ , oxidotitanium(2+)		
Tl	thallium	thallium	thallide	thallido
TlH ₂	-TlH ₂ , thallanyl			
TlH ₃	TlH ₃ , thallium trihydride [TlH ₃], thallane (parent hydride name), trihydridothallium			
Tm	thulium	thulium	thulide	thulido
U	uranium	uranium	uranide	uranido
UO ₂	UO ₂ , uranium dioxide	UO ₂ ⁺ , dioxidouranium(1+) [<i>not</i> uranyl(1+)] UO ₂ ²⁺ , dioxidouranium(2+) [<i>not</i> uranyl(2+)]		
V	vanadium	vanadium	vanadide	vanadido
VO	VO, vanadium(II) oxide, vanadium mon(o)oxide	VO ²⁺ , oxidovanadium(2+) (<i>not</i> vanadyl)		
VO ₂	VO ₂ , vanadium(IV) oxide, vanadium dioxide	VO ₂ ⁺ , dioxidovanadium(1+)		
W	tungsten	tungsten	tungstide	tungstido
Xe	xenon	xenon	xenonide	xenonido
Y	yttrium	yttrium	yttride	yttrido
Yb	ytterbium	ytterbium	ytterbide	ytterbido
Zn	zinc	zinc	zincide	zincido
Zr	zirconium	zirconium	zirconide	zirconido
ZrO	ZrO, zirconium(II) oxide	ZrO ²⁺ , oxidozirconium(2+)		

^a Where an element symbol occurs in the first column, the unmodified element name is listed in the second and third columns. The unmodified name is generally used when the element appears as an electropositive constituent in the construction of a stoichiometric name (Sections IR-5.2 and IR-5.4). Names of homoatomic cations consisting of the element are also constructed using the element name, adding multiplicative prefixes and charge numbers as applicable (Sections IR-5.3.2.1 to IR-5.3.2.3). In selected cases, examples are given in the Table of specific cation names, such as gold(1+), gold(3+); mercury(2+), dimercury(2+). In such cases, the unmodified element name appears with the qualifier '(general)'.

- ^b Where an element symbol occurs in the first column, the fourth column gives the element name appropriately modified with the ending 'ide' (argentide, americide, ferride, etc.). The 'ide' form of the element name is generally used when the element appears as an electronegative constituent in the construction of a stoichiometric name (Sections IR-5.2 and IR-5.4). Names of homoatomic anions consisting of the element in question are also constructed using this modified form, adding multiplicative prefixes and charge numbers as applicable (Sections IR-5.3.3.1 to IR-5.3.3.3). Examples are given in the Table of names of some specific anions, e.g. arsenide(3-), chloride(1-), oxide(2-), dioxide(2-). In certain cases, a particular anion has the 'ide' form itself as an accepted short name, e.g. arsenide, chloride, oxide. If specific anions are named, the 'ide' form of the element name with no further modification is given as the first entry in the fourth column, with the qualifier '(general)'.
- ^c Ligand names must be placed within enclosing marks whenever necessary to avoid ambiguity, cf. Section IR-9.2.2.3. Some ligand names must always be enclosed. For example, if 'dioxido' is cited as it must be enclosed so as to distinguish it from two 'oxido' ligands; if combined with a multiplicative prefix it must be enclosed because it starts with a multiplicative prefix itself. A ligand name such as 'nitridocarbonato' must always be enclosed to avoid interpreting it as two separate ligand names, 'nitrido' and 'carbonato'. In this table, however, these enclosing marks are omitted for the sake of clarity. Note that the ligand names given here with a charge number can generally also be used without if it is not desired to make any implication regarding the charge of the ligand. For example, the ligand name '[dioxido(•1-)]' may be used if one wishes explicitly to consider the ligand to be the species dioxido(•1-), whereas the ligand name '(dioxido)' can be used if no such implications are desirable.
- ^d The ending 'ide' in 'actinide' and 'lanthanide' indicates a negative ion. Therefore, 'actinoid' should be used as the collective name for the elements Ac, Th, Pa, U, Np, Pu, Am, Cm, Bk, Cf, Es, Fm, Md, No, Lr, and 'lanthanoid' as a collective name for the elements La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu (cf. Section IR-3.5).
- ^e The radical names in the present recommendations sometimes differ from those given in "Names for Inorganic Radicals", W.H. Koppenol, *Pure Appl. Chem.*, **72**, 437-446 (2000). Firstly, the exceptional status of anion radicals consisting of hydrogen and only one other element has been lifted. For example, the coordination-type additive name of $\text{BH}_3^{\bullet-}$ is 'trihydroborate(•1-)' (not 'boride'). Secondly, concatenation of ligand names, such as in 'hydridodioxido' (meaning the ligand 'dioxidanido'), which is otherwise never used in additive nomenclature, is not recommended here. Thirdly, additive names of dinuclear compounds are based here on selecting the most centrally placed atoms in the molecule as central atoms (see the general principles described in Section IR-7.1.2), e.g. $\text{NCSSCN}^{\bullet-}$ is named here 'bis[cyanidosulfur](S-S)(•1-)' rather than 'bis(nitridosulfidocarbonato)(S-S)(•1-)'.
- ^f Due to the strict adherence in these recommendations to the element sequence in Table VI, the order of oxygen and the elements chlorine, bromine and iodine, respectively, has been reversed relative to traditional names. This applies to binary stoichiometric names such as dioxigen chloride (cf. Section IR-5.2) and to additive names for the hypohalites, where the rules for selecting central atoms (Section IR-7.1.2) dictate the selection of oxygen rather than the halide. However, because of the additive names for the last three members of the series OX^- , XO_2^- , XO_3^- , XO_4^- ($\text{X} = \text{Cl}, \text{Br}, \text{I}$), namely dioxidohalogenate(1-), trioxidohalogenate(1-) and tetraoxidohalogenate(1-) (the halogen is chosen as the central atom because it has the central position in the structure), the additive names oxidochlorate(1-), oxidobromate(1-) and oxidoiodate(1-) are acceptable alternatives to the systematic 'oxygenate' names. Similar remarks apply to HOCl , $\text{HOCl}^{\bullet-}$, etc.

Table X Anion names, 'a' terms used in substitutive nomenclature and 'y' terms used in chains and rings nomenclature

<i>Element name</i>	<i>Anion name^a</i>	<i>'a' term</i>	<i>'y' term</i>
actinium	actinate	actina	actiny
aluminium	aluminate	alumina	aluminy
americium	americite	america	americ
antimony	antimonate	stiba ^b	stiby ^b
argon	argonate	argona	argony
arsenic	arsenate	arsa	arsy
astatine	astatate	astata	astaty
barium	barate	bara	bary
berkelium	berkelate	berkela	berkely
beryllium	beryllate	berylla	beryll
bismuth	bismuthate	bisma	bismy
bohrium	bohrate	bohra	bohry
boron	borate	bora	bory
bromine	bromate	broma	bromy
cadmium	cadmate	cadma	cadmy
caesium	caesate	caesa	caesy
calcium	calcate	calca	calcy
californium	californate	california	californy
carbon	carbonate	carba	carby
cerium	cerate	cera	cery
chlorine	chlorate	chlora	chlory
chromium	chromate	chroma	chromy
cobalt	cobaltate	cobalta	cobalty
copper	cuprate ^c	cupra ^c	cupry ^c
curium	curate	cura	cury
darmstadtium	darmstadtate	darmstadta	darmstadty
deuterium	deuterate	deutera	deutery
dubnium	dubnate	dubna	dubny
dysprosium	dysprosate	dysprosa	dysprosy
einsteinium	einsteinate	einsteina	einstein
erbium	erbate	erba	erby
europium	europate	europa	europy
fermium	fermate	ferma	fermy
fluorine	fluorate	fluora	fluory
francium	francate	franca	francy
gadolinium	gadolate	gadolina	gadolin
gallium	gallate	galla	gally
germanium	germanate	germa	germy
gold	aurate ^d	aura ^d	aury ^d
hafnium	hafnate	hafna	hafny
hassium	hassate	hassa	hassy
helium	helate	hela	hely
holmium	holmate	holma	holmy
hydrogen	hydrogenate	—	hydrony
indium	indate	inda	indy
iodine	iodate	ioda	iody
iridium	iridate	irida	iridy
iron	ferrate ^c	ferra ^c	ferry ^c
krypton	kryptonate	kryptona	kryptony
lanthanum	lanthanate	lanthana	lanthany
lawrencium	lawrencate	lawrenca	lawreny

Table X Continued

<i>Element name</i>	<i>Anion name^a</i>	<i>'a' term</i>	<i>'y' term</i>
lead	plumbate ^f	plumba ^f	plumby ^f
lithium	lithate	litha	lithy
lutetium	lutetate	luteta	lutety
magnesium	magnesate	magnesa	magnesy
manganese	manganate	mangana	mangany
meitnerium	meitnerate	meitnera	meitnery
mendelevium	mendelevate	mendeleva	mendelevy
mercury	mercurate	mercura	mercury
molybdenum	molybdate	molybda	molybdy
neodymium	neodymate	neodyma	neodymy
neon	neonate	neona	neony
neptunium	neptunate	neptuna	neptuny
nickel	nickelate	nickela	nickely
niobium	niobate	nioba	nioby
nitrogen	nitrate	aza ^g	azy ^g
nobelium	nobelate	nobela	nobely
osmium	osmate	osma	osmy
oxygen	oxygenate	oxa	oxy
palladium	palladate	pallada	pallady
phosphorus	phosphate	phospha	phosphy
platinum	platinate	platina	platiny
plutonium	plutonate	plutona	plutony
polonium	polonate	polona	polony
potassium	potassate	potassa	potassy
praseodymium	praseodymate	praseodyma	praseodymy
promethium	promethate	prometha	promethy
protactinium	protactinate	protactina	protactiny
protium	protate	prota	proty
radium	radate	rada	rady
radon	radonate	radona	radony
rhenium	rhenate	rhena	rheny
rhodium	rhodate	rhoda	rhody
roentgenium	roentgenate	roentgena	roentgeny
rubidium	rubidate	rubida	rubidy
ruthenium	ruthenate	ruthena	rutheny
rutherfordium	rutherfordate	rutherforda	rutherfordy
samarium	samarate	samara	samary
scandium	scandate	scanda	scandy
seaborgium	seaborgate	seaborga	seaborgy
selenium	selenate	selena	seleny
silicon	silicate	sila	sily
silver	argentate ^h	argenta ^h	argenty ^h
sodium	sodate	soda	sody
strontium	strontate	stronta	stronty
sulfur	sulfate	thia ⁱ	sulfy
tantalum	tantalate	tantala	tantaly
technetium	technetate	techneta	technety
tellurium	tellurate	tellura	tellury
terbium	terbate	terba	terby
thallium	thallate	thalla	thally
thorium	thorate	thora	thory
thulium	thulate	thula	thuly

Table X *Continued*

<i>Element name</i>	<i>Anion name^a</i>	<i>'a' term</i>	<i>'y' term</i>
tin	stannate ^j	stanna ^j	stanny ^j
titanium	titanate	titana	titany
tritium	tritrate	trita	trity
tungsten	tungstate	tungsta	tungsty ^k
uranium	uranate	urana	urany
vanadium	vanadate	vanada	vanady
xenon	xenonate	xenona	xenony
ytterbium	ytterbate	ytterba	ytterby
yttrium	yttrate	yttra	yttry
zinc	zincate	zinca	zincy
zirconium	zirconate	zircona	zircony

^a Modified element name used in additive names for heteroatomic anions containing the element as the central atom.

^b From the name stibium.

^c From the name cuprum.

^d From the name aurum.

^e From the name ferrum.

^f From the name plumbum.

^g From the name azote.

^h From the name argentum.

ⁱ From the name theion.

^j From the name stannum.

^k 'Wolframy' was used in "Nomenclature of Inorganic Chains and Ring Compounds", E.O. Fluck and R.S. Laitinen, *Pure Appl. Chem.*, **69**, 1659–1692 (1997)" and in Chapter II-5 of *Nomenclature of Inorganic Chemistry II, IUPAC Recommendations 2000*, eds. J.A. McCleverty and N.G. Connelly, Royal Society of Chemistry, 2001.

Subject Index

Element names, parent hydride names and systematic names derived using any of the nomenclature systems described in this book are, with very few exceptions, not included explicitly in this index. If a name or term is referred to in several places in the book, only the more informative references may be indexed.

A

'a' terms

- element ordering, 42–43
- names of organometallics of group
13–16 elements, 231
- table, 337–339
- use with parent hydrides
 - heteronuclear acyclic compounds,
94–95
 - heteronuclear monocyclic
compounds, 96–98
 - heteronuclear polycyclic compounds,
100–101
 - homonuclear monocyclic
compounds, 87–89
 - homonuclear polycyclic compounds,
89

Abbreviations of ligand names, 63,

153–154

construction, 63, 261

enclosing marks, 21

table, 261–268

Absolute configuration of chelate ring

conformations, 35–36, 191–193, 259

Absolute configuration of coordination

compounds, 185–198

C/A convention, 185, 187–191

octahedral species, 185, 189–190

pentagonal bipyramidal species, 191

see-saw species, 188

square pyramidal species, 188

tetrahedral species, 185–186

trigonal bipyramidal species, 187

trigonal prismatic species, 190

R/S convention, 185–187

tetrahedral species, 186

trigonal pyramidal species, 186

skew-lines convention, 185, 191–193

Acetylide, 293

Acidium, 132 (see footnote d)

Acids (see inorganic acids)

Actinide, 281

Actinoids (vs. actinides), 51–52,

336 (see footnote d)

metallocenes, 227

Acyclic parent hydrides, 86–87, 101

heteronuclear compounds, 94–95

chains and rings nomenclature, 95

homonuclear compounds, 86–87

with non-standard bonding numbers,
86–87

with standard bonding numbers, 86

Acyclic unsaturated hydrides, 87

Addition compounds, 12

additive names, 81

compositional nomenclature, 80–81

formulae, 12, 56, 62

use of centre dots, 28

hydrates, 80–81

indication of composition, 27, 31

names, 12, 80–81

order of components, 41, 80–81

stoichiometric descriptors, 33, 80–81

stoichiometric ratios, 23, 27

Additive nomenclature, 5, 7, 111–123

addition compounds, 81

- and arabic numerals, 38
 - chains and rings, 7, 118–122
 - choice of central atoms, 112
 - coordination compounds, 147–148, 150
 - general naming procedure, 149, 168
 - general characteristics, 111–112
 - heteronuclear chains, 95
 - heteropolyatomic anions, 74–75
 - heteropolyatomic cations, 71–72
 - inorganic acids, 7, 124, 126–134
 - functional replacement derivatives, 139–140
 - ions and radicals, 112
 - radical anions, 10–11
 - ligand representation, 112, 151
 - mononuclear entities, 113–114, 150
 - non-symmetrical dinuclear entities, 116–117, 166–167
 - oligonuclear compounds, 117–118, 167–174
 - order of prefixes, 16–17
 - organometallics, 7
 - of group 1 and 2 elements, 228–230, 232
 - of group 3–12 elements, 201–228, 232–233
 - polynuclear entities, 11, 114–118, 161–174
 - order of central atoms, 42, 162
 - comparison with substitutive names, 83–84, 113–118
 - for parent hydride derivatives, 102–103
 - symmetrical dinuclear entities, 114–116, 166
 - table of names, 280–336
- Affixes, 16
- geometrical, 259
 - structural, 259
- Agostic interactions, 224
- Aims of nomenclature, 3–4
- Alkali metals, 51
- Alkaline earth metals, 51
- Allotropes, 48–51
- Bravais lattices, 50
 - crystalline modifications, 49–51
 - Pearson notation, 49–51
 - with definite molecular formula, 49
 - with indefinite structure, 51
 - solid amorphous modifications, 51
- Alphabetical order
- atomic symbols in formulae, 58
 - central atoms in oxoacids, 125
 - ligand names in coordination compounds, 150, 168
 - ligands
 - bridging *vs.* terminal, 163, 211
 - derivatives of parent hydrides, 60
 - in formulae, 11, 40–41, 43–44, 153, 168
 - in polynuclear compounds, 161–166
 - nuclide symbols, 64
 - in salts, 40
- Aluminium *vs.* aluminum, 249
- Amide, 74, 106, 314
- Amido, 138, 314
- Amino, 109, 314
- Aminyl, 109, 314
- Ammine, 151, 314
- Ammonia and derived ions and radicals, 314
- Ammonio, 314
- Ammonium, 71, 105, 314
- Anion names
- additive names of heteropolyatomic anions, 74–75
 - ‘ate’ forms of element names, 337–339,
 - changed recommendations, 8, 10
 - compositional names, 8, 72–75
 - homopolyatomic anions, 73–74
 - coordination compounds, 162
 - of elements, 72–74
 - homoatomic anions, 336 (see footnote b)
 - endings, 72
 - of ligands, 10–11, 112, 151–152
 - oxoacids
 - hydrogen names, 137
 - simplified names, 137
 - from parent hydrides, 72, 106–108
 - homopolyatomic, 74
 - radicals, 10–11, 73–75

stoichiometric names, 72–75
substitutive names of heteropolyatomic
 anions, 74
 table, 280–336
Antimonic acid, 129
Antimonous acid, 129
Antimony compounds
 names of hydrides, 331
 oxoacids, 129
Antiphase boundary, 244
Aqua, 151, 306
Aquo, obsolete ligand name (see aqua)
Arabic numerals
 in formulae, 31–32
 as locants, 32, 38–40
 in names, 32–34
 and number of hydrogen atoms in
 boranes, 90, 99
arachno-boranes, 90–93, 99, 259
Arsenic acid, 129
Arsenic compounds
 oxoacids, 129
 table of names, 282–284
Arsenous acid, 129
Arsine, obsolete name for arsane, 85 (see
 footnote e)
Arsinic acid, 129
Arsinous acid, 129
Arsonic acid, 129
 derivatives, 125–126, 138
Arsonous acid, 129
Arsoric acid, 129
Arsorous acid, 129
Assemblies, in chains and rings
 nomenclature, 118–119
Asterisks, in formulae, 36
Atom names, 46–47
 of new elements, 47, 250
 tables, 248–250
Atomic number, 47–48
 nuclides, 32
 tables, 248–250
Atomic symbols, 46–47, 248–250
 and isotopic modification, 44, 48
 for new elements, 47, 250
 order in formulae, 58–63

 alphanumeric, 54, 58
 B vs. C, 60
 based on electronegativity, 58–59
 Hill order, 54, 60
 tables, 248–250
Auride, 285
Azide, 73, 318
Azinic acid, 128
Azonic acid, 128

B

Bidentate coordination, 146–147
 vs. didentate, 258
 and skew-lines convention, 191–193
Binary compounds and species
 charge numbers, 70, 73
 element ordering, 42
 formulae, 58–59
 multiplicative prefixes, 69–70
 name construction, 5
 oxidation number, 70
 separation of components, 30
 stoichiometric names, 69–70
 table of names, 280–336
Binuclear (see dinuclear)
Bipyramidal complexes
 absolute configuration, 190–191
 trigonal species, 187
 configuration index, 184–185
 specifying configuration, 184–185
 pentagonal species, 185
 trigonal species, 184
‘bis’ vs. ‘di’ as multiplicative prefixes, 37,
 76, 101, 113, 150–151, 258
Bis(bidentate) complexes, skew-lines
 convention, 191–193
Bis(tridentate) complexes, stereoisomers
 and priming convention, 195–196
Bond indicators, in line formulae,
 26–27
Bonds between central atoms, italicized
 symbols, 115
Boranes (see boron hydrides)
Borate, 127, 285
Boric acid, 127
Borinic acid, 12

- Boron compounds
 order of components of addition
 compounds, 12, 81
 oxoacids, 127, 137
 table of names, 285
- Boron hydrides
 bridging hydrogen atom distribution, 28, 35, 93–94
 heteroboranes, 99–100
catena prefix for chains, 91–92
 common structure types, 90
arachno, 90–93, 99, 259
closo, 90–92, 99–100, 259
hypho, 90, 259
klado, 90, 99–100
nido, 90–94, 99–100, 259
 relationships, 90–92
 cyclic compounds, 92
 cyclo prefix, 92
 Hantzsch–Widman system, 92
 electron counting in clusters, 90
 hydrogen substitution, 104–105
 indicated hydrogen, 35, 93–94
 number of boron atoms, 89
 number of hydrogen atoms, 34, 90
 in heteroboranes, 99
 numbering polyhedral clusters, 92–93
closo vs. *nido*, 92
 order of symbols in carbaboranes, 60
 positions of supplanting heteroatoms, 99–100
 replacement nomenclature, 99
 skeletal replacement, 98–100
 stoichiometric names, 89–90
 structural descriptor names, 90–92
 substituent groups, 104–105
- Boronic acid, 127
- Braces, in formulae and names, 24
- Bravais lattice, and allotrope names, 50
- Bridging index, 43–44
 organometallic compounds, 208–209
 polynuclear coordination compounds, 148, 163–165
- Bridging ligands
 citation vs. terminal ligands, 43–44, 163–164
 coordination compounds, 148
 polynuclear species, 163–165
 hydrogen in boranes, 28, 35, 93–94
 heteroboranes, 99–100
 substitution, 104–105
 in organometallic compounds, 203, 211–212
 bridge index, 43–44, 208–209
 vs. chelation, 203, 208–209
 position in names, 25
 in symmetrical dinuclear entities, 115–116
 use of μ symbol, 36, 163, 259
 multiplicity, 32–33
- Bromic acid, 131, 287
- Bromide, 287
- Bromido, 138, 287
- Bromine compounds and species, names, 287, 298, 319–321
- Bromo
 obsolete ligand name (see bromido)
 prefix in substitutive nomenclature, 101, 138, 287
- Bromous acid, 131, 287
- C**
- C/A convention and absolute configuration, 185, 187–191
 bipyramidal complexes, 190–191
 pentagonal species, 191
 trigonal species, 187
 general considerations, 187
 octahedral complexes, 189–190
 polydentate ligands, 190
 priming convention, 190
 see-saw centres, 188
 square pyramidal complexes, 188
 trigonal prismatic complexes, 190
- Caesium vs. cesium, 249
- Cahn, Ingold, Prelog (CIP) rules for
 coordination compounds, 44, 180, 194–195
 octahedral species, 189
 trigonal prismatic species, 190
- Carbaboranes, 99–100
 order of B and C in formulae, 60

- Carbene, 109, 291
Carbide, 287
Carbon compounds
 oxoacids, 127, 137
 table of names, 287–293
Carbonate, 127, 293
Carbonic acid, 127
Carbonyl, 112, 151, 154, 292
Casey, Evans, Powell (CEP) system for
 polynuclear compounds, 33,
 172–174
 descriptors, 172–173
Catena compounds, 118–119, 121–122,
 128, 133
catena prefix, 259
 and borane chains, 91–92
 and oxoacids, 127–129
Catenacycles, 118–119, 121–122
Catenate compounds, 121, 128–129
Catenium compounds, 121
Cation names
 additive names of heteropolyatomic
 cations, 71–72
 changed recommendations, 8
 compositional names, 70–72
 coordination compounds, 149
 heteropolyatomic species, 71–72
 homopolyatomic species, 71
 monoatomic species, 70–71
 of elements, 70–71
 and parent hydride names, 105–106
 radicals, 70–74
 stoichiometric names, 70–72
 substitutive names of heteropolyatomic
 cations, 71–72
 table, 280–336
Central atoms
 choice in additive names, 112
 in coordination entities, 145, 162
 in polynuclear organometallic
 compounds, 232–233
Central structural unit
 polynuclear clusters, 172–174
CEP (Casey, Evans, Powell) system for
 polynuclear compounds, 33, 172–174
 descriptors, 172–173
Cesium *vs.* caesium, 249
Chain compounds, formulae, 61
Chains and rings nomenclature, 7, 118–122
 assemblies, 118–119
 catena compounds, 118, 121–122, 128,
 133
 catenacycles, 118–119, 121–122
 catenate compounds, 121, 128–129
 catenium compounds, 121
 charge number, 121–122
 connectivity, 118
 construction of names, 121–122
 cyclate compounds, 121–122
 cycles, 118–119, 121–122, 129, 134
 cyclium compounds, 121
 heteronuclear acyclic parent hydrides,
 95
 locants, 119
 modules, 118
 nodal descriptors, 20, 29, 119–120
 oligonuclear compounds, 117
 oxoacids, 128–130, 133–134
 and polynuclear cluster, 174
 radicals, 121
 skeletal atoms, 121
 numbering, 42, 119
 order of citation, 41–42
 symmetrical dinuclear entities, 116
 ‘y’ terms, 121
 table, 337–339
Chalcogens, 51–52
Charge
 in formulae, 57–58
 indication for ions, 25–26, 70–75
 of nuclides, 47–48
 in solids, 27, 240–241
Charge number, 17
 in binary compounds, 70, 73
 chain and ring compounds, 121–122
 and coordination compounds, 152–153
 enclosing marks, 23
 and generalized stoichiometric names,
 77–78
 in hydrogen nomenclature, 12
 ions and radicals, 70–75
 and parent hydride names, 72

- Chelate rings
 - conformation and skew-lines
 - convention, 193
 - symbols for absolute configuration, 35–36, 259
- Chelating ligands
 - vs. bridging, 203, 208–209
 - in organometallics, 202–203, 209–210
 - κ convention, 210–211
- Chelation
 - in coordination compounds, 146
 - and κ convention, 156–160
 - vs. monodentate bonding, 147
 - in organometallic compounds, 202
- Chemical composition of solids, 237–238
- Chemical twinning, 244–245
- Chiral centres, 36
- Chloric acid, 295
- Chloride, 294
- Chlorido, 138, 294
- Chlorine compounds and species, names, 294–295, 299, 319, 312–322
- Chloro
 - obsolete ligand name (see chlorido)
 - prefix in substitutive nomenclature, 138, 294
- Chlorous acid, 131, 294
- Choice
 - between geometries for coordination compounds, 179
 - of central atom in additive names, 112
 - of nomenclature system, 7–9
 - of parent hydride names, 101
 - principal chain, 103–104
- Chromium oxides, names, 295–296
- Chromocene, 225
- CIP (Cahn, Ingold, Prelog) rules for
 - coordination compounds, 44, 180, 194–195
 - octahedral species, 189
 - trigonal prismatic species, 190
- cis* prefix, 67, 179, 259
 - and octahedral geometry, 182
 - and square planar geometry, 180
- cis*-bis(bidentate) complexes, absolute configuration and skew-lines
 - convention, 191–193
- closo*-boranes, 90–92, 99–100, 259
- Clusters
 - boranes, 90
 - numbering, 92–93
 - structural descriptors, 170, 172–174
 - central structural units, 172–174
- Cobaltocene, 225
- Colons
 - and central atom locants, 164
 - in names, 28
- Commas in formulae and names, 29
- Commensurate structures, 242–243
- Composition of non-stoichiometric compounds, 21
- Compositional nomenclature, 5–6, 68–82
 - addition compounds, 80–81
 - generalized stoichiometric names, 75–80
 - vs. hydrogen nomenclature for inorganic acids, 136
 - of ions, 70–75
 - heteropolyatomic anions, 74–75
 - heteropolyatomic cations, 71–72
 - homopolyatomic anions, 8, 73–74
 - homopolyatomic cations, 71
 - monoatomic anions, 72–73
 - monoatomic cations, 70–71
 - order of constituents, 41
 - of radicals, 70–75
 - anions, 73–75
 - cations, 70–72
 - stoichiometric names, 68–70
 - binary compounds, 69–70
 - elements, 69
- Configuration of coordination entities, 149, 174–193
 - absolute configuration, 185–198
 - coordination geometry, 175–179
 - ligand priority, 193–198
 - relative configuration, 179–185
- Configuration index, 33–34, 144, 179–180
 - bipyramidal systems, 184–185
 - CIP rules, 180

- and diastereoisomers, 179–180
- octahedral systems, 182–183
 - reference axis, 182
- priming convention, 195–198
- priority numbers, 180–181, 193–195
- see-saw systems, 185
- square planar systems, 180–181
- square pyramidal systems, 183–184
- T-shaped systems, 185
- trigonal prismatic systems, 190
- Connectivity
 - in chain and ring compounds, 118
 - in organometallic compounds, 201–203, 216
- Constitution of coordination compounds, 149–174
 - formulae, 153–154
 - names, 150–153
 - polynuclear species, 161–174
 - specifying donor atoms, 154–161
- Coordination compounds and entities, 142–199
 - absolute configuration, 185–198
 - C/A* convention, 185, 187–191
 - distinguishing enantiomers, 185–193
 - general considerations, 185
 - octahedral species, 189–190
 - R/S* convention, 185–186
 - see-saw species, 188
 - skew-lines convention, 185, 191–193
 - square pyramidal species, 188
 - tetrahedral species, 186
 - trigonal bipyramidal species, 187
 - trigonal pyramidal species, 186
 - CIP rules, 44, 180, 194–195
 - configuration, 149, 174–193
 - general considerations, 174–175
 - polyhedral symbols, 175–178
 - configuration index, 33–34, 144, 179–180
 - priming convention, 195–198
 - constitution, 149–174
 - definitions, 144–148
 - dinuclear species, 166–167
 - symmetrical, 166
 - unsymmetrical, 166–167
 - formulae, 59–61, 153–154
 - construction, 59
 - enclosing marks, 18–19, 153
 - and ionic charge, 153–154
 - ligand abbreviations, 154
 - oxidation number, 153–154
 - sequence of symbols, 153
 - square brackets, 113–117
 - κ convention, 155–160
 - and η convention, 161
 - use of superscripts, 156
 - ligand priority, 193–198
 - priority numbers, 180–181, 193–195
 - names, 144, 150–153
 - numbering of ligands, 150–151
 - order of ligands and central atoms, 150
 - representing ligands, 151–152
 - naming procedure, 149–150, 168
 - flowchart, 149–150
 - polyhedral symbols, 33–34, 144, 175–179
 - polynuclear clusters, 172–174
 - central structural unit, 172–174
 - CEP descriptors, 172–174
 - polynuclear compounds, 161–174
 - bridging ligands, 148, 163–165
 - metal–metal bonding, 165–166
 - specifying configuration, 179–185
 - bipyramidal species, 184–185
 - configuration index, 179–180
 - distinguishing diastereoisomers, 179–185
 - general methods, 179–180
 - octahedral species, 182–183
 - square planar species, 180–181
 - square pyramidal species, 183–184
 - T-shaped species, 185
 - specifying donor atoms, 154–161
 - italicized element symbols, 155–156
 - trinuclear species, 167–172
- Coordination nomenclature, 7, 147–148
- Coordination number
 - in organometallics, 201–202
 - polyhedral symbols, 176
- Coordination polyhedra, 145–146, 176–178